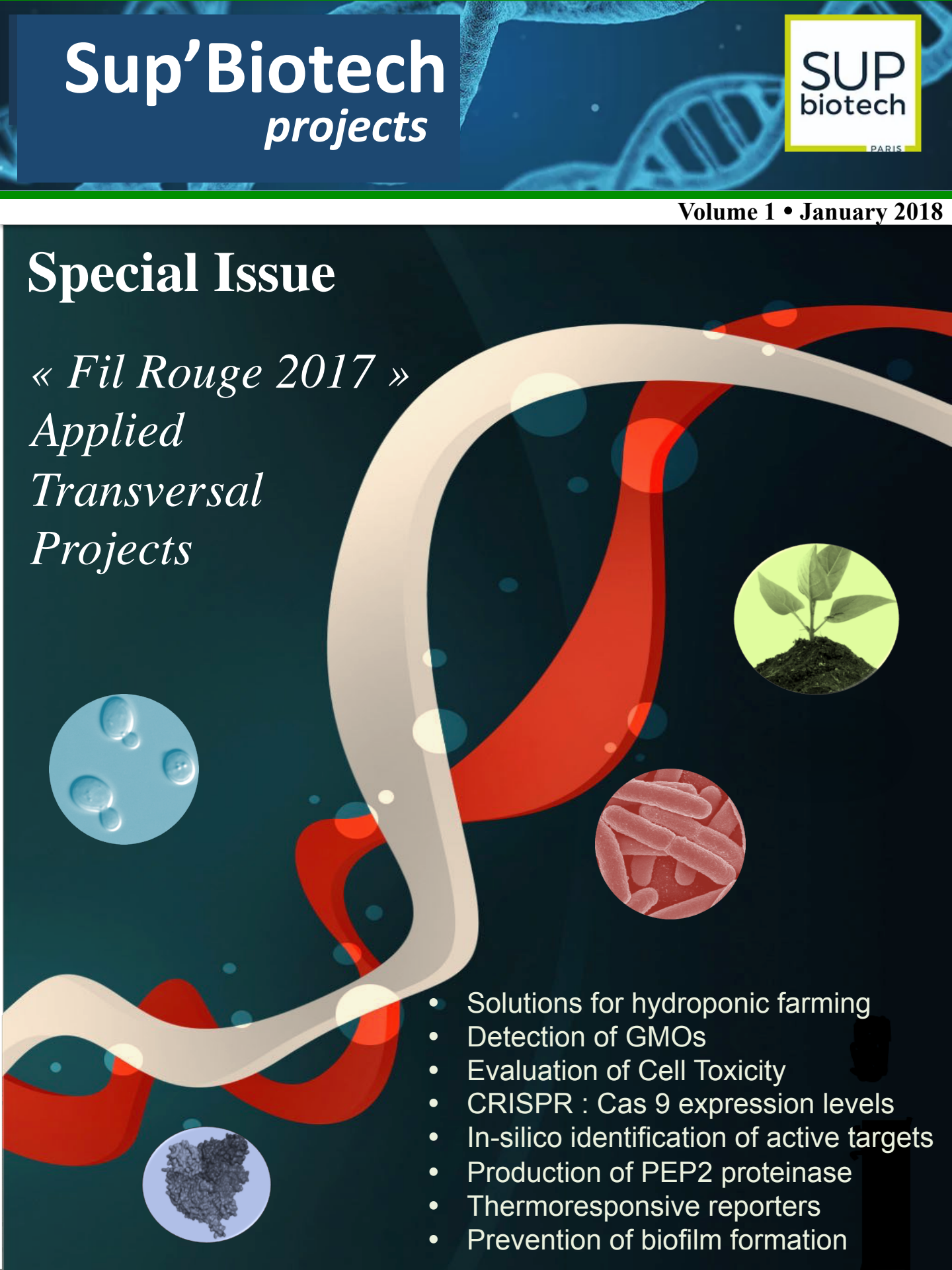


Special Issue

« *Fil Rouge 2017* »
*Applied
Transversal
Projects*

- 
- Solutions for hydroponic farming
 - Detection of GMOs
 - Evaluation of Cell Toxicity
 - CRISPR : Cas 9 expression levels
 - In-silico identification of active targets
 - Production of PEP2 proteinase
 - Thermoresponsive reporters
 - Prevention of biofilm formation

EDITORIAL

A new online journal : Sup'Biotech Projects

We are proud to present the first Issue of Sup'Biotech Projects, an online journal dedicated to the publication of the results of student projects in the format of scientific articles. Work published in the Sup'Biotech Projects online journal is supervised by academic tutors, and when possible, articles are peer reviewed by Sup'Biotech students.

The primary aim of SBPJ is to provide a platform to validate the acquisition of scientific redacting skills throughout the scholarship of Sup'Biotech students. SBPJ is also an excellent vector for communicating the results acquired by Sup'Biotech students during their long-term projects.

About Applied Transversal Projects ("Projets Fil Rouge")

Sup'Biotech is an engineering school, with a 5-year program during which graduates acquire the skills and know-how necessary to work as an engineer in the field of biotechnology. Spanning the 4th and 5th year of academic education, students are required to conduct a project with subjects predefined by a team of academic tutors. Starting with a broad definition of the subject, the students first need to determine which specific aims they wish to work on. The students are then allowed to use the laboratory facilities of the school, and time is assigned to complete these projects (4 independent weeks throughout the year). Students need to design protocols, provide a list of reagents to order, and organize their worktime in the laboratory in order for the project to be completed during the assigned time.

The aim of Applied Transversal Projects is to place students in the position of an engineer at the very beginning of a project, in a startup environment. Tutors are present in the laboratory to provide help but do not otherwise supervise or direct the project.

During the 2017 year, 48 students were recruited for the R&D Applied Transversal Projects, and 8 projects were led under the supervision of 8 tutors. We would like to thank the tutors for the work accomplished during this year, and of course thank all the students who, through their hard work and motivation have managed to obtain and write up the results published in this first issue of Sup'Biotech Projects Journal.

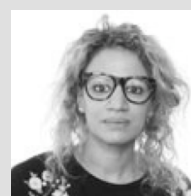
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Project : title of article	Academic tutors
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Detection of GMOs in food products using PCR	Frank Yates Rafika Jarray
Toxicity assessment on HeLa and HEK 293 cells after treatment with Acetaminophen and 5- fluorouracil	Anabelle Planques
Quantify the Cas9 level of expression after transfection in human cells	Rafika Jarray Ferid Nassor
Identification of active targets in bioassays of Ivermectin similar compounds	Jean Yves Trosset
Production and purification of a His-tagged PEP2 protein in Pichia pastoris	Jacqueline Bert
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The posters which were presented by all groups during the special “Fil Rouge Poster Session” are available online at <http://www.recherche.supbiotech.fr/sup-biotech-projects>

Development of a nutritive solution for eco-sourced hydroponic farming: identification and characterisation of the main microorganisms present in compost

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Abstract

Answering the actual water waste issues, a new way of farming develops itself: hydropony. It is an aboveground way of farming in which the ground is replaced by an inert substrate and plant roots float on fertilized water. Composting is the natural process of aerobic decomposition of organic matter producing natural fertilizers and is often used in agriculture. However, the compost's bacterial population is not well-known, so the composting process remains misunderstood. This study consists in the identification and characterisation of the bacteria allowing the compost phenomena, to determine their use in the fertilizing capacities of the compost. It would help moving on the researches for a biological hydroponic nutritive solution.

The first step of bacterial identification from compost samples was a phenotypic approach. The strains identification was assessed with identification percentages and comparison with literature.

In this study, two compost samples of different compositions but at the same composting phase were withdrawn. Twenty isolated colonies were analysed allowing the identification of fifteen different species (three were recurrent). Those species are from different origins and produce different nutrients useful for the fertilizing effects. Identifications will be confirmed with the sequencing approach and sequences alignments analysis. Bacterial population from other compost phases and compositions may be assessed to complete this study.

Keywords: bacterial identification, compost, fertilizer, hydropony, microorganisms, nutritive solution

Introduction

Nowadays, water waste issues have led agriculture to find out new farming ways to spare it. Hydroponic farming is an aboveground way of farming in which the soil is replaced by an inert substrate (clay balls). Plant roots float on mineral-rich water and oxygen that circulate in a closed circuit (**Figure 1**). This water is a nutritive solution that contains the necessary elements for plant growth¹. The advantages of hydroponic farming are that plants are no longer sensitive to drought and flood; the soil quality no longer impacts the crops and it is adapted for farming in urban areas. Hydroponic yields in tomato plants are 5 times higher than classic soil farming². The control of plants nutrition allows the control of growth factors. Thanks to the use of a closed system, 70 to 90% of the water is saved, with the absence of evaporation and infiltration³. Finally, the soil quality is preserved because the nutritive solution is directly absorbed by the plant, without going in the soil.

To grow, besides the need of light, heat and moisture, plants need several essential elements⁴. (1) Nitrogen has a role in quality and yield of productions. It is

transformed into ammonium by microorganisms located in the soil to be usable by plants. (2) Phosphorus has a role in plant growth, allowing efficient rooting, good resistance to drought and is involved in fruit maturation. (3) Potassium has a role in production, transportation and stocking of sugars within the plant. (4) Sulphur is constitutive of many proteins and involved in the composition of 3 essential amino acids. (5) Magnesium has a role in the plant's vital functions, especially photosynthesis and by extension the yield and quality of plants. (6) Calcium is constitutive of cellular walls and fruit conservation. It also allows to maintain the soil's or nutritive medium physical properties and pH. (7) Oligo-elements (boron, copper, iron, chlorine, cobalt, manganese, molybdenum, silicon and zinc) have a crucial role in oxidation-reduction reactions, but are absorbed in very small quantities with a threshold of efficiency very close to the one of toxicity.

Those elements can be brought by fertilizers, that are often used in agriculture to help plant growth. Those fertilizers can be natural (organic) or chemical (minerals) and bring plants essential elements to help their growth, and consequently improve the yield.

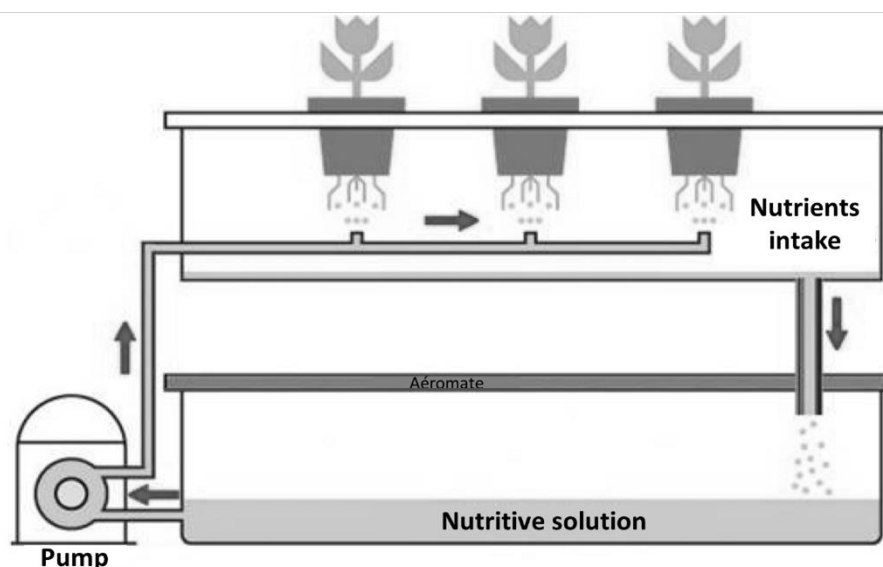


Figure 1. Schematic representation of hydroponic farming
(Modified with authorization from aeromate-71.websself.net/)

Natural fertilizers can be produced through organic wastes decomposition. The natural process of aerobic decomposition of organic matter, such as peelings, herbs, dead leaves, egg shells, is composting. The composting product, compost, is rich in nutrients and can then be used as a natural fertilizer and soil amendment in agriculture to help growing crops. The degradation and decomposition occur thanks to the action of various organisms such as bacteria, fungi, insects...⁵

The decomposition process consists in four main stages that differ in physico-chemical conditions such as temperature, pH but also in the organisms' populations inside. The 4 stages are the (i) mesophilic, (ii) thermophilic, (iii) cooling and (iv) decomposition phases.

(i) The mesophilic phase is the composting initial phase. During it, bacteria and fungi are active, allowing the decomposition of simple molecules (sugar or lipids) and of some polymers (proteins or starch). There is a fast increase of temperature within the centre of the compost, up to 35°C within hours or days, with a strong bacterial multiplication and, during this phase, oxygen is important. There is a production of water vapour and of CO₂ associated to a diminution of the C/N balance and an acidification of the compost.

(ii) The thermophilic phase is the hottest phase (>60°C). The heat, mostly present in the heart of the compost, results in the replacement of mesophilic microorganisms by thermophiles ones. This stops the fungi development but allows the growth of actinomycetes. This phase consists mainly in the decomposition of carbon chains. Since the temperature is higher, the decomposition process is faster. At the end of this phase, the compost will have free nitrogen (NH₄⁺ and NH₃). Due to decomposition, the compost decreases in volume. This

phase ends when the temperature no longer increases, even if the oxygen supply is increased.

(iii) During the cooling phase, there is a decrease of bacteriological activity due to a lack of easily degradable organic matter. The produced heat is smaller than the dissipating heat, resulting in a slow decrease of the compost heat, that drops towards 40-50°C. This led to the arrival of new organisms such as fungi, macro fauna composed of earthworms, sow bugs, manure worms and actinomycetes that will degrade cellulose, hemicellulose and lignin. During the cooling phase, humus begins to be formed by incorporating nitrogen into complex compost molecules.

(iv) Finally, the decomposition phase is the slowest one, the duration varies according to the types of materials inserted. During it, it is mostly the macro fauna that is important. There is a production of minerals that are combined with the degradation products of cellulose and lignin, and form humus. The humus is stable, which means that it can no longer decompose. The temperature is about 30°C but continues to decrease to reach the surrounding temperature.

In the compost, we find macro and microorganisms that can be common or dependent of the stages and the stages' temperatures⁶. The macro-organisms are mostly common insects that we found on the gardens: ants, millipede, snails, slugs, nematode, mites, spiders, fly, earthworm... The microorganisms can be bacteria, fungi... The bacterial populations have not been precisely studied, but it has been shown that mesophilic phase presents *Pseudomonas*, *Bacillus*, *Flavobacterium* and *Clostridium* families whereas the thermophilic phase, hotter, presents *Bacillus* and *Thermus* families^{7, 8}.

Bacteria can be divided into 4 main functional groups.

(i) Decomposers are charged to degrade the organic masses located in the soil to extract nutrients and carbon-based compound easy to digest (simple sugar). Aerobic and anaerobic bacteria fit within this category.

(ii) Mutualistic bacteria form a partnership with plants. The most well-known are those helping to fix nitrogen on the roots. *Rhizobium* is a mutualistic subgroup, consisting in gram – bacilli form bacteria, which forms a symbiosis with the plant roots and furnishes the plant with ammonium obtained from the transformation of free atmospheric nitrogen within nodules and help the plant growth.

(iii) Pathogens are responsible for plant diseases, they are balanced by other bacterial species within the compost, with some of them directly helping to prevent diseases.

(iv) Lithotrophs (chemoautotrophs) bacteria get their energy from non-carbonic sources (sulphur, nitrogen, etc.) and are involved in the recycling of sulphur and nitrogen.

Bacterial identification can be performed by several techniques. The two mains are first a phenotypic

approach, by performing galleries of tests to assess the biochemical characteristics of the studied strains. We can determine families thanks to identification trees depending on microscopic characteristics, and two basic tests (catalase and oxidase), leading then to the use of specific species galleries for a more accurate identification (**Figure 2**). Then, identification can be performed following a sequencing approach, by sequencing the 16s DNA sequences of the strains and then comparing them to studied genomes to identify specifically the studied species⁹.

For the development of a nutritive solution for eco-sourced hydroponic farming, we wish to identify and characterize the main bacterial species that are present in compost. We firstly followed the phenotypic approach for the identification, but the sequencing approach is ongoing to confirm the phenotypic identifications. Due to lack of time, we will assess this step next year. Then, bibliographic researches allowed the characterization of the identified strains to determine their use in the compost.

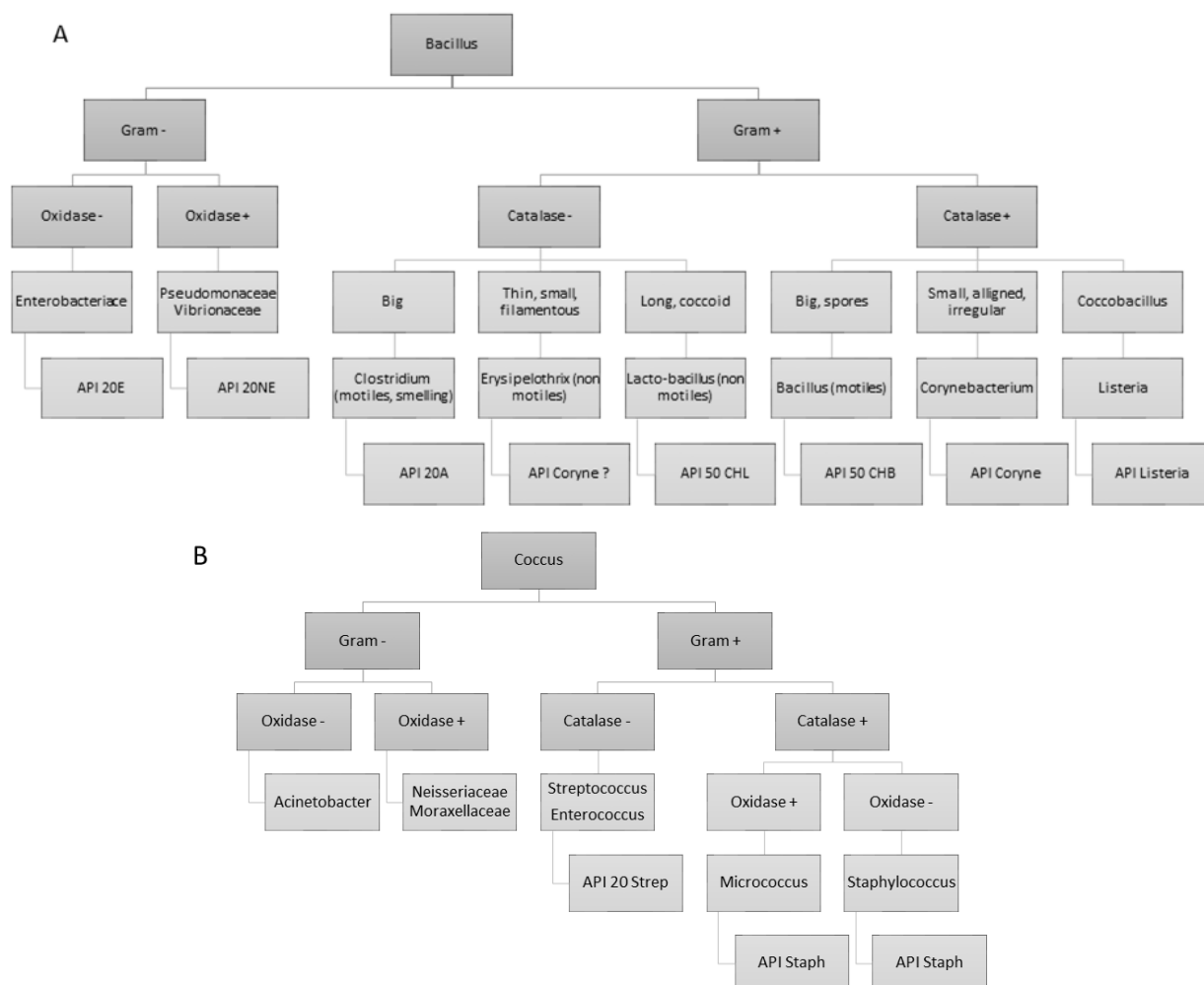


Figure 2. Identification trees to determine the families of A. bacillus or B. coccus bacteria depending on their membrane characteristics (Gram + or -), on their reaction to basic biochemical tests (catalase and oxidase) and on their microscopic morphology. Associated API galleries when existing for further identification is also mentioned.

Material and Methods

Compost samples. Two different compost samples were studied. The first one was collected from a building compost (mostly household wastes, such as fruits and vegetables peelings) and the second one from an external compost (mostly green wastes, such as leaves and herbs). Samples were both in mesophilic stages (the first one) at the time of the collection to compare the population at the same stage from two compost composition.

Culture medium. Cultures were done on LB medium, liquid and agar (10g tryptone, 10g NaCl, 5g yeast extract for 1L in water, 15g agar if necessary). Prepared LB plates were stored at 4°C, prepared LB liquid medium was stored at room temperature (RT).

Sample preparation. Compost samples were diluted in sterile PBS (Phosphate Buffered Saline), spread on LB agar plates and incubated for 48 to 96 hours at 30 and 37°C. Isolated colonies were then transplanted to new LB plates and incubated at corresponding temperatures for isolation checking and further analysis.

Once analysed, bacteria samples were cryopreserved in 20% sterile glycerol (one colony) or in final concentration of 25% glycerol (0.5mL of thick bacteria suspension in 1mL final). Obtained cryostock is stored at -80°C.

Wet mount observation. Shape, regroups and motility of bacteria is assessed with a wet mount observation. A drop of bacteria suspension is placed on a microscope slide and covered with a cover slip and then the observation is realised by microscopy at sense *80.

Gram staining. Shape, regroups and membrane characteristics is assessed with a Gram staining observation. An isolated colony is spread on a drop of water on a microscope slide and then is dried thanks to the flame of a Bunsen burner. The staining is done with three different dyes and washes with distilled water between all steps: (1) crystal violet (staining of the peptidoglycan cell wall of Gram positive bacteria) for 30 seconds, (2) lugol (binding of crystal violet dye) for 30 seconds, discoloration with ethanol, (3) fuchsin (counterstaining) for 30 seconds. The observation is realised by microscopy at lenses *80 and *200 with submersion oil.

Catalase test. An isolated colony is stirred into a drop of hydrogen peroxide 10 volumes. The immediate presence of bubbles indicates the production of catalase by the bacteria, thus being catalase positive.

Oxidase test. An isolated colony is spread on a strip previously humidified with oxidase reagent. A purple/violet staining within 30 second indicates the

production of oxidase by the bacteria, thus being oxidase +.

API galleries analysis. Specific species galleries used were API® galleries (for Analytical Profile Index), from Biomérieux®, according to the kit instructions. API® 20E, API® 20NE, API® Coryne, API® 50CHB and API® Listeria kits were used. API® analyse was performed thanks to the API® identifier available on UPBM website (lab.upbm.org). The identification percentage is automatically given by the API® identifier thanks to similarities to known strains characteristics.

DNA extraction, purification and checking. DNA extraction and purification were realised using a Bacterial Genomic DNA Mini-prep Kit, from Sigma, according to the kit instructions, with a RNase A treatment step as proposed.

The checking was done with a spectroscopy dosage and an electrophoresis. The spectroscopy measure, to determine the purity and concentration of the DNA samples, was realised using a µDrop™ plate, with deposits of 5µL, from ThermoFisher, and the absorbance was read at 230, 260, 280 and 320 nm with elution buffer from DNA purification kit as a blank. A230 and A320 are used to check the presence of solvents and contaminants, the ratio A260/A280 indicates the presence of DNA when comprised between 1,8 and 2 and finally A260 is used to determine the DNA concentration (A260*50*20 gives the DNA concentration in µg/mL). The electrophoresis, to determine the size and quality of the DNA samples, was realized with a 0.8% agarose gel in 1X TAE buffer. Samples were prepared with DNA Gel Loading Dye 6X (ThermoFisher) to have a final concentration of 1X (18µL were loaded) and 2-log DNA ladder (BioLabs®) was used. Electrophoresis was run for 45 minutes at 100V plus 20 minutes at 110V, 150mA. Gel and DNA fragments were observed under UV light.

Results

Bacteria isolation

The first part of the work was a large bacteria isolation from the compost samples on LB medium, to have a standard nutritive medium, on which many species can grow. After compost dilution and incubation at selected temperatures, isolated colonies (173 strains) from the first compost sample were observed to note their morphology and transplanted in new plates. Then, Gram staining analyses were done to note the bacterium morphology (shape, regroups and membrane characteristics) (**Supplementary Table 1**).

The most common type of bacteria in the first compost sample was bacillus Gram -, representing 71% of the studied population. Bacillus Gram + were also largely

detected, with 17%. However, coccus species were underrepresented, with only 5% of the population, almost all being Gram -, except one Gram + (**Figure 3**). In a second time, only the more recurrent colonies have been isolated from the second compost sample (10 strains) and 10 other strains from the first compost sample have been selected, depending on their apparent frequencies (the most common have been chosen). Among the selected strains that were specifically studied, almost all were bacillus (90%), with 50% of the population that were bacillus Gram - and 40% bacillus Gram + (**Figure 4**).

Family identification

Macroscopic (colony) and microscopic (bacterium) observations were realized, with wet mount study and Gram staining, followed by basic biochemical assays (catalase for Gram + and oxidase for Gram - bacteria) to allow the determination of possible families thanks to identification trees (**Figure 2**) (**Table 1**).

Species identification

Then, with family identifications, associated galleries were performed to identify the species of the selected strains. The interpretation was done after 24 and 48 hours, with probabilities percentages for each interpretation and each gallery (**Table 2**).

Within the chosen bacteria population, fifteen different species were identified with API galleries. Some strains appear to be identical in pairs (*Comamonas testosteroni*/Ps. *Alcaligenes*) or triplets (*Corynebacterium ulcerans*, *Aeromonas salmonicida* ssp. *salmonicida*) but most of them occurred only once (**Table 3**).

Identification checking

Then, to validate the species identifications, the genome sequences of the 20 strains should have been analysed. DNA extraction and purification were performed to send the samples to a sequencing laboratory that will perform a Next Generation Sequencing (NGS) and then will allow the strain identification thanks to sequences alignments with MUSCLE/Clustal OMEGA (global alignment). In case of specific subspecies or in the absence of a 100-99.99% matchup, a second alignment with BLAST (local alignment) will be realized to refine the results by identifying specific sequences within the genome.

Discussion

Studied bacteria microscopic characteristics

The first large bacteria isolation and study allowed the determination of principal characteristics' prevalence among the compost bacteria population.

Strains identifications and specificities

For each identified strain, bibliographic researches were done to determine their origin, their pathogenicity and if their genome was already sequenced (and where it can be found) to prepare the future identification through NGS (**Table 3**).

(1) *Aeromonas salmonicida* ssp. *salmonicida* are Gram negative bacteria, living in lightly salted or freshwater. They are responsible of the furunculosis of salmon and trout, an epidemic infection occurring in aquacultures. Contrary to other *Aeromonas salmonicida* species, this strain is not opportunistic and can cause diseases in healthy fishes¹⁰. The three samples identified as *Aeromonas salmonicida* ssp. *salmonicida* were all isolated from the second compost sample. Its presence can then be explained in such quantity if some pieces of fish with this infection were thrown in the composter, allowing the bacteria to spread and take over other bacteria species.

(2) *Corynebacterium ulcerans* are rod-shaped, aerobic, Gram positive bacteria. They are associated with mastitis in cattle. They have previously been shown to degrade crude oil¹¹.

(3) *Comamonas testosteroni*/Ps. *Alcaligenes* are environmental Gram-negative bacteria, commonly found in soil. They have been used in remediation approaches, with ability to degrade polycyclic aromatic hydrocarbons (Ps. *Alcaligenes*)¹².

(4) *Aeromonas hydrophila*/caviae are straight rods with rounded ends Gram negative bacteria. They are found in several seafood (oysters, mussels, shells, shrimps) but also in poultry, meat, raw milk, ice cream and raw vegetables, contaminated by the intestinal portorage of animal or by polluted waters. They can cause gastric issues after the infection, as well as several types of diseases for aquatic creatures¹³. The presence of this strain in the compost could be caused by the addition of contaminated seafood or other meats remains in the composter.

(5) *Arcanobacterium pyogenes* are nonmotile, facultatively anaerobic Gram-positive bacteria. They can be found in gastrointestinal and upper respiratory tracts of cattle, goats, horses, musk deer, pigs and sheep¹⁴¹⁵.

(6) *Bacillus cereus* are motile, facultative anaerobic, beta haemolytic, rod shaped Gram-positive bacteria. They can be found in soil and non-refrigerated cooked aliments. Some strains are harmful while others can play the role of probiotic by competing with *Salmonella* and *Campylobacter* species in the gut microbiota. Some strains are used as microbial additives in composters to enhance the composting process¹⁶, explaining its presence in the compost sample.

Table 1. Colony and bacterium morphologies and biochemical tests results for all 20 strains, with possible bacterium families determined thanks to identification trees.

Compost sample	Strain reference				Colony morphology			Bacterium morphology (wet mount)			Bacterium morphology (Gram)			Biochemical tests	Possible bacterium families
	Temperature	Dilution	Dish	Strain	Size	Shape	Colour	Shape	Motility	Regroups	Shape	Gram	Regroups		
1	30°C	1/100	2	4	small	irregular (halo)	translucent	bacillus short	no		bacillus short	-	heap	Oxidase +	Pseudomonaceae / Vibrionaceae
1	30°C	1/100	3	1	medium	irregular (halo)	opaque	bacillus small	no		bacillus short	+		Catalase +	Corynebacterium
1	30°C	1/1000	1	1	medium	irregular	opaque	bacillus	yes		bacillus	+	long chains	Catalase +	Bacillus / Corynebacterium
1	30°C	1/1000	2	4	small	irregular (halo)	opaque	bacillus small	yes		short bacillus/coccus	-		Oxidase +	Pseudomonaceae / Vibrionaceae
1	30°C	1/1000	3	11	medium	irregular (halo)	opaque	bacillus short + diplococcus	no		bacillus	-	chain	Oxidase +	Pseudomonaceae / Vibrionaceae
2	30°C	10 ⁻⁶	1	1	medium	irregular	opaque, translucent halo	diplobacillus	yes		bacillus	-	heap chains	Oxidase +	Pseudomonaceae / Vibrionaceae
2	30°C	10 ⁻⁶	1	2	big	cloudy	opaque	bacillus	yes		bacillus	+	duo	Catalase +	Bacillus / Corynebacterium
2	30°C	10 ⁻⁷	1	1	big	cloudy	opaque	bacillus	yes		bacillus	-	small chains	Oxidase +	Pseudomonaceae / Vibrionaceae
2	30°C	10 ⁻⁷	1	2	big	cloudy	opaque white	bacillus	yes		bacillus	+	big in duo and small isolated	Catalase +	Bacillus / Corynebacterium
2	30°C	10 ⁻⁷	1	4	big	cloudy	opaque white	bacillus	yes	duo	bacillus	-		Oxidase +	Pseudomonaceae / Vibrionaceae
1	37°C	1/100	3	12	small	round	translucent	bacillus small	no		bacillus	-		Oxidase -	Enterobacteriaceae
1	37°C	1/1000	1	1	big	irregular	opaque	bacillus	no	small groups	bacillus	+	long chains	Oxidase +	Bacillus / Corynebacterium
1	37°C	1/1000	1	4	big	flower	translucent (white center)	bacillus longs motiles/ bacillus short non motiles	yes/no		bacillus	+		Oxidase -	Bacillus / Corynebacterium
1	37°C	1/1000	1	17	big	flower	translucent (white center)	bacillus	yes		cocco-bacillus	-	heap	Oxidase +	Pseudomonaceae / Vibrionaceae
1	37°C	1/1000	1	23	big	round irregular	opaque	bacillus	no		bacillus	-	long chains	Oxidase +	Pseudomonaceae / Vibrionaceae
2	37°C	10 ⁻⁶	1	1	small	round	translucent	bacillus	yes		bacillus	-	short	Oxidase -	Enterobacteriaceae
2	37°C	10 ⁻⁶	1	2	grande	irregular	opaque	bacillus	yes		bacillus	+	chains	Catalase + weak	Corynebacterium
2	37°C	10 ⁻⁷	1	1	big	cloudy	opaque	bacillus	yes	duo	bacillus	-	big, thickset, in chains	Catalase +/- Oxidase +	Pseudomonaceae / Vibrionaceae
2	37°C	10 ⁻⁷	1	2	small	irregular	translucent	bacillus	yes		bacillus	-	big and small, small chains	Catalase +/- Oxidase +	Pseudomonaceae / Vibrionaceae
2	37°C	10 ⁻⁷	1	3	medium	irregular	opaque with halo	bacillus	yes		bacillus	+		Catalase +	Corynebacterium

Table 2. Performed API galleries and obtained identifications for all 20 strains, with the probable species determined.

Compost sample	Strain reference				Possible bacterium families	First API gallery	Result 1	Result 2	Second API gallery	Result 1	Result 2
	Temperature	Dilution	Dish	Strain							
1	30°C	1/100	2	4	Pseudomonaceae / Vibrionaceae	API 20NE	Chryseobacterium indologenes 100% (24h)	Burkholderia cepacia 100% (48h)			
1	30°C	1/100	3	1	Corynebacterium	API Coryne	Corynebacterium ulcerans 100%				
1	30°C	1/1000	1	1	Bacillus / Corynebacterium	API Coryne	Corynebacterium ulcerans 100%		API 50CHB	Geobacillus thermoglucosidarius 99,8%	Bacillus cereus 1 97,6% (48h)
1	30°C	1/1000	2	4	Pseudomonaceae / Vibrionaceae	API 20NE	Vibrio parahaemolyticus 72,5%		API Listeria	-	
1	30°C	1/1000	3	11	Pseudomonaceae / Vibrionaceae	API 20NE	Burkholderia cepacia 99,4%		API Listeria	Listeria ivanovii 100%	
2	30°C	10 ⁻⁶	1	1	Pseudomonaceae / Vibrionaceae	API 20NE	Shewanella putrefaciens group 81,9% (24h)	Aeromonas hydrophila/caviae 100% (48h)			
2	30°C	10 ⁻⁶	1	2	Bacillus / Corynebacterium	API Coryne	-		API 50CHB	Bacillus cereus 2 53,6%	Bacillus mycoides 43,6%
2	30°C	10 ⁻⁷	1	1	Pseudomonaceae / Vibrionaceae	API 20NE	Aeromonas salmonicida ssp salmonicida 100%				
2	30°C	10 ⁻⁷	1	2	Bacillus / Corynebacterium	API Coryne	Corynebacterium ulcerans 99,9%		API 50CHB	Bacillus cereus 2 53,6%	Bacillus mycoides 43,6%
2	30°C	10 ⁻⁷	1	4	Pseudomonaceae / Vibrionaceae	API 20NE	Aeromonas salmonicida ssp salmonicida 100%				
1	37°C	1/100	3	12	Enterobacteriaceae	API 20E	Proteus vulgaris group 62,7%	Proteus penneri 36,2%			
1	37°C	1/1000	1	1	Bacillus / Corynebacterium	API Coryne	Corynebacterium ulcerans 100%		API 50CHB	Bacillus mycoides 93,1%	Bacillus cereus 1 100% (48h)
1	37°C	1/1000	1	4	Bacillus / Corynebacterium	API Coryne	Corynebacterium ulcerans 100%		API 50CHB	(Incubation issues)	
1	37°C	1/1000	1	17	Pseudomonaceae / Vibrionaceae	API 20NE	Ralstonia pickettii 99,1%		API Listeria	-	
1	37°C	1/1000	1	23	Pseudomonaceae / Vibrionaceae	API 20NE	Comamonas testosteroni/ Ps. Alcaligenes 85,3% (24h)	Pseudomonas aeruginosa 95% (48h)			
2	37°C	10 ⁻⁶	1	1	Enterobacteriaceae	API 20E	Proteus mirabilis 99,9%				
2	37°C	10 ⁻⁶	1	2	Corynebacterium	API Coryne	Arcanobacterium pyogenes 63,2%	Corynebacterium ulcerans			
2	37°C	10 ⁻⁷	1	1	Pseudomonaceae / Vibrionaceae	API 20NE	Aeromonas salmonicida ssp salmonicida 100%				
2	37°C	10 ⁻⁷	1	2	Pseudomonaceae / Vibrionaceae	API 20NE	Comamonas testosteroni/ Ps. Alcaligenes 99,4% (24h)	Shewanella putrefaciens group 70% (48h)			
2	37°C	10 ⁻⁷	1	3	Corynebacterium	API Coryne	Corynebacterium ulcerans 100%				

Table 2. End									
Compost sample	Strain reference			Result 3	Result 4	Third API gallery	Result 1	Result 2	Probable identification
	Temperature	Dilution	Dish Strain						
1	30°C	1/100	2 4						Chryseobacterium indologenes
1	30°C	1/100	3 1						Corynebacterium ulcerans
1	30°C	1/1000	1 1			API 20E Bacillus	Bacillus cereus 1 77,7%		Geobacillus thermoglucosidiasus
1	30°C	1/1000	2 4						Vibrio parahaemolyticus
1	30°C	1/1000	3 11						Burkholderia cepacia
2	30°C	10 ⁻⁶	1 1						Aeromonas hydrophila/caviae
2	30°C	10 ⁻⁶	1 2	Bacillus cereus 1 64,7% (48h)	Bacillus mycoides 35% (48h)	API 20E Bacillus	Bacillus thuringensis 61%	Aneurinibacillus aneurinolyticus 30%	Bacillus cereus 2
2	30°C	10 ⁻⁷	1 1						Aeromonas salmonicida ssp salmonicida
2	30°C	10 ⁻⁷	1 2	Bacillus mycoides 70,8% (48h)	Bacillus cereus 1 29,2% (48h)	API 20E Bacillus	Bacillus choshinensis/brevis 91,2%	Bacillus mycoid 43,6%	Bacillus mycoides
2	30°C	10 ⁻⁷	1 4						Aeromonas salmonicida ssp salmonicida
1	37°C	1/100	3 12						Proteus vulgaris group
1	37°C	1/1000	1 1			API 20E Bacillus	-		Corynebacterium ulcerans
1	37°C	1/1000	1 4			API 20E Bacillus	Bacillus licheniformis 99%		Bacillus licheniformis
1	37°C	1/1000	1 17						Ralstonia pickettii
1	37°C	1/1000	1 23						Comamonas testosteroni/ Ps. Alcaligenes
2	37°C	10 ⁻⁶	1 1						Proteus mirabilis
2	37°C	10 ⁻⁶	1 2						Arcanobacterium pyogenes
2	37°C	10 ⁻⁷	1 1						Aeromonas salmonicida ssp salmonicida
2	37°C	10 ⁻⁷	1 2						Comamonas testosteroni/ Ps. Alcaligenes
2	37°C	10 ⁻⁷	1 3						Corynebacterium ulcerans

(7) *Bacillus licheniformis* are mesophilic Gram-positive bacteria commonly found in soil and birds feathers. Due to their high capacity of alkaline serine protease secretion, they have become major bacteria in industrial enzyme production. They have then been subjected to numerous genetic modifications to increase their production or to produce other elements, such as silver nanocrystals. For composting, it has already been used as an accelerant of the composting process^{17, 18}.

(8) *Bacillus mycoides* are spore forming Gram positive bacteria. They are naturally common in the soil and creating rhizoid colonies. They can produce acid from glucose and hydrolyse starch. They are often used in pesticides for their plant pathogenic agents' growth inhibition abilities^{19, 20}.

(9) *Burkholderia cepacia* are lactose-nonfermenting, catalase producing, Gram negative bacteria. They possess an antagonistic activity toward fungi (*Rhizoctonia solani*, *Sclerotium rolfii*, *Pythium spp.*, *Fusarium spp.*, and *Phytophthora capsici*)^{21, 22}.

(10) *Chryseobacterium indologenes* are yellow pigmented, filamentous, non-motile, rod shaped Gram-negative bacteria. They can be found in soil, plants, foodstuffs and water sources²³.

(11) *Geobacillus thermoglucosidasius* are thermophilic (45-70°C) Gram positive eubacteria. They can convert pre-treated lignocellulosic material LCM into ethanol. This specie was first discovered in Japanese soil in 1983²⁴.

(12) *Proteus mirabilis* are facultative anaerobic Gram-negative bacillus. They are very present in water and soils and were described as one of the most commonly found organisms in plant compost²⁵.

(13) *Proteus vulgaris* group are rod-shaped, indole positive, hydrogen sulphide producing, catalase positive Gram-negative bacteria. They can be found in faecal matter, soil and water^{26, 27}.

(14) *Ralstonia pickettii* are rod-shaped Gram-negative bacteria. They can be found in soil, lakes and rivers, as well as in environment with high metal contamination. They currently present a heavy problem for hospitals as they have disastrous effects on humans with poor health, causing nosocomial infections. They can degrade biaryl compounds, such as biphenyl and 4-chlorobiphenyl^{28, 29}.

(15) *Vibrio parahaemolyticus* are straight rod shaped, curved as a comma, halophilic (growth in 0.5 to 10% NaCl), oxidase positive, saccharose negative, aerobic-anaerobic facultative Gram-negative bacteria. They are found in coastal waters worldwide and are present in sediments, fishes, plankton and crustacea. They can cause gastroenteritis infections in humans.³⁰

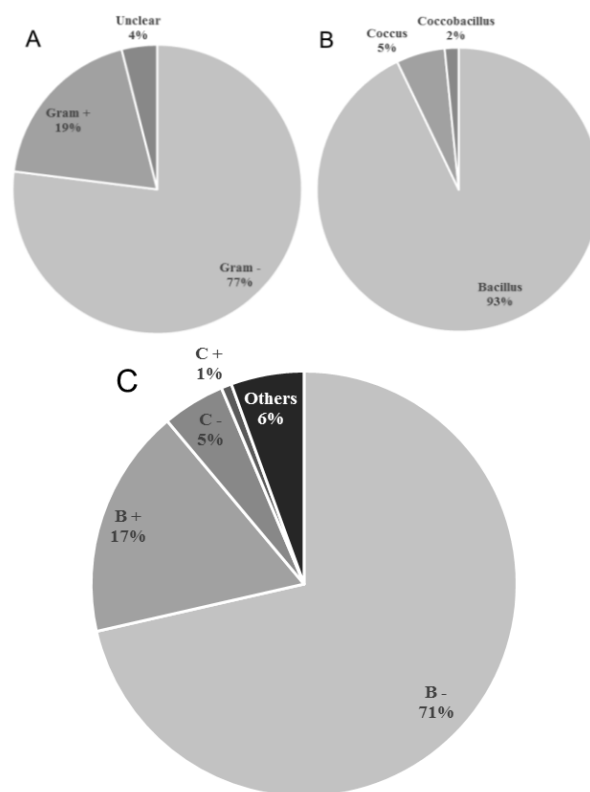


Figure 3. Pie charts representing the repartition of A. Gram characteristics, B. bacteria shapes or C. both microscopic characteristics, among the studied bacterial population from the first compost sample (>120 strains).

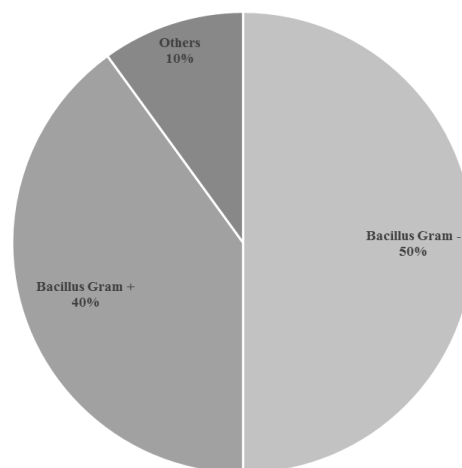


Figure 4. Pie chart representing the repartition of bacillus Gram +, bacillus Gram - and others (coccus, coccobacillus) among the chosen bacteria population (20 strains).

Table 3. Summary of the identified strains and their occurrences.

Identified strains	Occurrence	Sequenced genome available on
<i>Aeromonas salmonicida</i> ssp <i>salmonicida</i>	3	NCBI
<i>Corynebacterium ulcerans</i>	3	NCBI
<i>Comamonas testosteroni</i> /Ps. <i>Alcaligenes</i>	2	NCBI for both
<i>Aeromonas hydrophila</i> /caviae	1	NCBI (non approved genomes for <i>A. caviae</i>)
<i>Arcanobacterium pyogenes</i>	1	NCBI
<i>Bacillus cereus</i> 2	1	NCBI
<i>Bacillus licheniformis</i>	1	NCBI
<i>Bacillus mycoides</i>	1	NCBI
<i>Burkholderia cepacia</i>	1	NCBI
<i>Chryseobacterium indologenes</i>	1	NCBI
<i>Geobacillus thermoglucosidiasus</i>	1	EMBL-EBI
<i>Proteus mirabilis</i>	1	NCBI
<i>Proteus vulgaris</i> group	1	NCBI
<i>Ralstonia pickettii</i>	1	NCBI
<i>Vibrio parahaemolyticus</i>	1	NCBI

Our work allowed the probable identification of bacterial species present in compost. This identification will be confirmed with a future NGS thanks to the previously extracted DNA samples. By comparing the obtained sequences to the known genomes, it will be possible to identify precisely the species of the samples. In a second time, it could be interesting to determine the bacterial population from different types of compost, of known compositions, and from different stages of compost (here, the study was on mesophilic phases compost samples). This step would assess the reproducibility of the bacteria identification in compost. Then, the study of the metabolic characteristics of the commonly found bacteria and of the bacterial population dynamics would help in the determination of

ideal inputs for a compost. Inputs could go through composting process in bio fermenters, to reduce the needed time to obtain usable compost. This ideal compost can then be used in agriculture, allowing the reduction of matter loss but also can be transformed to be used in bioponic agriculture.

Finally, knowing the concentration of essential elements that is produced in such ideal compost would allow to precisely create a nutritive solution for bioponic agriculture.

The study of other macro-organisms in compost and their action could be interesting for the understanding of the first steps of composting process.

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Detection of GMOs in food products using PCR

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This project aimed at developing a method to detect genetic modifications, specifically in GMO (genetically modified organism) maize (*Zea mays*). The MON810 maize has been chosen as a model since it is most often used to combat crop loss due to insects. It has been genetically modified with a gene coming from the bacterium *Bacillus thuringiensis* and coding for the cry1Ab toxin, efficient against insects. Therefore, the strategy of GMO detection is to identify the cry1Ab gene. The method of detection was based on DNA extraction and the supposition that the cry1Ab gene could be detected by amplification through PCR (Polymerase Chain Reaction). Several primers were designed to realize the amplification: primers targeting the cry1Ab gene and control primers, targeting reference genes for *Zea Mays* (hmg gene, NCBI Acc. n° AJ131373) and plant (chloroplast gene, NCBI Acc n°X15901, Z00044). After PCR, the genetic material was then quantified by gel electrophoresis to confirm the presence of the target sequences. In parallel, a preparation of a monoculture of *Bacillus thuringiensis* was produced from a bio-insecticide. The sequence responsible for 'cry' protein synthesis was extracted from the BT spore and used as a positive control.

Nowadays, 32% of maize (*Zea mays*) crops are genetically modified worldwide, making it the second most genetically modified species behind soy [1]. Most of the genetically modified maize strains express resistance to insect pests. For example, the MON810 corn was designed to combat crop loss due to insects in the Lepidoptera order (butterflies and moths), including the European Corn Borer [2]. It has been genetically modified with a gene coming from the bacterium *Bacillus thuringiensis* (BT) and coding for the toxin cry1Ab, efficient against insects [3].

Bacillus thuringiensis is a Gram-positive, aerobic, endospore-forming bacterium. The insecticidal properties of BT are due to crystalline protein inclusions produced during sporulation [4]. The genetic modification of maize aims at providing it the ability to produce these insecticidal proteins. Indeed, when they are ingested by insect larvae, a protoxin is dissolved and activated

under alkaline conditions present in the midgut of target insects, releasing the cry1AB active toxin. It then binds to a specific receptor in the insect's midgut epithelial cells resulting in pore formation in the membrane which eventually leads to disruption of osmotic balance, cell lysis and death [5].

This paper describes a method to detect genetic modifications, specifically in maize, through the identification of the cry1Ab gene. The method of detection is based on DNA extraction with CTAB, a powerful surfactant, and amplification through PCR (Polymerase Chain Reaction). Additionally, a preparation of a monoculture of *Bacillus thuringiensis kurstaki* was produced from a bio-insecticide to extract the sequence responsible for the cry protein synthesis from the BT spore which can be used as a positive control for the presence of the cry1ab fragment in maize.

MATERIALS AND METHODS

Maize types. Several types of maize were used: a French raw corn on the cob coming from the Auchan supermarket, bought the 3th of December (named MS) and another one coming directly from a French field in the Seine-et-Marne department 77 (named MC).

Culture and identification of *Bacillus thuringiensis* (BT) from the insecticide. The bacteria were extracted from a commercial bio-insecticide (Delfin® Jardin). The acetate selection method was used to selectively grow *Bacillus thuringiensis* [6]. Briefly, 0,5g of bio-insecticide was inoculated in LB broth buffered with sodium acetate (0.25 M, pH 6.8) in a 125-ml flask. This mix was placed at 30°C and 250 rpm for 4 hours. A 1-ml aliquot was then heated to 65°C for 10 min in a pre-warmed 5-ml test tube, and undiluted aliquots were spread on LB-agar plates and incubated for 48 h at 30°C. Bacteria were then stored at 4°C for future experimentations.

For identification, bacteria were gram-stained using classical protocol from Bartholomew *et al.* [7].

Genomic DNA extraction. The preparation of CTAB (Cetyl trimethylammonium bromide) buffer required for the maize & leaf DNA extraction involves preparation of TE (Tris-EDTA) buffer 1X. To prepare the TE buffer, Trizma-Hydrochloride (Tris-HCl, Sigma, T5941) and Ethylenediaminetetraacetic acid disodium (EDTA, Sigma, E5134) were used at pH 8.

The CTAB buffer was prepared with 0.5g of CTAB (H6269, Sigma), 2.05g of NaCl (S3014, Sigma) and 0.25g of PVP (Polyvinylpyrrolidone, PVP40, Sigma) in 25 mL of TE buffer 1X (previously prepared).

200mg of maize or leaf was ground with a mortar and pestle, then gently dispersed in 500mL of CTAB extraction buffer. The mixture was placed in a 65°C water bath for

5 minutes. Following this step, 200 µl of chloroformisoamyl-alcohol (C0549, Sigma) were added. After centrifugation (12 000 rpm, 5 min), the upper aqueous phase was discarded. 200µL of isopropanol were added to precipitate the DNA. After centrifugation (12 000 rpm, 15 min), the isopropanol is removed and the pellet was washed with 200µL of ethanol 70%. After centrifugation (12 000 rpm, 5 min), the ethanol was completely removed, and the pellet was re-suspended in 30 µL of TE overnight.

Regarding the bacterial genomic DNA extraction, it was followed as described in Valicente *et al.* [8]. After the extraction, samples were stored at -20°C until further use.

Quantification of extracted DNA. The quantification of DNA after extraction was realized by two methods: spectrophotometry and agarose gel electrophoresis. Spectrophotometry was conducted using a Microdrop (Thermo Scientific µDrop Plate, AN-MRµDrop_0911). In brief, 3 µL of the DNA samples in the microdrop plate was deposited, in comparison with 3 µL of TE buffer (blank). The ratio A260/A280 gives an estimate of the protein contamination of the sample. For a good quality sample, the value should be between 1.8 and 2.0. The ratio A260/A230 estimates the chemical contamination such as phenols, carbohydrates or a high salt concentration. The expected 260/230nm ratio is approximately 2. The DNA concentration can be calculated by using this formula: [DNA] (µg/mL) = Abs260 x 50 µg /mL x (10mm/0,5mm) = Abs260 x 50 x 20. In parallel, a 1 % agarose gel electrophoresis was performed with TAE buffer 1X and 0,5 of 1Kb DNA ladder, at 120V for one hour. In parallel, a 1 % agarose gel electrophoresis was performed with TAE buffer 1X and 0,5 of 1Kb DNA ladder, at 120V for one hour.

Table 1. Sequences of primers designed to amplify plant, maize, and bacterial regions of DNA (Eurofins Genomics)

Name	Sequence (5' -> 3')	% GC	Tm (°C)
Singh-plantf	CGAAATCGGTAGACGCTACG (20)	55	59.4
Singh-plantr	GGGGATAGAGGGACTTGAAC (20)	55	59.4
Singh-maizef	GAAATCCCTGAGCGAGTCGGTA (22)	54.5	62.1
Singh-maizer	GCGATGGCCTTGTTGTTGTACTCGA (25)	52	64.6
Singh-Cry1abf	CGCAGGCCCTTCAACATCGGTATC (24)	58.3	66.1
Singh-Cry1abr	GGTCGGCGCCCAACAACAAGA (21)	61.9	63.7

Primer Design. Three pairs of primers (Table 1) were ordered to Eurofins following the primers design from Singh, C. K., *et al.* [9], to amplify a 550 bp fragment of a plant gene (chloroplast gene, NCBI Acc n°X15901, Z00044), a 175 bp fragment of a maize gene (hmga gene, NCBI Acc. n° AJ131373) and a 1533 bp fragment of the cry1AB gene.

Polymerase Chain Reaction. An annealing temperature optimization for each primer set was performed using the temperature gradient setting. Each 25 µL PCR reaction contained: 1 µM forward and reverse primers, 250 ng DNA, 0.25mM dNTP mixture (Taq Ozyme), 1.25U DNA Taq Polymerase (Taq Ozyme), 1X PCR Amplification buffer (RB), 1.5 mM MgCl₂ and sterile distilled water. PCR settings are listed in Table 2.

Table 2. PCR conditions for amplification of extracted DNA

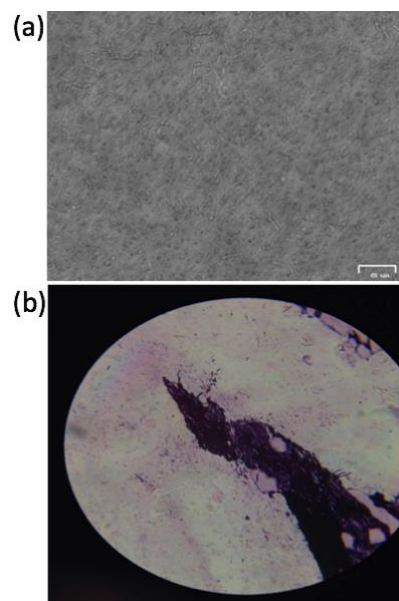
	Temperature (°C)	Time (s)
1	94.0	120
2	94.0	30
3	Gradient 52.5 – 60.0°C	30
4	72.0	60
Steps 2-4 were repeated for 30 cycles		
5	72.0	300
6	10.0	Pause

DNA electrophoretic analysis. 10µL of PCR DNA samples were deposited with

loading buffer 6X and analyzed in 1% agarose gel with TAE buffer 1X, in comparison with 1kb and 100bp DNA ladders. The gel was run at 120 V for one hour. DNA was then observed under the UV light.

RESULTS

Identification of the biological material. Bacteria BT extracted from the biological pesticide were cultured on LB agar plate as described in the methods section.

**Fig 1.** Fresh state *Bacillus thuringiensis* observed with optical microscope X1000 (a) Gram stained *Bacillus thuringiensis* observed with optical microscope X400 (b)

Bacteria were first observed without any coloration to test their mobility and shape. The bacillus shape was confirmed. Figure 1a showed colonial and microscopic morphology identical to *B. thuringiensis*. Figure 1b shows result of the gram- staining, bacillus shape and purple coloration were observed displaying the gram-positive structure expected for *Bacillus thuringiensis*.

Assessment of the quantity of extracted DNA by Microdrop. Three replicates were used for each sample. To examine the purity of the DNA extract, the UV absorption at 230, 260, and 280nm were measured and the A260/280 and A260/230 ratios were calculated (Table 3). The two types of maize did not have a significant effect on the DNA concentration, there were high concentration of DNA in all tubes. Regarding the ratios of contaminations, there were no protein contamination in the samples (the value of A260/A280 was between 1.8 and 2.0). However, the ration A260/230 of the maize from the field (MC) sample suggested the presence of chemical contamination such as phenols, carbohydrates or a high salt concentration.

Concerning the bacterial genomic DNA extraction, the sample was not contaminated by proteins but indicated the presence of co-purified contaminants (phenols, salt or carbohydrates).

Table 3. DNA quantification by Microdrop (MS: maize from the supermarket; MC: maize from the field)

	MS	MC	<i>Bacillus thuringiensis</i>
[DNA] (µg/mL)	1 831	1 395	238
A260/A280	1,93	1,71	1,87
A260/A230	2,09	1,39	1,79

Visualizing extracted DNA by agarose gel electrophoresis. Three replicates were used for each sample of maize and two tree's leaves as positive controls. The extracted maize DNAs are presented in Figure 2. The gel electrophoresis revealed a single, high molecular weight DNA band up to 1kb for the MS (maize from the supermarket) and the MC (maize from the field) samples. The visualization confirms the presence of DNA.

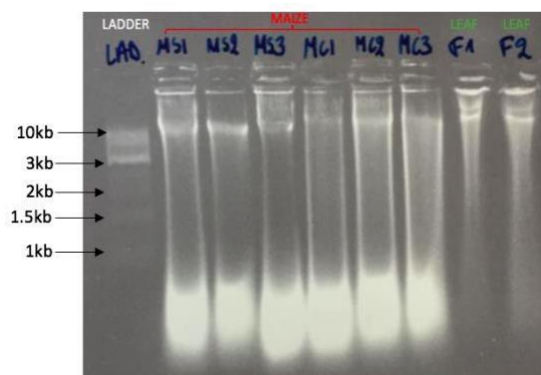


Fig 2. Electrophoresis of extracted maize DNA on 1% agarose gel. MS: Maize coming from the supermarket - MC: Maize coming directly from the field - Ladder: 1kb

The extracted bacteria DNA is presented in the two wells in Figure 3. The gel electrophoresis revealed an expected single, high molecular weight DNA band up to 10kb. Therefore, the visualization verified the presence of DNA in the sample.

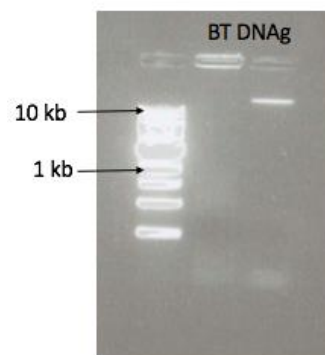


Fig 3. Agarose gel (1%) results showing total DNA extracted from *Bacillus thuringiensis*

PCR Optimization. The optimization of the annealing temperature for maize primers and plant primers was done with DNA from the MS and the MC sample. The primer sets Singh-maizef/Singh-maizer hybridized well within the annealing temperature range of 56-60°C. However, the primer sets were optimal at 60°C (data not shown). The primer sets Singh- plantf/Singh-plantr performed well within the annealing temperature range of 52- 56,8°C with an optimal annealing temperature at 56,8°C (data not shown).

Amplification of a plant gene (chloroplast) and a maize gene (hmga). Two replicates were used for each sample, one positive control (the leaf) and one negative control (TE buffer). The maize primers and the plant primers were used to confirm the presence and quality of DNA extracted from the maize sample. The maize primers and the plant primers successfully amplified the expected regions from the MS and MC DNA samples (Figure 4). Indeed, a band is observed at 175bp which corresponds to the maize fragment (Figure 4a). Regarding the plant gene (chloroplast gene), a band is observed at 550bp for the two different maize samples representing the specific gene (Figure 4b). Therefore, the results confirm the presence of maize DNA in the MS and MC samples.

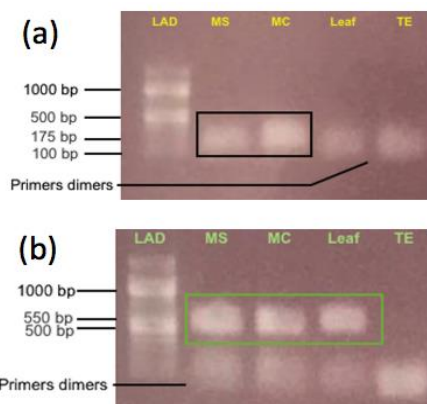


Fig 4. PCR amplification of a maize gene (a) and a plant gene (b)

Amplification of the cry1AB gene.

Three different samples were used: maize DNA from the supermarket (MS), maize DNA from the field (MC) and the bacteria DNA (BT). Three positive control (maize DNA with maize primers for the hmga gene) and a negative control (TE buffer) were utilized. The amplification is presented in Figure 5. All the positive controls were as expected, a 175 bp fragment is presented. However, the 1533 bp fragment expected in the samples is not observed. Therefore, it seems that the maize samples and the bacteria did not have the cry1AB sequence.

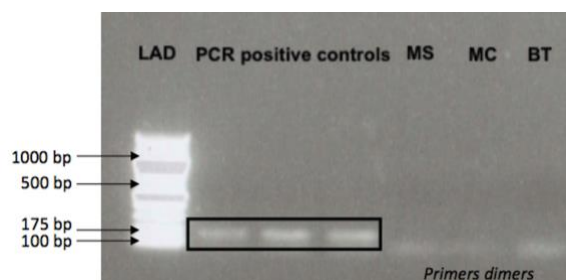


Fig 5. The agarose gel analysis of the amplification products, using the primers specific for cry1AB gene

DISCUSSION

This work aimed at developing a method to detect GMO, specifically in maize, by detecting the presence of the gene cry1Ab. The method of detection is based on DNA extraction and amplification through PCR (Polymerase Chain Reaction), with specific primers that target the cry1Ab gene. Successful PCR products were obtained but a small amount of primers dimers was also observed, suggesting that concentration maize should be optimized, indeed, an overload of primers is suspected. The first step was the maize DNA extraction using the CTAB method.

Cetyltrimethylammonium bromide (CTAB) is a powerful cationic surfactant that has been established as the best detergent to use during the extraction/isolation of highly polymerized DNA from plant material. This detergent simultaneously solubilizes the plant cell wall and lipid membranes of internal organelles, and denatures proteins. It thus promotes the cell lysis [10]. However, some difficulties were encountered, making the DNA extraction challenging to reproduce. Indeed, plant tissues are rich in complex polysaccharides and secondary metabolites like polyphenols. These metabolites interfere and co-precipitate with DNA during the isolation procedure. Thus, some other chemicals like Polyvinylpyrrolidone (PVP) were used to minimize the interference effects of these metabolites [11]. On another hand, chloroformisoamyl-alcohol was used to simultaneously remove glass beads, cell wall debris, denatured proteins, and most polysaccharides. Following these steps, isopropanol was added to precipitate the DNA. The bands obtained after migration of genomic DNA confirmed that this procedure can be validated and allows to have correct concentrations of maize DNA.

Bacteria extraction was performed from the germinated spore of *Bacillus Thuringiensis*, leading to difficulties. Bacteria spore are a resistant form and do not display same characteristics and physiology as the native one, and so do not have the same extraction protocol. In the current literature, there are various existing methods which did not seem to present difficulties. An acetate culturing protocol allowed us to isolate and germinate the spores after several attempts. The DNA extraction in Valicente et al. [8] also presented some imperfections since the DNA yield was not important and with impurities. Proteinase K was suspected to be the cause

of these impurities since it's an enzyme having digestive activity of proteins [12]. Proteinase K step has been optimized by heating at 55°C instead of a possible inactivation at 65°C and could be done in 5 hours instead of an overnight culture. Spectrophotometric assay indicated an increase yield by 4 and less contaminants, allowing to suppose that proteinase K is more active at 55°C.

Microscope observation with and without gram coloration confirmed partially the nature of our bacteria. Indeed, since the bacterial support is from a commercial bio-pesticide (Delfin® Jardin), identification of the strain could have been investigated more precisely. Common techniques are available such as 16S-rDNA [13], RFLP (Restriction Fragment Length Polymorphism)[14], and RAPD (Random Amplified Polymorphic DNA) [15]. These PCR-based techniques are dependent on DNA extraction which has been difficult to proceed and so we were unable to provide a precise and specific identification. However, RFLP and RAPD are known for a lack of reproducibility and reliability, producing false positive and false negative [16]. API Gallery is also a simple and well-documented method to identify specific micro-organism [17], but in our case *B. thuringiensis* spp. *kurstaki* would not have been completely identifiable by this method since there is a lack of specificity in this assay. In Logan and Berkley [17], *B. thuringiensis* API signature is not even distinct from *B. cereus* and *B. mycoides* which are also considered as plant resistance-promoting bacteria.

Successful PCR products were obtained with maize and plant primers, validating the correct PCR protocol. However, no amplifications were observed for the gene cry1Ab. Several facts could explain these negative results. First, we are not totally sure

of the bacteria type as it comes from a commercially available insecticide. Then, the gene may have been modified from the native *B. thuringiensis* gene in the process of obtaining the GMO, which could explain the non-specificity of the primers chosen. For instance, as described in Jabeen R. et al. [17], codon optimization has probably been done to modify the maize DNA. However, DNA sequencing could be done to know the location of mutagenesis points.

FUTURE EXPERIMENTS

More strain identification techniques such as 16S-rDNA should be attempted to identify precisely the bacteria. This technique is reliable enough to identify the genus, the species and even the subspecies of the bacteria tested.

To definitely validate our PCR method to detect GMO in maize, several experiments can be performed. For example, other genetically modified maize could be tested, with the same control primers (maize and plant). Moreover, as we did not obtain any amplification after PCR with cry1ab primers, the use of other sets of primers to detect the cry1ab sequence could also constitute further studies. They could be designed after bacterial DNA sequencing.

CONCLUSION

In this study, we used the PCR method to detect maize GMO. The results showed that the technique could be a potential step in a prototype Do-It-Yourself for the diagnostic of GMO.

ACKNOWLEDGEMENT

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Toxicity assessment on HeLa and HEK 293 cells after treatment with Acetaminophen and 5-fluorouracil

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“Fil Rouge” Project, 2017

Context: Acetaminophen is a commonly used pain killer, which is available without prescription. 5-fluorouracil is an anticancer molecule which is a heavy treatment with a strict regulatory access. These drugs are known to induce apoptosis in some cells type. For instance, their cytotoxicity and mechanism of action has been deciphered in HeLa cell, a well characterized cell line. However, other used cell lines as HEK 293, an embryonic kidney cell-derived cell line very used in cell biology researches, lack some insights about the effect of acetaminophen compared to an anti-cancer drug, 5-fluorouracil, on cell death.

Methods: Cytotoxicity of acetaminophen and 5-fluorouracil on the human embryonic kidney cell line HEK 293 was assessed by carrying out XTT viability assay. In a second time, the cell death mechanism, apoptosis or necrosis, was investigated by performing an AO/BE and Hoechst staining. Finally, a western-blot analysis was done to evaluate the extrinsic or intrinsic character of the apoptosis. HeLa cell line was used as a control, as this cell line is already well known, both in terms of cellular mechanisms and of toxicity.

Results: Hoechst and XTT viability assay demonstrated both 5-fluorouracil and acetaminophen treatment induced cell death by both apoptosis or necrosis for the two cell lines.

However, HEK 293 displayed a better relative viability compared to HeLa. The 5-fluorouracil toxicity increases with incubation time and is maximal at 72h for 10^{-5}M and 10^{-4}M respectively.

Results suggest that acetaminophen maximal effective concentration was reached, as the results were not different between the two different chosen concentrations.

Keywords: acetaminophen, 5-fluorouracil, toxicity, HeLa cells, HEK 293 cells

I - Introduction

We focused on the toxicity of two distinct molecules:

- The first one is an anticancer molecule, the 5-fluorouracil, which is commonly used as an anti-cancer treatment.

5-fluorouracil is used for colon, pancreas, esophagus, stomach or breast cancer [1] and came into medical use in 1962. This treatment is listed on the World Health Organization's List of Essential Medicines, which are the most effective and safe medicines used in a health system. Its mode of action is based on the incorporation into RNA under UTP form and the blocking of the thymidylate synthase. This drug is often among the most used cures in cancer treatment, so we aimed at studying its impact on cells during a treatment.

- The second one is the acetaminophen, which is a pain-killer. This drug is easy to access and can be used by anyone.

Acetaminophen is the most commonly used medication for pain in the United States and Europe, and is normally safe at recommended doses [7]. However, at too high doses, it can result in liver failure [9]. Its effect on liver is well known and apoptosis is one of the cellular mechanism resulting from a potential acetaminophen overdose. [3]

The goal here was to study the toxicological impact of these two drugs *in vitro*, in order to evaluate the mechanism induced by an easy-to-access medicine. 5-fluorouracil is taken as a control treatment, as it is a common anti-cancer treatment and the chosen cell lines are both cancerous [5]. The cell death mechanisms are assessed in HEK 293 cell line and compared to HeLa cell line, which metabolism is already well known [8].

The HEK 293 cell line has been chosen because of its ease to maintain in culture and well

described [12]. However, they are not as documented as HeLa cells, and some cellular responses as apoptosis under cytotoxic drugs remain unclear [6].

II - Material and Methods

a. Chemicals and Reagents

Acetaminophen was purchased as “Doliprane” (SANOFI, FRANCE) tablet. Tablets were reduced to powder, and solubilized in DMEM medium. 5-fluorouracil, DMEM/F12 (HEPES, L-Glutamine), streptomycin, penicillin, acridine orange, ethidium bromide, TGS buffer (with SDS and ethanol 10%), RIPA buffer and Bradford reagent were purchased from Sigma Aldrich (St. Louis, MO, USA). XTT kit, Hoechst 33342 (Trihydrochloride), DAPI (4',6-Diamidino-2-Phenylindole, Dihydrochloride) and DAB staining kit were purchased from ThermoFisher (Waltham, Massachusetts, United States).

Rabbit anti-human cleaved caspase 8, rabbit anti-human caspase 8, rabbit anti-human cleaved caspase 9 and rabbit anti-human caspase 9 were purchased from Cell Signaling Technology (Danvers, Massachusetts, United States).

b. Cell lines and treatments

The human cervical cancer cell line HeLa and human embryonic kidney cell line (HEK 293) were obtained from CEA (Commissariat à l'Energie Atomique et aux énergies alternatives). The cells were cultured in B10 Petri dishes or 96-wells microplate (Dominique Dutscher, France) containing DMEM supplemented with 10% of FBS, 1% of amino acids, 1% of pyruvate sodium, 100 U/ml of penicillin and 100 U/ml of streptomycin at 37°C in a humidified atmosphere containing 5% CO₂. For XTT viability test, 5.10³ HeLa cells/well or 2.5.10³ HEK 293 cells/well were seeded. Cells were plated 24h before treatments for adhering. Then cells were treated for the time indicated on the figures in the presence or absence of

different acetaminophen and 5-fluorouracil concentrations (acetaminophen: 4mg/ml, 8mg/ml [4], 5-fluorouracil: $5 \cdot 10^{-5}$ M, 10^{-4} M [5]).

c. XTT viability test

The cytotoxic effect was measured by using ThermoFisher XTT Cell Viability Assay kit X6493. Cells were plated into 96-wells plates and cultured in a humidified 5% CO₂-containing atmosphere at 37°C for 24h. 10 µL of PMS solution (10 mM) were added to the XTT solution immediately before labeling cells. 50 µL of XTT/PMS solution (4 mg/ml) was added to each well, and the plates were incubated for 2h at 37°C. The absorbance at 450 nm was measured using a Multiskan GO microplate reader from ThermoFisher Scientific.

d. Acridine Orange/BET assay

This staining is used to detect cells apoptosis and necrosis. A 1:1 dye mix was prepared with acridine orange (ThermoFisher) and ethidium bromide (ThermoFisher). Cells were trypsinized and transferred to a 96-wells plate before performing the staining. Cells were counted with a ZOE microscope.

e. Hoechst staining

This staining is used to detect cells apoptosis. Cells were incubated 24h and 30h with test solutions in a 96-wells plate in a humidified 5% CO₂-containing atmosphere at 37°C. After incubation, the supernatant was removed and the staining solution was added in each well and incubated 10 minutes with protection against light. After this treatment the staining solution was removed and wells washed 3 times with PBS before imaging the cells with an excitation wavelength at 350 nm and a reading wavelength at 461 nm.

f. Western-blot

HeLa and HEK 293 cells proteins were extracted using RIPA buffer with centrifugation at 14 000g for 15 min at 4°C. Proteins samples were loaded in a 10% SDS-PAGE electrophoresis. A Coomassie blue staining was realized to confirm the presence of protein on the electrophoresis gel, then they were transferred on a nitrocellulose membrane at 25V at 4°C overnight. After blocking with 5% non-fat milk for 1 h at room temperature, the membranes were incubated with cleaved caspase 8 (1:1000), caspase 8 (1:1000), cleaved caspase 9 (1:1000), caspase 9 (1:1000) antibodies for 1 hour at room temperature. The membranes were then washed with the TBS three times for 5 min each time. The membranes were then incubated with anti-rabbit (1:1000) antibody for 1 h at room temperature. The membranes were revealed using 10mg of DAB reagent in 10ml of PBS with 10µl of H₂O₂.

g. Statistical analysis

Cellular viability by XTT and Hoechst assay results were analyzed with GraphPad (GraphPad Prism Software). Two-way analysis of variance (Anova) was used for statistical comparison. The data presented are the mean ± standard deviation of a duplicate. $P < 0.05$ was considered to indicate a statistically significant result.

III - Results

Acetaminophen and 5-fluorouracil induce cell death on HEK 293 and HeLa cells lines.

XTT analysis demonstrated significant reduced cell viability of HeLa cells treated with acetaminophen at 24h ($P < 0.001$, Fig. 1A) but no effect with 5-fluorouracil. HEK 293 were not affected by any treatment at 24h. After 30h of treatment, a significant cell viability decrease was observed for both treatments on HeLa cells ($P < 0.05$, Fig. 1B) but no significant change in

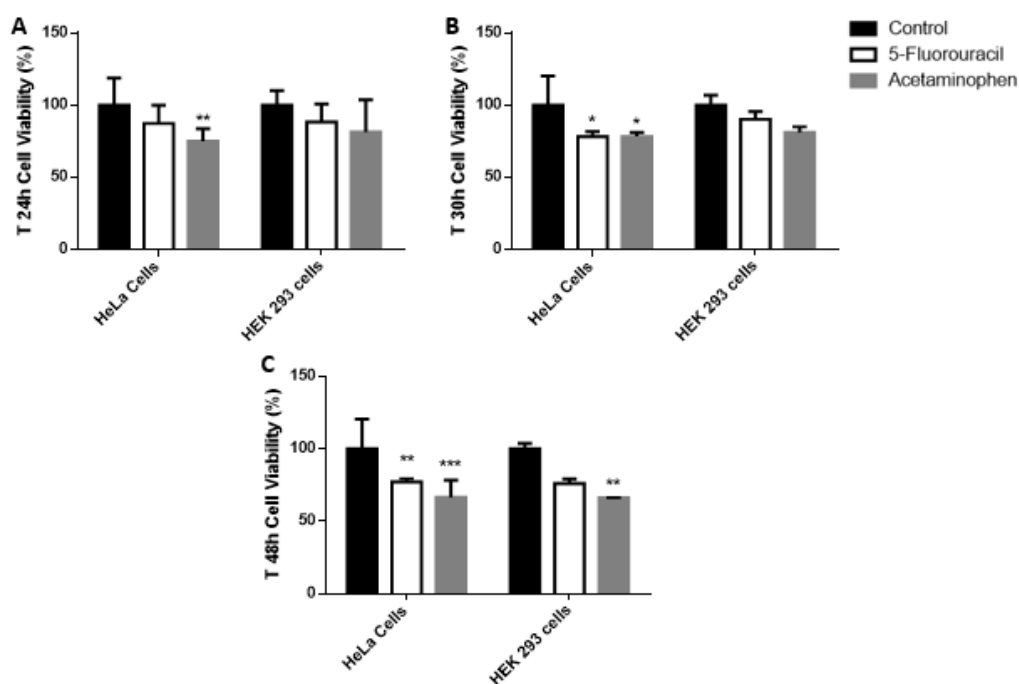


Figure 1: Cell viability of HeLa and HEK 293 cells lines treated with acetaminophen or 5-fluorouracil decreases with incubation time. (A) Cell viability analysis for 24h of incubation. (B) Cell viability analysis for 30h of incubation. (C) Cell viability analysis for 48h incubation time. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ compared with the control group.

HEK 293 cells compared to the control group. After 48h of treatments, cell death was also observed on HeLa cells ($P < 0.01$, Fig. 1C) at similar level that for the 30h incubation results. HEK 293 cells viability was significantly affected by the acetaminophen treatment. HEK 293 seemed to be affected by the 5-fluorouracil but the results obtained were not significant ($P < 0.01$, Fig. 1C). Therefore, HeLa cells are more sensitive than HEK 293 cells for both, 5 fluorouracil and acetaminophen treatments, at all observed times. In addition, drugs-induced cell viability reduction is faster in HeLa cell line compared to HEK 293 cells.

Acetaminophen induce necrosis and 5-fluorouracil induce apoptosis of HeLa cells.

Acridine orange permeates all cells and stains the nuclei in green. Ethidium bromide is taken in cells when membrane integrity is lost and stains nucleus red by dominating the green staining. Consequently, the living cells present a normal green nucleus. The early apoptotic cells have a bright green nucleus because of

pyknosis: the chromatin condense in the nucleus. The late apoptotic cells present a bright orange chromatin due to loss of membrane integrity. Finally, an orange nucleus is observed for necrotic cells. Necrotic cells and late apoptotic cells are both stained in orange but can be differentiated because of their nuclei shapes. Indeed, necrotic cells keep a normal spheric nucleus shape whereas late apoptotic cells have a condensed and fragmented nucleus shape.

Acridine Orange/BET assay seemed to show a necrotic death for HeLa cells after 24h of treatment with acetaminophen (Fig. 2A). On the opposite, results for 5-fluorouracil (Fig. 2B) tend to show an apoptotic cell death.

Acetaminophen and 5-fluorouracil induce a greater apoptosis rate on HeLa cells than on HEK 293 cells.

The Hoechst 33342 staining (2'-[4-ethoxyphenyl]-5-[4-methyl-1-piperazinyl] 2,5'-bi-1H-benzimidazole trihydrochloride trihydrate) is a cell-permeable DNA stain that is

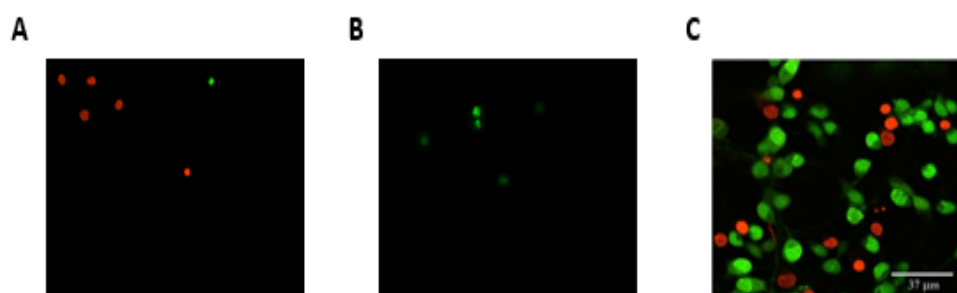


Figure 2: Acridine Orange/BET assay results compared to expected results. (A) Acridine Orange/BET assay on HeLa cells treated with acetaminophen after 24h. (B) Acridine Orange/BET assay on HeLa cells treated with 5-fluorouracil after 24h (C) Expected results [2].

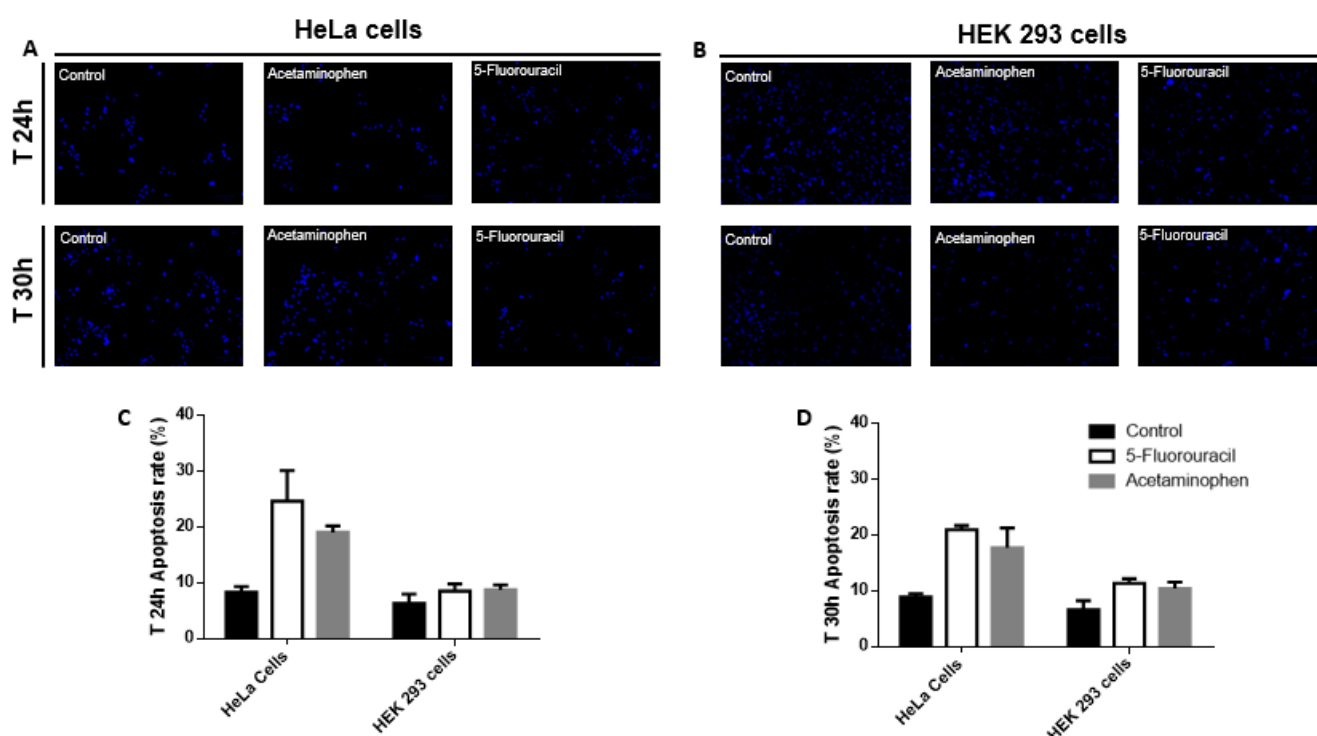


Figure 3: Apoptosis rate in HEK 293 cell lines. (A) Hoechst staining in HeLa cells with acetaminophen and 5-fluorouracil after 24h and 30h. (B) Hoechst staining in HEK 293 cells with acetaminophen and 5-fluorouracil after 24h and 30h. (C) Apoptosis rate after 24h incubation time. (D) Apoptosis rate after 30h incubation time.

excited by ultraviolet light and emits blue fluorescence at 460-490 nm. It binds preferentially to adenine-thymine (A-T) regions of double stranded DNA. The Hoechst 33342 dye was used to distinguish condensed pyknotic nuclei in apoptotic cells (Fig. 3A, 3B). Results show a greater apoptosis rate for Acetaminophen and 5-fluorouracil of HeLa cells than for HEK 293 cells after 24h of treatments (Fig. 3C). This result is confirmed after 30h of incubation with the two different treatments, the apoptosis rate of HeLa cells is

still higher than for HEK 293 cells (Fig. 3D). Those observations are consistent with the viability assay.

A table containing the failed experiments and the explanations has been made and placed in appendix.

IV - Discussion

Preliminary results seemed to indicate that HEK 293 and HeLa cells have different reactions according to the two treatments.

These results could have been interpreted as the fact that HEK 293 cell line have a better resistance than the HeLa one. We hypothesized that it was probably due to a specific renal metabolism related to HEK 293. Another hypothesis is that the growth rate of HEK 293 cells, higher than the growth rate of HeLa cells, leads to an artificial resistance with higher survival rate of HEK 293 cells. Moreover, the statistical analysis of the results tends to confirm this hypothesis.

Also, the death mechanism was determined by Hoechst staining, according to each treatment: acetaminophen seems to induce necrosis rather than apoptosis in both cell lines, showing that this drug presents uncontrolled side-effects. However, 5-fluorouracil provokes more apoptosis than necrosis, which is its primary function. Indeed, an anti-cancer drug is supposed to induce apoptosis in cancerous cells [14]. Acridine Orange/BET assay did not show any accurate results to confirm this observation. Increasing the well's cell density might improve the staining. The extrinsic or intrinsic pathways of apoptosis might be assessed by western blot, with 4 types of antibodies: caspase 8, cleaved caspase 8, caspase 9 and cleaved caspase 9. As the caspase 8 is related to membrane death-mediated apoptotic pathway (extrinsic) and that caspase 9 is related to mitochondrial apoptotic pathway (intrinsic), then the detection of cleaved (activated) caspases would have indicated a predominance of one way rather than other one.

Nevertheless, the Western Blot did not show any conclusive result. The electrophoresis step was let overnight at 25V, which made the proteins migrate over the membrane as any of them was observable. The Coomassie Blue still indicated the presence of proteins as a control.

Therefore, further investigations on apoptotic pathway in HEK 293 is needed, both in acetaminophen-mediated and 5-fluorouracil-mediated cell death.

A previous study deciphered that HeLa acetaminophen-mediated cell death went through caspase-3 activation and release of cytochrome c. [16] As cytochrome c is a major compound in the formation of apoptosome (Apaf-1/caspase-9), their results seem to indicate that HeLa underwent mitochondrial apoptosis (intrinsic) rather than extrinsic [16]. Indeed, extrinsic pathway is characterized by the activation of caspase 8 through Fas receptor while intrinsic pathway is characterized by the formation of apoptosome (Fig. 4). Also, they demonstrated that another major factor of acetaminophen-mediated apoptosis in different cultured cells (as HeLa cells) is a deoxyribonuclease 1-like 3 (DNAS1L3) [16]. HeLa cells are DNAS1L3-deficient, and this endonuclease has been transfected in cells in the study, hypothetically enabling them to undergo this way of apoptosis. It is therefore possible that HeLa cells in our own study showed more necrosis than apoptosis, unfortunately, the experiment did not produce any valuable data. Indeed, they were not transfected with DNAS1L3 and so could not undergo an efficient apoptosis. HEK 293 are also DNAS1L3 deficient, so the same hypothesis could be applied. In order to really compare acetaminophen toxicity with 5-fluorouracil, cell lines transfected with DNAS1L3 caspase 3-dependent are needed to assess apoptosis.

Also, another previous study [15] demonstrated that 5-fluorouracil induced both extrinsic, in Apo-1/CD95 fashion, and intrinsic pathway, via the cytochrome C release. The predominance of either way however has not been deciphered. Therefore, upregulation of genes/proteins implied in intrinsic or extrinsic pathway could be expected in HEK 293 5-fluorouracil-mediated cell death. However, determining dominance of one pathway would require a successful quantification of cleaved caspase 8 and cleaved

caspase 9. The study also showed that HeLa 5-fluorouracil-mediated cell death implied activation of pro-apoptotic factor (as p53) but also reduction of viral oncogenic genes expression in cells, like E6/E7 genes of the HPV virus. As the HEK 293 are embryonic kidney cells transfected with adenovirus, study of the influence of 5-fluorouracil on viral genes in HEK 293 could give some insights about its mechanism of action and targets for viral-mediated cancer.

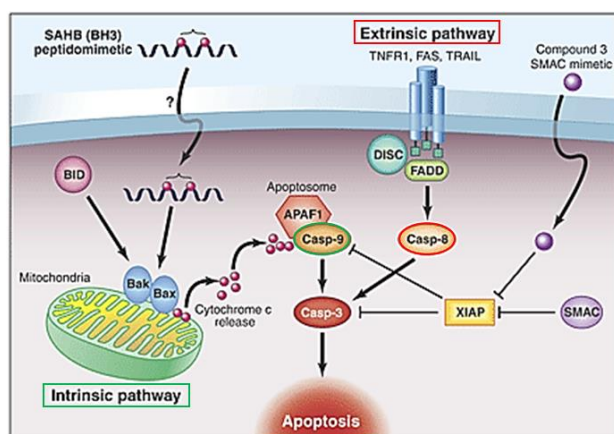


Figure 4: caspase 9 is involved in intrinsic pathway and caspase 8 in extrinsic pathway [13].

V – Conclusion

To conclude this study, several observations have been made: acetaminophen and 5-fluorouracil induce different cell death rate on HEK 293 and HeLa cells lines, suggesting a better resistance of the HEK 293 cells in both cases. This may be an advantage concerning the acetaminophen but a possible drawback for the 5-fluorouracil. Indeed, if cells show a resistance to the treatment, this would slow the recovery. As there are two different cell types, cervix cells for the HeLa and renal cells for the HEK 293, the cell type might be a determinant factor. Further investigations need to be conducted in order to assess the significance of the comparison. It is already well-known that acetaminophen is toxic for the liver cells [10] and the results showed here that acetaminophen is also toxic for both HeLa and HEK 293 cell

lines and so cervix and renal cells. However, our results on acetaminophen-induced cell death might have been disrupted as the HeLa cell line used in this study did not display some factors that seem to be involved in acetaminophen-induced apoptosis and then, they rather undergo necrosis. A better characterization of apoptosis/necrosis ratio in each test could be reassessed with an Acridine Orange/BET assay. Then, the involved ways of apoptosis could be investigated by successfully carrying out a caspase-antibody assay. This can be done both by achieving protein quantification (other techniques than Bradford assay could be more efficient) and Western Blot. To go further, the evaluation of apoptosis ways could be coupled with semi-quantitative RT-PCR analysis of candidate genes and also by using caspase-3/DNAS1L3-transfected cell lines.

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Quantify the Cas9 level of expression after transfection in human cells

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The use of RNA-guided Cas9 nuclease associated with the Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR), derived from microbial immunity, has become a novel promising method for targeted gene editing in any species including humans. Its efficiency highly depends on the Cas9 delivery system in cells, directly correlated with its expression pattern. Uncertainties remain towards the duration of Cas9 expression within the cell following its plasmidic delivery. The aim of our study is to evaluate the expression level of Cas9 protein over time following its transient transfection by a plasmid-based method via a cationic polymer-based molecule in HeLa cells. Cas9 RNA extracted by a previous year team was used as a measure due to poor result of RNA extraction this year. The Cas9 RNA cell content appeared equivalent at 48h and 72h post-transfection. A Coomassie Blue and a Western Blot were performed to quantify the Cas9 protein compared to a BSA range by using the Gelix One®G230 Version 4:1 (biostep®, Burkhardtsdorf, Germany) software. The Cas9 protein was directly detected in cells after 24h and 48h in a similar way, suggesting a long-lasting Cas9 expression. More experiments have to be realized in the future as the measure of the Cas9 expression on a wider time range (from 24h to 7 days for example) to allow a more precise insight on the Cas9 expression compared to other delivery techniques and evaluate which appears optimal for an efficient genome editing.

Keywords

CRISPR; Cas9; recombinant plasmid; cell transfection; gene expression; protein synthesis

Upon more than a decade of basic research into the Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR) repetitive elements, derived from a microbial adaptive immune system, the CRISPR-Cas9 tool was finally fully developed for application in mammalian cells in 2013 (Cong L. et al, 2013 & Mali P. et al, 2013). The CRISPR-based adaptive immune systems in prokaryotes rely on the insertion of an invader sequence copy within the CRISPR loci, and the subsequent synthesis of short transcripts named crRNA. These non-coding RNAs serve as guides for targeting

the CRISPR-associated RNA-guided Cas endonucleases to invading nucleic acids and silencing them at the DNA or RNA level (Terns M.P. and Terns R.M., 2011).

Among the diverse Cas proteins existing in microbial immune systems, the site-specific Cas9 nuclease recently appeared as a promising basis for developing a novel genetic engineering tool. By replacing the guide sequence by a particular sequence of interest matching the target, genomic modifications can be performed at any precise location in the genome of numerous eukaryotic organisms

including humans. After the targeted gene is silenced due to the ability of Cas9 to induce a Double Strand Break (DSB), the activation of a DNA damage repair pathway allows a re-ligation process, resulting either in the gene knock-out or the replacement of a defective gene by its functional analogue. The presence of errors in this repairing process results in the gene modification (Zhang F. et al, 2013). The ease and efficiency of genome editing allowed by this technique was the leading reason for its global adoption by the scientific community, and constituted a significant turning point in genetic engineering (Mali P. et al, 2013).

The reconstitution of the RNA-guided nuclease function of Cas9 in mammalian cells requires the delivery of two essential elements: Cas9 protein and RNA components including the guide RNA (Zhang F. et al, 2013). The growing interest towards this technology justifies the search for optimized heterologous Cas9 delivery and expression methods, constituting a limiting step in genome editing efficiency (DeWitt MA et al, 2017). Two parameters must be taken into account: the Cas9 format (plasmid DNA, mRNA or protein) and the delivery method (electroporation, ribonucleoprotein complex, lipid-mediated transfection, lentiviral delivery...). Transient transfection using chemical compounds is the most common strategy due to its low toxicity for the cells. Precedent studies have suggested that Cas9 nuclease remained longer in the cell when delivered in the form of a plasmid, and therefore induced more off-target mutations (OT) (Liang X. et al, 2015). Indeed, Cas9 appearance in cells transfected in those conditions followed an increasing pattern over at least 72h. On the contrary, when delivered as RNA or protein, Cas9 expression was globally lower and reached a maximum as early as four hours post transfection. However, uncertainties remain towards the exact duration of Cas9

expression within the cell following its plasmidic delivery (Liang X. et al, 2015).

In the present study, we investigated the expression level of Cas9 after a DNA transfection to precise its persistence in human cells and further determine the optimal time for a guide RNA parallel delivery. We measured Cas9 RNA and protein levels after 24h, 48h and 72h following its transient transfection by a plasmid-based method via a cationic polymer-based molecule.

MATERIALS AND METHODS

Materials. HeLa cells, pSpCas9(BB)-2A-GFP (PX458), pEBV-YFP(2347), Gibco® 10% Fetal Bovine Serum (FBS), Microdrop™ Plate and water bath from Thermo Fisher Scientific, Dulbecco's Phosphate Buffered Saline (PBS), Trypsin-Versene EDTA, 10cm Culture dish, 12-well plate, Zoe™ Fluorescent Cell Imager, Thermal cycler, BioRad iScript™ cDNA Synthesis kit, Mini-PROTEAN Tetra Cell, PrecisionPlus Protein Kaleidoscope Standard and 1kb ladder from Biorad, Malassez Cell and lamella, JetPrime^R kit from polyplus transfection, Agarose gel 1%, Buffer 3.1 (100 mM NaCl, 50 mM Tris-HCl, 10 mM MgCl₂, 100 µg/ml BSA, pH 7.9@25°C) from NEB, BglII restriction enzyme, PvuI restriction enzyme, EcoRV restriction enzyme, NotI restriction enzyme, Loading dye, RNeasy Mini Kit from Qiagen, Dimethylsulfoxide (DMSO), Taq'Ozyme kit, 4X Laemmli Buffer, Centrifuge MiniSpin from Eppendorf, β-mercaptoethanol, Heat Blocker from Stuart, (Bovine Serum Albumine powder (BSA) 96%, Acrylamide/Bis-Acrylamide 30%, Tris Base, TEMED, Coomassie Blue, Decoloration solution, Nitrocellulose

membranes, Whatman blotting papers, Western-Blot grade skimmed-milk powder, Tween 20, Anti-CRISPR/Cas9 antibody, Mouse Monoclonal clone, 3-3' Diaminobenzidine, HRP conjugated Anti-mouse IgG concentrate, Dulbecco's Modified Eagle Medium (DMEM) and Non-Essential Amino Acid solution from Sigma-Aldrich, Gelix One®G230 Version 4:1 (biostep®, Burkhardtsdorf, Germany).

Methods.

Mammalian cell cultures. The cell line was obtained from ATCC. HeLa cells were cultured in Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% Fetal Bovine Serum (FBS, Gibco™, ThermoFisher Scientific), 1% Non-Essentials Amino Acids (100X-M7145, Sigma) and 1% L-Glutamin-Penicilin-Streptomycin solution (G1146, Sigma) in 10 cm culture dishes. Cells were passaged 1 time and seeded in a 12-well plate 24h prior to transfection.

Cell transfection. HeLa cells were transfected using the cationic polymer-based jetPRIME transfection kit (PolyPlus-transfection) 24 hours after the first passage, at a density of 200 000 cells per well. For one well of a 12-well culture plate, 0.75µg of DNA was diluted into 75µL of jetPRIME buffer and 1,5µL of jetPRIME reagent was added before a 10min incubation at RT. 200µL of transfection mix was distributed in each

well and replaced by growth medium after 4h. Cells were transferred to -80°C either 24h or 48h post-transfection and stored until the extraction step.

Plasmids used. pSpCas9-2A-GFP plasmid was used and provided by Addgene as the Cas9 expression vector in mammalian cells (Zhang F. et al, 2013).

pEBV-YFP(2347) plasmid was also used, as a positive transfection control and a negative Cas9 expression control. This plasmid has been given by Dr. Denis Biard (CEA/DRF/Jacob/SEPIA).

2.5 Plasmid digestion. The Cas9 plasmid was digested with 3 different restriction enzymes. The digestion mix was prepared as follows (Table 1).

Table 1: Composition of digestion mixes

Digestion	Mix digestion
Digestion 1	Buffer 3.1, BglII restriction enzyme, PvuI HF restriction enzyme, native plasmid and distilled water.
Digestion 2	Buffer 3.1, EcoRV restriction enzyme, NotI restriction enzyme, native plasmid and distilled water.
Digestion 3	Buffer 3.1, NotI restriction enzyme, PvuI restriction enzyme, native plasmid and distilled water.

After 1 hour of incubation at 37°C water bath, loading mixes were prepared as follows (Table 2).

Table 2: Composition of loading mixes for digestion product migration along with the molecular weight of the bands theoretically expected.

Samples	Mix	Expected bands*
Ladder 1kb	Ladder, loading dye, distilled water.	
Native plasmid	Native plasmid, loading dye, distilled water.	9289 bp
Digestion 1	Digestion mix 1, loading dye.	6406 bp and 2883 bp
Digestion 2	Digestion mix 2, loading dye.	5964 bp and 3325 bp
Digestion 3	Digestion mix 3, loading dye.	7809 bp and 1480 bp

*The expected sizes were predicted using the ApE software.

Mixes were loaded onto the 1% agarose gel, and the migration was run at 110 volts for 30 minutes.

RNA extraction. 200 000 transfected cells were thawed on ice. Total RNA was extracted using the silica membrane-based RNeasy Mini Kit (Qiagen) technology. Cell samples were lysed and homogenized in the presence of a highly denaturing guanidine-thiocyanate-containing buffer (RLT). 70% ethanol was added to the homogenized lysate, and 700µL of the sample was applied to the RNeasy Mini spin column placed in a 2mL collection tube. Three washing steps were performed using RW1 and RPE buffers, along with two 15s and one 2min centrifugation at 10 000 rpm. Attached RNA was eluted in RNase-free water and stored at -80°C until the reverse-transcription step.

Spectrophotometry measurements. RNA content and quality were assessed by spectrophotometric measurements using a µdrop plate (Thermo Fisher Scientific).

Either 2µL or 5µL of samples were pipettes onto the plate and absorbances were measured at 230 nm, 260 nm and 280 nm on a Multiskan GO Microplate Spectrophotometer (Thermo Fisher Scientific). RNase-free water was used as a blank, and its values were automatically subtracted from the absorbance obtained for the samples. A260/A230 and A260/A280 ratios were automatically calculated, and the concentration in µg/mL was calculated using the formula $A_{260} \times 20 \times 40$.

Cas9 cDNA amplification by RT-PCR.

Reverse transcription (RT) of RNA has been performed with BioRad iScript™ cDNA Synthesis kit. A mix of RNA template, Nuclease-free water, 5X iscript reaction mix, iscript Reverse Transcriptase has been prepared. The mix has been run in a thermocycler (BioRad) according to the following program (Table 3).

Table 3: Conditions for reverse-transcription of RNA

Annealing	25°C 5 minutes
Extension	42°C 30 minutes
Reverse transcriptase's stop	85°C 5 minutes
Conservation	10°C Infinite

Polymerase Chain Reaction (PCR) has been performed with Taq'Ozyme kit. The mix was prepared with Nuclease-free water, Reaction Buffer 10X, MgCl₂, dNTP PreMux, Upstream primer, Downstream primer, Taq' Ozyme and DNA matrix. The mix has been run in a thermocycler (BioRad) according to the following program (Table 4).

Table 4: PCR conditions for amplification of cDNA

Initial Denaturation		95°C 2 minutes
Cycle X35	Denaturation	95°C 30 secondes
	Annealing	55 or 58°C 30 secondes
	Extension	72°C 1 minute
Final Extension		72°C 5 minutes
Conservation		4°C Infinite

The couple of primers was used from a previous study to amplify the Cas9 gene

(Senturk S. et al, 2017). One primer couple was used to amplify the TBP domestic gene. An overview of the primers used along with the lengths of the subsequent fragments generated are presented in Table 5.

Table 5: Sequences of primers to amplify Cas9 and TBP regions of cDNA expression products, along with the corresponding fragment length.

Name	Sequence	%GC	TM (°C)	Amplicon length (bp)
SpCas9-F	5'-CCCAAGAGGAACAGCGATAA-3'	47	53	220
SpCas9-R	5'-TTGGCTTCCAGAAAGTCGAT-3'	45	55	
TBP-F	5'-CTCACAGGTCAAAGGTTTAC-3'	45	52	90
TBP-R	5'-GCTGAGGTTGCAGGAATTGA-3'	50	57	

RNA/DNA electrophoresis on agarose gel. The 1% agarose gel was prepared in Tris Acetate-EDTA (TAE) buffer and heated for 2 min 30. 3μL Ethidium Bromide solution (BET) (Sigma-Aldrich) was added in 100 mL. For each RNA/DNA extract, 5 volumes were added with 1 volume of 6X gel loading dye (NEB). Migration was run at 110V for 30 min and revelation was made under UV-light.

Protein isolation and Western Blot analysis. Proteins were extracted using Laemmli Buffer. Cell lysates were resuspended in 2X Laemmli Buffer (1% Bromophenol Blue, 1.5M Tris-HCL pH 6.8, glycerol, SDS) and heat at 95°C during 5 minutes in a Heat Blocker. Then, the tube has been centrifuged at maximum speed for 1 minute. The supernatant containing proteins have been transferred in a new tube.

The goal was to load 25 µg of total protein cell lysate in each well. Due to the quantification measurements problems connected to Laemmli buffer we could not obtain a precise quantification of the proteins. We had to estimate the proteins quantification by BSA range (Figure 7). Eventually, 20 µL of sample was loaded per well to SDS PAGE for the electrophoresis corresponding to the equivalent of the protein content of 200000 cells. Proteins were then transferred onto a nitrocellulose membrane. Afterword the amount of the protein was quantified from the picture (Figure 7) by software Gelix One®G230 Version 4:1 (biostep®, Burkhardtsdorf, Germany). In comparison with a BSA range the amount of protein was calculated with a calibration curve and mathematical equations (Figure 8).

Western blot procedure was conducted under a specific condition for 2 hours. Immunodetection was made by incubation with Primary antibodies (Cas9 (7A9-3A3) Mouse mAb #14697 AbCam) at a 1:1000 dilution in blocking buffer consists 5% skimmed milk powder in PBS + Tween 20 for 1 hour. After washing, the membrane was incubated for another 1 hour with Secondary antibodies (HRP-conjugated Anti-Mouse IgG Concentrate (Item I2)) that were diluted to 1:1000 in blocking buffer consists 5% skimmed milk powder in PBS + Tween 20.

The secondary antibodies conjugated with Horseradish peroxidase (HRP) were used in order to obtain good bands visualization. The final coloration was made by the activation of HRP by the 3, 3'-diaminobenzidineHydrogen (DAB), a substrate of HRP that yields to a brown

precipitate in presence of HRP and peroxide.

RESULTS

Cas9 plasmid verification. Before transfecting HeLa cells with the pSpCas9-2A-GFP plasmid, a selective digestion using pre-identified restriction enzymes was performed to verify its integrity. The digestion products were migrated on an agarose gel and revealed under UV-light (Figure 1).

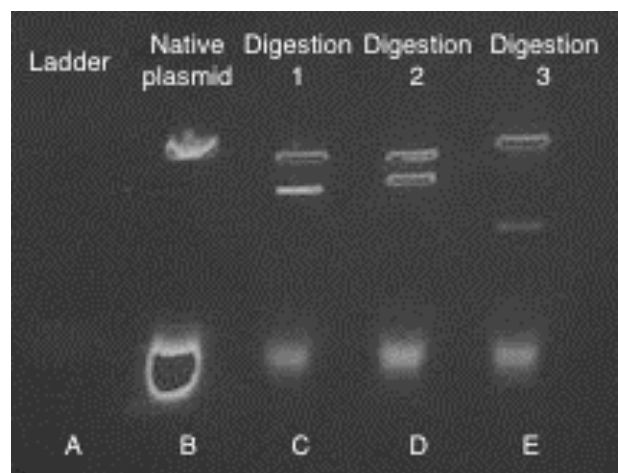


Fig 1: Digestion of the pSpCas9-2A-GFP plasmid using several restriction enzyme couples. (A) 1kb ladder (B) Native pSpCas9-2A-GFP plasmid (C) Native plasmid digested with BglII and PvuI-HF (Digestion 1) (D) Native plasmid digested with EcoRV and NotI (Digestion 2) (E) Native plasmid digested with NotI and PvuI (Digestion 3).

The ladder was not observable on the gel so the exact size of the bands obtained could not be determined. However, the relative heights of the bands present for each sample appeared consistent with the different sizes theoretically expected. For the native plasmid, the single band obtained appeared thicker than those of the digestion fragments suggesting a proper plasmid integrity before digestion. Based on this assumption, and given the plasmid

molecular weight is known to be 9288 bp, this band could serve as a reference for the approximation of digestion fragment sizes. The digestion 1 gave 2 bands close to those given by the digestion 2 as expected. (6406 bp and 2883 bp for Dig.1 and 5964 bp and 3325 bp for Dig.2). The two fragments obtained following digestion 3 were respectively higher (7809 bp) and lower (1480 bp) than those obtained with the previous digestions matching the theoretical expectations.

HeLa cells transfection. HeLa cells were transfected with the pSpCas9-2A-GFP plasmid and observed under a fluorescent microscope to verify the proper plasmid insertion and expression compared to a positive control (pEBV-YFP(2347)).

Non-transfected HeLa cells were not visible under the ZoeTM Fluorescent Cell Imager due to the absence of the GFP according to the experiment.

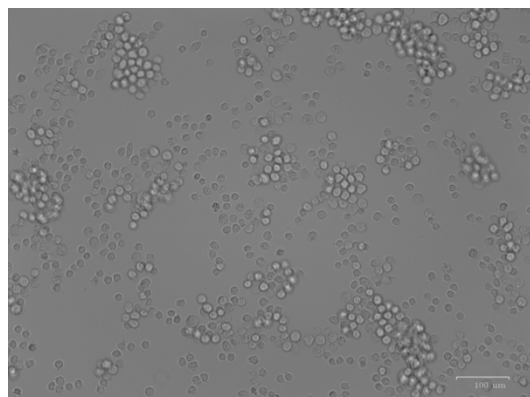
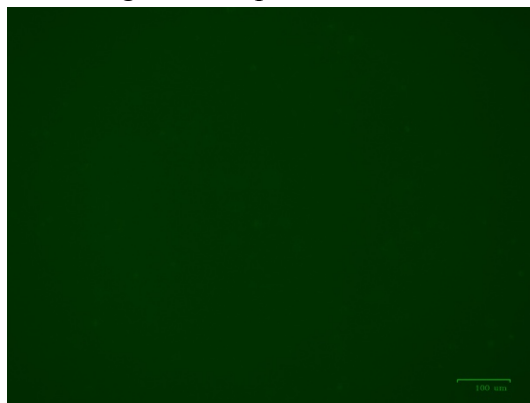


Fig 2: Non-transfected HeLa cells observed under ZoeTM Fluorescent Cell Imager.

Visual aspect of HeLa cells was observed 48h post-transfection with the pSpCas9-2A-GFP plasmid. Some cells clearly appeared fluorescent, indicating that the Cas9-GFP plasmid was properly inserted and expressed within those cells.

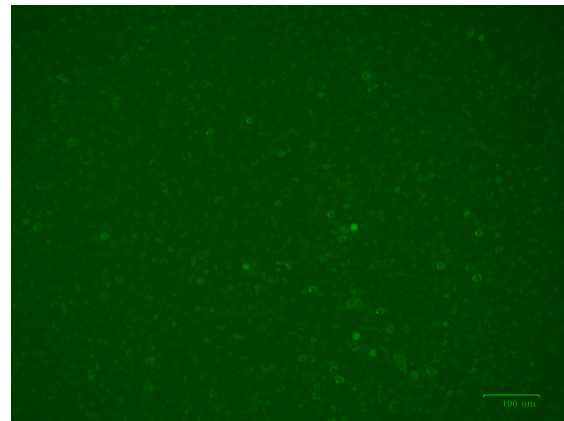


Fig 3: HeLa cells 48h post-transfection with pSpCas9-2A-GFP observed under ZoeTM Fluorescent Cell Imager.

Visual aspect of HeLa cells was observed 48h post-transfection with the pEBV-YFP(2347) plasmid. Some cells appeared fluorescent, indicating that the pEBV-YFP plasmid was properly inserted and expressed within the cells.

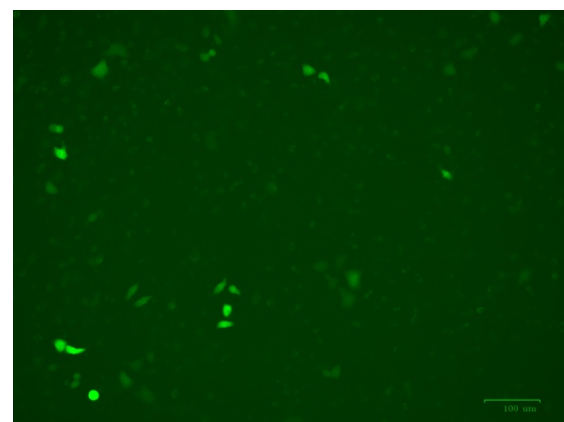


Fig 4: HeLa cells 48h post-transfection with pEBV-YFP(2347) observed under ZoeTM Fluorescent Cell Imager.

Verification of extracted RNA aspect.

- **Determination of extracted RNA quantity and quality**

Total RNA content was extracted from HeLa cells either 24h or 48h post-transfection in order to further assess the Cas9 level of expression at those times. Before amplifying the Cas9 gene, these extracts were submitted to a spectrophotometer analysis to verify the respective RNA concentrations obtained, as well as the eventual presence of protein and solvent contaminations. The absorbance was measured at 230nm, 260nm and 280nm. Sample quality was assessed based on the calculation of different ratios, and RNA concentration was deduced from the A_{260} value. Results from RNA extracts obtained in 2017 in comparison with those obtained in 2016 by another team are presented in Table 6.

Table 6: Determination of RNA content and quality by spectrophotometer measurements (NT = Non-transfected cells)

Sample	Average A_{260nm}/A_{230nm}	Average A_{260nm}/A_{280nm}	Concentration ($\mu g/mL$)	
p2347	0,12	2,51	27,06	2017 samples (N=2)
NT	0,68	2,05	23,58	
Cas9 24h	0,21	2,51	104,28	
Cas9 48h	0,36	2,38	15,88	
NT	1,74	1,94	603,20	2016 samples (N=1)
Cas9 48h	2,13	1,97	651,52	
Cas9 72h	1,53	1,88	353,52	

The A_{260} values obtained for the 2017 samples were too low ($<0,15$) to allow a relevant RNA concentration calculation. The values obtained were therefore not significant. These low A_{260} values obtained suggest a poor content in nucleic acids which means that the RNA was contaminated. On the contrary, the 2016 samples show a more satisfactory RNA

concentration in accordance with the higher A_{260} values obtained.

The A_{260}/A_{280} ratio values should be around 2,0 ($\pm 0,1$) to qualify an RNA extract as pure. We obtained good values for the 2016 samples, but only one 2017 sample value was comprised in this range over the four measured (Ctrl 48h). The others are slightly higher but still acceptable.

The A_{260}/A_{230} ratio values should be equal to 2,0. In our samples, it is the case for only one 2016 sample measured (Cas9 48h). Globally, the 2016 samples displayed a value closer to this range whereas the 2017 samples are drastically lower. This strongly suggests the presence of a compound absorbing at 230 nm and therefore a solvent contamination.

- **Verification of RNA integrity on 1% agarose gel**

Due to the unexpected results obtained for the RNA content verification by spectrophotometry, all 2017 extracts were migrated on the agarose gel to visually evaluate the aspect of the total nucleic acid content. For each sample, an estimated quantity of 300 ng of RNA was loaded on the gel. Results are presented in Figure 5.

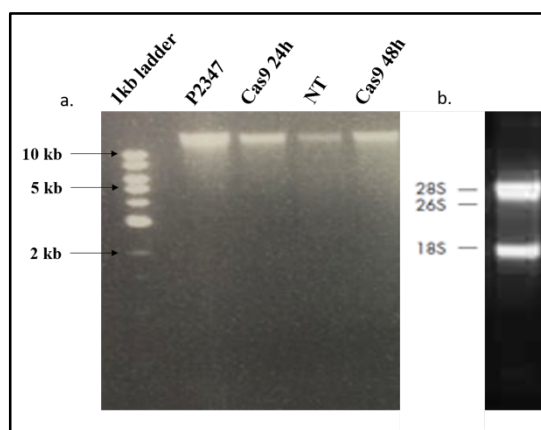


Fig 5: Molecular weight distribution of nucleic acids contained in RNA extracts a. 2017 samples; b. Theoretical results expected.

The migration results did not correspond to the theoretical expectations. Total human RNA should include two bright bands at 5 kb and 1.9 kb, respectively corresponding to the 28S and 18S ribosomal RNA. Only one band was visible for all samples, at a molecular weight above 10kb. The bands obtained were bright and well-defined; no smear was significantly observable. This molecular weight seems to indicate the presence of DNA, rather than RNA, in our samples. In addition to this important DNA contamination, almost no RNA was obtained which means that our RNA extraction did not work properly.

Cas9 gene amplification by RT-PCR. In order to estimate the quantity of Cas9 expression in the transfected mammalian cells at different times, the corresponding gene was amplified from the RNA extracted using a RT-PCR strategy. This test could not be performed with our RNA extracts, due to the absence of RNA in our samples associated with an important DNA contamination. RNA extracted by another team in 2016 were submitted to selective RT-PCR for Cas9 gene and the resulting products were migrated on agarose gel for a subsequent expression estimation by comparison with TBP housekeeping gene (Figure 6).

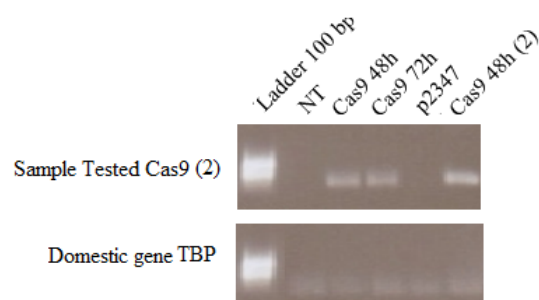


Fig 6: RT-PCR results after Cas9 amplification on 1% agarose gel compared to the domestic gene TBP.

Once band was observable in each well containing RT-PCR products of the cells transfected with the Cas9 containing plasmid. The ladder was highly compact and did not allow a precise molecular weight determination for the bands obtained. However, their molecular weight appeared to approximately match the theoretical expectation of 220 bp, corresponding to the size of the expected amplicon. This result shows the expression of Cas9 in transfected cells. The band thickness and fluorescence did not seem to significantly differ between 48h and 72h post-transfection. Both NT and p2347 controls were negative. Despite the presence of TBP amplification products, the results obtained were not clear enough to allow a relevant estimation of the relative Cas9 gene expression compared to TBP.

Coomassie Blue. A Coomassie Blue was performed in order to verify the presence of Cas9 protein in our samples. Our samples were in presence with Laemmli buffer which prevents us from colorimetric dosage such as Bradford dosage because of the blue color of this buffer. That's why, a BSA range in five dilutions was performed as a reference in order to obtain accurate protein measurements. We performed the protein quantification from the picture (Figure 7) by software Gelix One®G230 Version 4:1 (biostep®, Burkhardtsdorf, Germany). The protein quantity was calculated with a standard curve and a mathematical equation in two repetitions after a densitometric analysis (Figure 8). The average amount of the

protein in our samples from the two mathematical equations are equal to 0,1982 mg/ml. For the BSA range, we saw on the gel that the higher the concentration, the larger was the band. According to the gel, we obtained one band in each well for the BSA range between 50 and 75kDa which is in agreement with the expected size for BSA (66,430kDa). The results obtained for the Cas9 and control samples indicate the total input of the proteins extracted through the presence of multiple bands at different weights along the migration axis. The Cas9 protein, which molecular weight is 158,441 kDa, is not clearly visible on the gel. However, a particularly large band is present for all transfected and non-transfected samples at around 60 kDa. This should correspond to an over-expressed protein encoded within the HeLa cell genome. We could suppose that it corresponds to tubulin, which is implicated in the cell division process and is therefore highly expressed in cancerous cell lines.

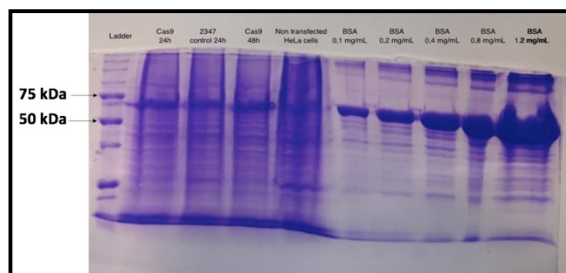


Fig 7: Coomassie Blue of Cas 9 protein expression compared to BSA range.

Well 1: PrecisionPlus Protein Cat#161-0363 Biorad, Well 2: Cas9 protein expression in HeLa cells after 24 hours transfection, Well 3: HeLa Cells after 24 hours transfection with pEB-GFP (2347) plasmid, Well 4: Cas9 protein expression in HeLa cells after 48 hours transfection, Well 5: Non transfected HeLa cells, Well 6: BSA at the concentration of 0,1mg/mL, Well 7: BSA at the

concentration of 0,2 mg/mL, Well 8: BSA at the concentration of 0,4mg/mL, Well 9: BSA at the concentration of 0,8mg/mL, Well 10: BSA at the concentration of 1,2mg/mL.

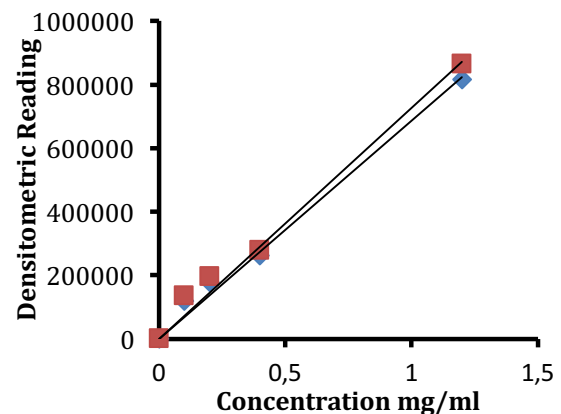


Fig 8: Curves of densitometric readings of the same BSA range gel in two repetitions. Mathematical equations: red squares curve $y = 685499x$, blue dots curve $y = 726545x$.

Western Blot analysis. In order to total intracellular proteins visualization before the Western Blot assay we decided to perform a Coomassie Blue (figure 9, A). It is hard to observe any band on the electrophoresis Gel after Coomassie Blue by the naked eyes. To obtain precise indication of Cas9 protein we decided to do a Western Blot analysis, which is an immunodetection technique using a specific antibody against Cas9 protein. The Cas9 protein weight is approximately 160 kDa. (Figure 9, Both of the Cas9 proteins bands are clearly visible, line 2 and 4) which confirms an efficient plasmid transformation and expression. However, band 4 that indicates the Cas9 protein level in the HeLa cells after 24 hours of the expression is thicker. This can prove that the 24 hours post-transfection expression of Cas9 is more effective than after 48 hours.

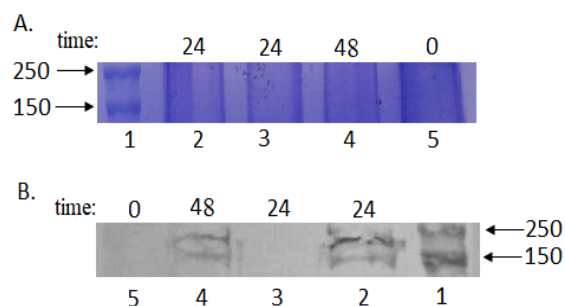


Fig 9: Detection of the Cas9 protein from the extracted total intracellular proteins. A - Coomassie Blue gel colouration analysis and B - Western Blot analysis. Line 1 was protein ladder (Precision Plus Protein™ Kaleidoscope™ Prestained Protein Standards cat #161-0375), line 2 was the Cas9 protein level in the HeLa cells after 24 hours of the expression, line 3 shows control sample, HeLa cells transfected by pEB-GFP (2347) plasmid without Cas9 sequence after 24 hours, line 4 shows the Cas9 protein level in the HeLa cells after 48 hours of the expression, line 5 shows our negative control sample, corresponding to non-transfected HeLa cells.

DISCUSSION

The pSpCas9-2A-GFP plasmid was successfully used to deliver the Cas9 protein into human cells. The observation of fluorescence under the ZOE Fluorescent Imager proved the success of the transfection step as well as the proper plasmid expression. The gene expression level and the quantity of protein synthesized were evaluated at different times post-transfection.

The total RNA content was extracted from the transfected cells either 24h or 48h post-transfection. The results obtained for the RNA analysis showed a poor RNA content along with a high DNA contamination, suggesting that the extraction did not work properly. The RNA extraction was performed twice year-on-year by different operators, and the same poor results were obtained. Various

factors could explain this unexpectedly low RNA content. The extraction protocol followed is efficient for isolating RNA with a size superior to 200 bp. Both ribosomal 18S and 28S RNAs, as well as the Cas9 RNA, are largely above this limit. The presence of degraded RNA in the final extracts is unlikely as no smear was visible at the bottom of the migration gel. It therefore appears that simply no RNA was extracted. According to the different steps performed, the problem could come from the homogenization step which is supposed to induce the inactivation of RNases. An abnormal RNase activity due to an inefficient lysis buffer could result in RNA degradation in the extracts obtained. Once degraded, the resulting RNA fragments could possibly be too small to be retained in the silica membrane leading to their leakage in the flow-through. The abnormal presence of DNA could come from a contamination of the centrifuge tube with the remaining flow-through. We decided for the rest of our experiment to work on the RNA extracted by the group from the previous year as the amount of RNA obtained then was exploitable.

The Cas9 amplification results were consistent with our expectations. The presence of bands for Cas9 transfected cells confirms the presence of Cas9 RNA and therefore the Cas9 gene expression at every time post-transfection tested. It also validates both Cas9-specific primers. For the housekeeping genes, we can identify a band in every well for TBP. Our primers were validated for TBP only. After 30 cycles, we must be at saturation, yet the diffuse aspect of the bands disallows us to quantify for sure the expression of Cas9. This might be due to an inadequate gel texture, or to an old TAE, and could not

lead to a precise and relevant expression quantification. However, we were able to evaluate the relative expression of Cas9 over time. The expression did not seem to significantly differ between 48h and 72h post-transfection. Despite the absence of results for its expression 24h post-transfection, the results strongly suggest the long-lasting Cas9 expression until 72h in cells.

The Cas9 protein was detected by our Western Blot analysis. The presence of bands at the expected size of 160 kDa corresponding to the Cas9 protein (158,441 kDa) was confirmed by the protein extraction and the immunologic detection using Primary antibodies. Moreover, we obtained no band for our two controls (24 hours post-transfection with pEB-GFP (2347) plasmid, and HeLa cells without transfection. The two bands obtained at 24 hours and 48 hours post-transfection appeared globally similar, despite a slightly thicker band at 24h. This suggests a better protein synthesis after 24h, but duplicates have to be performed to confirm this hypothesis. Furthermore, in our Western Blot analysis we observed 2 bands in both 24 and 48 hours samples. We suggest that the higher band, around 200 kDa, illustrates a result of post-transcriptional attachment to the Cas9 protein. Above mentioned attachment could be a reason for higher molecular weight and in the consequence of the slower protein migration rate.

Altogether, those results confirm that Cas9 delivery through a recombinant plasmid can be efficiently used for Cas9 expression in mammalian cells. The protein appears equally expressed from 48h to 72h post-transfection, with a

slightly lower synthesis at 24h. In the context of CRISPR-Cas9 use for gene editing, the nuclease is intended to perform its activity quickly and in a limited period of time in order to prevent OT mutations. The results obtained in the present study for the Cas9 expression pattern suggest that delivering Cas9 by a plasmid transfection in human cells, when used in combination with a guide RNA inserted into this pSpCas9-2A-GFP backbone, does not appear as an optimal method for this use. Indeed, this method allows an optimal Cas9 expression after 48h only and the long-lasting of the protein over time leading to potential off-target effects. If used prior to the parallel delivery of a guide RNA, this should be injected at 48h post-transfection, when the Cas9 expression appears maximal. However, further experiments should be driven in order to precise those assumptions and better evaluate the Cas9 expression pattern in comparison with other delivery strategies.

FUTURE EXPERIMENTS

All steps have been realized only once, so more experiments about the quantification of Cas9 protein should be done for the repeatability of our results. The transfection of HeLa cells with the pSpCas9-2A-GFP plasmid have to be performed again at 24, 48 and 72 hours to precisely quantify Cas9 expression at those times after plasmid transfection and try to assess any potential difference among time. It would also be interesting to measure the Cas9 expression on a wider time range (from 24h to 7 days for example) using a different type of cell lines. It could help to find the most

efficient time duration of the expression and the most optimal cell line for that kind of experiments. Here, the protein expression only was measured. Cas9 efficiency in genome editing should also be monitored over time in parallel, as its active state only raises potential off-target issues.

Moreover, the plasmid verification could be improved using other restriction enzymes.

The RNA extraction step was the most important difficulty encountered. Improvements should be done in the design of this experiment by adopting several strategies: avoid using a kit already used, working in all sterile necessary conditions, and eventually performing an additional DNase digestion after RNA purification.

Once those steps are optimized, we should compare the expression duration of Cas9 when delivered in a plasmid, and as the protein directly. This could be a great help in determining the optimal Cas9 delivery method for use in genetic engineering, depending on the editing and the cell type targeted.

ACKNOWLEDGMENTS

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Identification of active targets in bioassays of Ivermectin similar compounds

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Jean-Yves TROSSET

Abstract:

Phenotypic-based screening strategy is getting popular in pharmaceutical research to jumpstart drug discovery effort with already active compound on disease related phenotype. However, these screening techniques use no “a priori” knowledge on the mechanism of action i.e. the protein targets the drug candidate interfering with. As such there is a need for an *in silico* target identification protocol from which a target profile of an cell-active compound can be deduced from. This approach is based on compound similarity with compounds already screened against known targets, meaning higher the similarity, better the prediction. We tested our methods on known drugs for which the target is known. We then applied our software for the identification of nuclear targets involved in antibiotics resistant pathway for the polyketide antibiotics Ivermectin. We successfully predict the existence of such nuclear receptors and provide insights for such biological pathways using PubChem Bioassay database

Introduction:

Computational biology or bioinformatics is the science of using biological data to develop algorithms and relations among various biological systems. There are large amounts of chemical and biological data available, this is why it is important for a large range of applications to be able to sort and organize these data.

We focused on the ability to find quickly the known targets of a given small molecule (ligand), for application in drug discovery and assessments of toxicity. In this article, we propose an informatic method to automatize target research from a ligand query. Indeed, it is possible to find this information manually by browsing the different databases websites and using their search tools manually but this can be a tedious work due to the great amount and variance of results.

In this review, the investigated approach is interesting to complement phenotypic screening, used in drug discovery, which has been getting a lot of support during the last years [1]. Phenotypic screening can identify active compounds against a disease-related phenotype but does not give any indication of the target or the drug's action mechanism. One of the main challenges of early-stage drug discovery is to find this mechanism of action. Therefore, being able to find quickly the known targets of such compound may help to understand it's way of action.

Experimental target-based bioassay data from public domain database such as PubChem [2] has been used in order to find all target activities of a specific compound. Combining it with similarity search tool from a query molecule allows to broaden research to proteins that are known to be targets to similar compounds.

Potential targets of interest represent all targets that are shown by the bioassays to be

active when confronted with the query and similar compounds.

We aim to test this *in silico* target finding approach in continuation of a project concerning drug resistance with respect to a natural polyketide, Ivermectin (IVM), a well-known antiparasitic drug with a broad spectrum of activity, high efficacy as well as a wide margin of safety [3].

Ivermectin binds selectively and with high affinity to glutamate-gated chloride ion channels in invertebrate muscle and nerve cells of the microfilaria. This binding causes an increase in the permeability of the cell membrane to chloride ions and results in hyperpolarization of the cell, leading to paralysis and death of the parasite. Ivermectin is also known to bind nuclear receptor in human and parasites [4]. Our previous work was to try to determine the targets of IVM, more specifically to understand its binding with nuclear receptors. The purpose of the work presented in this article is to identify active targets for a given compound, taking IVM and its similar compounds as an example.

Materials and Methods:

Our goal is to speed up the search of targets by automatizing it informatically. In order to do this, we created a python code that does all the following steps automatically starting from an input of the user that we will call the query.

As preliminary work, we tried to look for the targets manually, to be able to know the steps we would have to take to create our program. During our research, we noticed that PubChem's search tool's URL addresses were built very specifically. Each URL includes a constant part and a variable one. The variable part defines the kind of search tool that was used and the item ID (compound, bioassay, etc.) used as the query. We took advantage of this to be able to access all the pages automatically in our python program by adding directly the text string corresponding to the

search tool we wanted to use, and the ID of our query compound into the URL. We call this method logical URL completion.

The program starts with the input of a small target molecule in SMILE format. SMILE format is the conventional way of writing molecules in bioinformatics. In order to broaden the search and get more potential results, we first use the similarity search.

During our approach to validate the model, we searched IVM similar compounds in PubChem database to find other compounds SMILE similar to the query.

The first thing to do is to find the ID of the compound for which we want to find targets, so we can use it as a query for the search tools. Afterwards, logical URL completion has been utilized, as described above, to access the list of similar compounds using PubChem's similarity search tool. Urllib library has been used to download the data (Annexe).

With a similar approach, we downloaded all the bioassays in PubChem, using the bioassay search tool for each of the similar compounds found with the previous step.

The next step consists in the filtering of all the downloaded bioassays by the python program which keeps in a separate file only the relevant bioassays displaying an activity toward a target, represented by "non NULL" and "active target".

Finally, we got a bioassay file which only contains active targets of IVM and similar compounds.

An interesting part during the research was the origin of the organism. Indeed, the origin of the target's organism might be a criteria for the selection of a drug compound. We may want to find only organism-specific targets, or to keep the broad results. To deal with this parameter, we decided to add it into the bioassays file. It is performed by a simple function that can either take all the results or select them only according to the wanted origin.

After collecting all bioassays of IVM similar compounds which correspond to the

criteria, we searched the gene of each active bioassay. Targets' gene data has been stored, which could help for developing new drugs. This data allow to find orthologous genes to the target's genes. Orthologous genes are genes in different species that evolved from a common ancestral gene by speciation. Normally, orthologs retain the same function in the course of evolution. Identification of orthologs is critical for reliable prediction of gene function

in newly sequenced genomes. Finding orthologous genes can help to find uses of drugs known to work in some specific organisms into organisms in which this drug would have a similar target and mechanism of action.

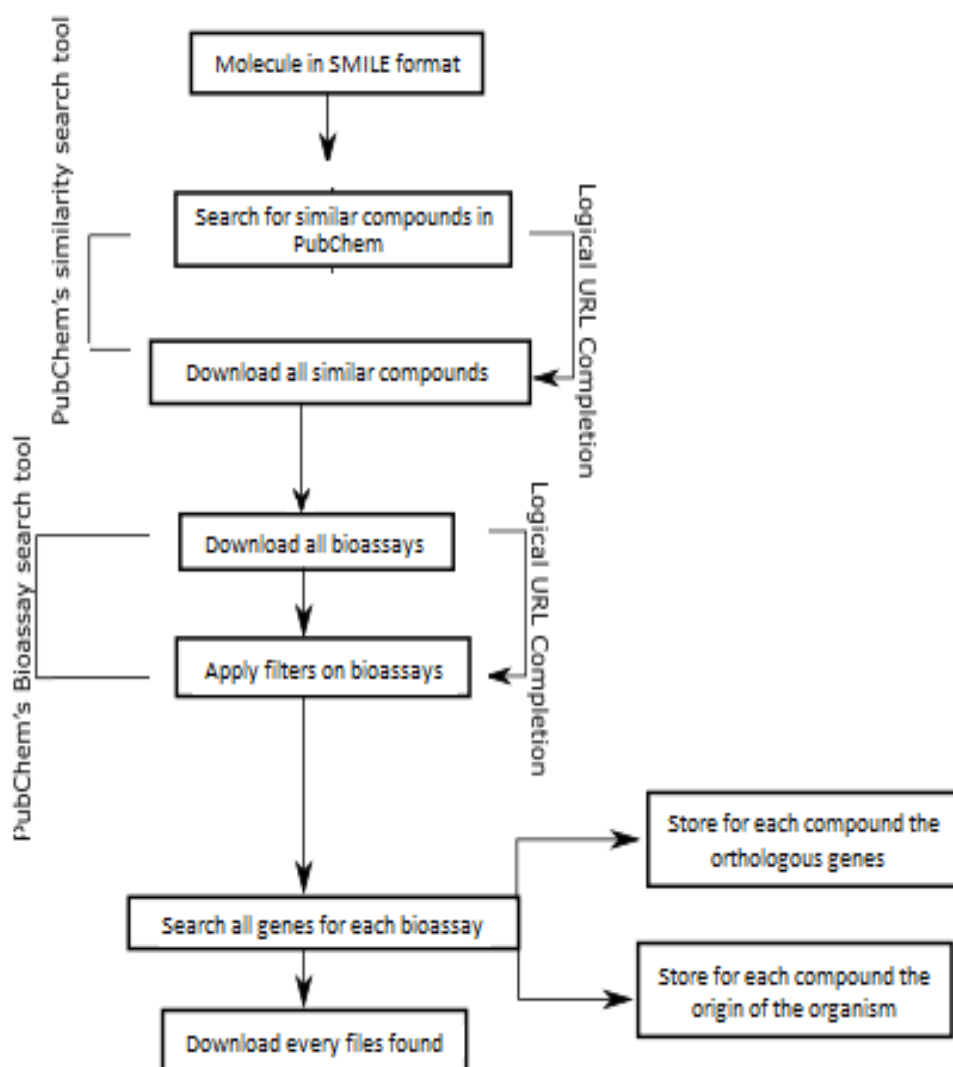


Figure 1: Schematic functions of the script

Moreover, we used a PDB tool to identify targets based on their 3D structure and their documented ligands into the database. To do so we used the XML query the PDB is based on.

The XML query is a formatted in a way to ask the system a specific question. So, for each compound found, a query was formulated in order to look for 3D similar compound with a percentage of similarity that can be set by the user depending on his research. For our tests, we set the similarity at 95%. The output is then the protein-ligand complexes (including these compounds) documented in the PDB. Thus, the output is a list of protein targets of the compounds.

Results:

The program we made has been tested with the IVM molecule as a query.

Once the program has been fed with IVM's SMILE, it returned a folder containing

all the similar compounds in PubChem named by their compound ID. Each of these compound folders contain a list of bioassays, each one giving information about one target of that specific compound. Each bioassay gives the following information: activity of the compound, activity value, cid, sid, aid, aid name, target name and target URL.

For each one of these compounds we looked for 3D structures and protein-ligand complexes in the PDB as explained previously.

So, for each compound a list of PDB id was saved along with the Structure Data File of the compound.

It is very interesting to see that by using this platform and this method we find newly studied compound for IVM resistance like the Farnesoid X Receptor [5] (pdb ID: 4WVD).

On addition, all active bioassays from PubChem are stored in a CSV file. Here in this particular example the file is named: all_active_bioassay.csv file (Figure 2 and 4).

☐ Nom	Modifié le	Type	Taille
☒ compounds_files	06/12/2017 09:07	Dossier de fichiers	
CID_9812710.txt	06/12/2017 08:08	Document texte	33 Ko
list_of_similar_smiles.csv	06/12/2017 08:13	Fichier CSV Microsoft...	14 Ko
☐ Nom	Modifié le	Type	Taille
3085416	06/12/2017 09:07	Dossier de fichiers	
4330618	06/12/2017 09:07	Dossier de fichiers	
6071412	06/12/2017 09:07	Dossier de fichiers	
6321424	06/12/2017 09:07	Dossier de fichiers	
6419971	06/12/2017 09:07	Dossier de fichiers	
6427057	06/12/2017 09:07	Dossier de fichiers	
6435110	06/12/2017 09:07	Dossier de fichiers	
6440492	06/12/2017 09:07	Dossier de fichiers	
9812710	06/12/2017 09:07	Dossier de fichiers	
11957587	06/12/2017 09:07	Dossier de fichiers	
16760011	06/12/2017 09:07	Dossier de fichiers	
42648499	06/12/2017 09:07	Dossier de fichiers	
45114068	06/12/2017 09:07	Dossier de fichiers	
46936176	06/12/2017 09:07	Dossier de fichiers	
53384911	06/12/2017 09:07	Dossier de fichiers	
56603652	06/12/2017 09:07	Dossier de fichiers	
71308642	06/12/2017 09:07	Dossier de fichiers	
71749648	06/12/2017 09:07	Dossier de fichiers	
92023844	06/12/2017 09:07	Dossier de fichiers	
122172888	06/12/2017 09:07	Dossier de fichiers	
all_active_bioassay.csv	06/12/2017 08:13	Fichier CSV Microsoft...	26 Ko

Figure 2: Compounds folders named by compound ID containing bioassays for each of them



Figure 3: **A:** Example folder containing the files downloaded for one compound. **B:** PDB id of targets of compounds with 95% similarity to IVM (IVM included)

1	activity,acvalue,cid,sid,aid,aidname,targetname,targeturlActive,1.9953,11957587,50106505,1815,qHTS for differential inhibitors of proliferation of Plasmodium falciparum line 7G8, NULL, NULL				
2	Active,1.122,11957587,50106505,1816,qHTS for differential inhibitors of proliferation of Plasmodium falciparum line GB4, NULL, NULL				
3	Active,11957587,50106505,1822,qHTS for inhibitors of Plasmodium falciparum proliferation: Summary, NULL, NULL				
4	Active,0.7943,11957587,50106505,1876,qHTS for differential inhibitors of proliferation of Plasmodium falciparum line 3D7, NULL, NULL				
5	Active,2.5119,11957587,50106505,1877,qHTS for differential inhibitors of proliferation of Plasmodium falciparum line D10, NULL, NULL				
6	Active,1.7783,11957587,50106505,1882,qHTS for differential inhibitors of proliferation of Plasmodium falciparum line Dd2, NULL, NULL				
7	Active,1.11957587,50106505,1883,qHTS for differential inhibitors of proliferation of Plasmodium falciparum line W2, NULL, NULL				
8	Active,1.122,11957587,50106505,1886,qHTS for differential inhibitors of proliferation of Plasmodium falciparum line HB3, NULL, NULL				
9	Active,7.0795,11957587,50106505,2326,qHTS Assay for Inhibitors of Influenza NS1 Protein Function, NULL, NULL				
10	Active,0.1169,11957587,50106505,488745,"Quantitative high throughput screen for delayed death inhibitors of the malarial parasite plasmodium, 96 hour incubation", NULL, NULL				
11	Active,1.6511,11957587,50106505,488752,"Quantitative high throughput screen for delayed death inhibitors of the malarial parasite plasmodium, 48 hour incubation", NULL, NULL				
12	Active,6.3096,11957587,50106505,540276,qHTS for inhibitors of binding or entry into cells for Marburg Virus, NULL, NULL				
13	Active,6.3096,11957587,50106505,540276,qHTS for inhibitors of binding or entry into cells for Marburg Virus, NULL, NULL				
14	Active,11957587,268738266,1159580,"The chemical genetic matrix (CGM) dataset as reported in Wildenhain et al. (2015) Prediction of Synergism from Chemical-Genetic Interactions by Machine Learning. Cell Systems Volume 1, Issue 6, p383-395", NULL, NULL				
15	Active,11.8832,11957587,312344818,1224819,qHTS assay to identify small molecule antagonists of the estrogen related receptor (ERR) signaling pathway, ESRR, https://pubchem.ncbi.nlm.nih.gov/target/gene/ESRR				
16	Active,13.3332,11957587,312344818,1224821,qHTS assay to identify small molecule antagonists of the estrogen related receptor signaling pathway with the pleiotropic PPARgamma coactivator (PGC), ESRR, https://pubchem.ncbi.nlm.nih.gov/target/gene/ESRR				
17	Active,18.8336,11957587,312344818,1224822,qHTS assay to identify small molecule antagonists of the estrogen related receptor signaling pathway with the pleiotropic PPARgamma coactivator (PGC) ### cell viability counter screen, NULL, NULL				
18	Active,,16760011,26756590,504749,qHTS profiling for inhibitors of Plasmodium falciparum proliferation, NULL, NULL				
19	Active,,16760011,26756590,504749,qHTS profiling for inhibitors of Plasmodium falciparum proliferation, NULL, NULL				
20	Active,,16760011,26756590,504749,qHTS profiling for inhibitors of Plasmodium falciparum proliferation, NULL, NULL				
21	Active,,16760011,26756590,504749,qHTS profiling for inhibitors of Plasmodium falciparum proliferation, NULL, NULL				
22	Active,,16760011,26756590,504749,qHTS profiling for inhibitors of Plasmodium falciparum proliferation, NULL, NULL				
23	Active,,16760011,26756590,504749,qHTS profiling for inhibitors of Plasmodium falciparum proliferation, NULL, NULL				
24	Active,,16760011,26756590,504749,qHTS profiling for inhibitors of Plasmodium falciparum proliferation, NULL, NULL				
25	Active,,16760011,26756590,504749,qHTS profiling for inhibitors of Plasmodium falciparum proliferation, NULL, NULL				
26	Active,,16760011,26756590,504749,qHTS profiling for inhibitors of Plasmodium falciparum proliferation, NULL, NULL				
27	Active,,16760011,26756590,504749,qHTS profiling for inhibitors of Plasmodium falciparum proliferation, NULL, NULL				
28	Active,,16760011,26756590,504749,qHTS profiling for inhibitors of Plasmodium falciparum proliferation, NULL, NULL				
29	Active,,16760011,26756590,504749,qHTS profiling for inhibitors of Plasmodium falciparum proliferation, NULL, NULL				
30	Active,,16760011,26756590,504749,qHTS profiling for inhibitors of Plasmodium falciparum proliferation, NULL, NULL				
31	Active,,16760011,26756590,504749,qHTS profiling for inhibitors of Plasmodium falciparum proliferation, NULL, NULL				
32	Active,,16760011,26756590,504749,qHTS profiling for inhibitors of Plasmodium falciparum proliferation, NULL, NULL				

Figure 4: Extract of CSV file containing active bioassays

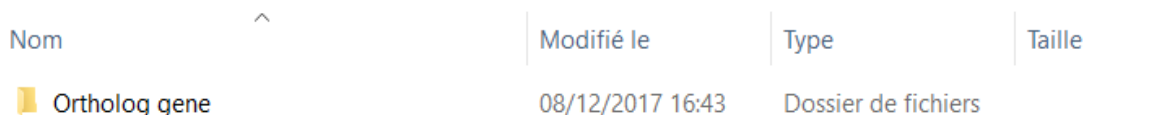


Figure 5: Orthologous genes folder

As represented in Figure 3, where only relevant bioassays were kept, bioassays results can be filtered and sorted. We were interested in finding targets, so we filtered bioassays and only kept the one showing the activity of the compound toward a target. In order to select relevant active bioassays, we only keep bioassay for which target activities is written as “active” and target name as “non null”.

Orthologous genes are also obtained, as explained previously, in a folder containing

every orthologous genes found during the computation.

Discussion:

We finally tested our program with IVM as a query. The results we expected were forecasted by manual research on databases and background from our previous work done last year about the targets of IVM.

The manually searched IVM similar compounds that we expected were present in our program's output among other results. Thus, our results are in adequation with our expectations.

We downloaded the bioassays of all IVM similar compounds for the selected criteria. We decided to sort and select all active and non-null targets manually and we put all these data in a file. Again, those results were confirmed by our program which selected the same non-null active targets. We were able to find in these results the FXR receptor (PDB id: 4WVD) [6] and *C. elegans* glutamate-gated chloride channel (PDB id: 3RHW) [7], or in other words the targets we worked on during our previous work about Ivermectin. Thus, once again, our expectations are in adequation with our results.

We didn't have any background on orthologous gene, so we only checked our results versus a quick manual search. Results appeared to be as expected, but further confrontation might be done in the future. In fact, it often gives more results than those obtained manually because as humans, we are not as methodical as this tool. This is also a reason why we must ensure that all our data is

well stored and ordered as it should remain easy for the user to interpret the results and navigate in the files.

The results we presented were expected and were to be expected as our automatized tool only searches for known data in databases. We only fetch existing and already known data, so there was little surprise to have in the outcome of our program considering it working as intended. This tool now still allows us to quickly fetch and find among a tremendous amount of data every known targets to any drug, and compounds similar to it, and quickly find about orthologous genes to deduce similar targets.

During this work we focused on the results present in PubChem's bioassay library but also implemented a path to search into the Protein Data Bank as well. We can then manually confront the results coming from each platform. In the future, we intend to improve our tool to make our extracted data more homogeneous in order to fuse all the results and automatize a comparison so that results are flagged depending whether they come from either or both the PubChem and the PDB. Our expected final tool would then in a very simplified way look like figure 6.

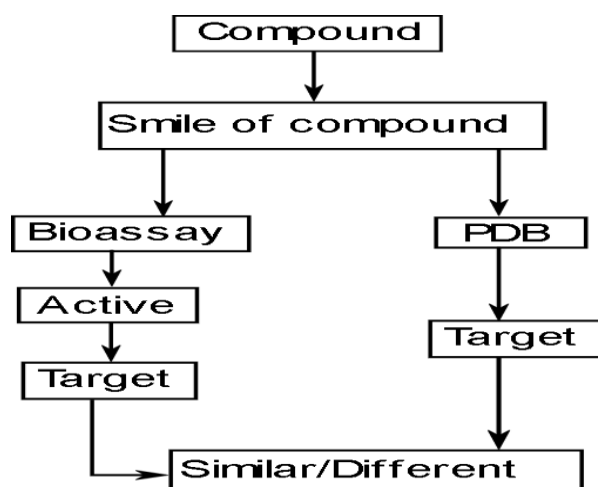


Figure 6: Simplified schematic of the program

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Production and purification of a His-tagged PEP2 protein in *Pichia pastoris*

HERAUD Jules, PHILIP Axel, SARROT Pierre, TODERASC Ruxandra

Applied Transversal Project, R&D - Sup'Biotech

This project aims at producing a recombinant proteinase in a yeast expression system and more precisely in *Pichia pastoris*. This yeast had already been transformed previously, so we produced and induced a batch of it. The production of the recombinant proteins went well and was a success, that was verified by SDS-PAGE. After that, we wanted to purify those His-tagged protein, and also tried to prove that the genomic construction was still present inside the yeast. The purification step failed, after performing it several times, and we think that the Ni-NTA marbles may be the cause of this failure. Regarding the genomic verification, we correctly extracted the genomic DNA of the yeast, but the PCR step with specially designed primers failed too. Those weeks of experimentation led us to think that the purification beads were not efficient anymore, and the PCR step has to be improved by searching better annealing temperatures for instance. Finally, an application for the purified PEP2 had been thought.

Introduction

Several different fungi species are pathogenous. Among them, *Aspergillus fumigatus* (*A. fumigatus*) is one of the most well-known. This saprophytic strain is also one of the most commonly found in ambient air⁽¹⁾. Humans and animals are constantly breathing its spores, that are fortunately eliminated by immune system in healthy bodies. However, in rare cases, invasive aspergillosis and Allergic BronchoPulmonary Aspergillosis Aspergillosis (ABPA) are observed in healthy systems. In immunodeficient patients, an infection can often lead to serious and fatal consequences. Many studies were carried out to better understand the biology of *A. fumigatus* and the pathologies it causes. Thanks to recombinant antigens, a diagnostic test of aspergillosis in immunodeficient and healthy patients have been developed⁽²⁾. Markers were directed toward a ribonuclease (18-kDa RNU), a catalase (360-kDa CAT) and a peptidase (88-kDa DDPV).

An aspartic peptidase PEP2, coming from *A. fumigatus* has been previously transformed into *Pichia pastoris* (*P. pastoris*). This transformed strain was received from Reichard, et al. They chose this protein because there is evidence that additional peptidases than the ones already known are produced by *A. fumigatus*. This 39 kDa protein is present in a small quantity and is not secreted by the fungus in native conditions⁽³⁾. As its name suggests, it hydrolyzes proteins.

A. fumigatus cannot be manipulated in laboratories available at school for the project, because of the high risk of infection. Furthermore, many people are potentially in direct contact with manipulation spaces. To produce the recombinant PEP2, *P. pastoris* expression system was used. *P. pastoris* is a well-known yeast strain in biotechnologies, and widely used for its expression system function. This strain allows the production of high amounts of recombinant proteins, with a highly regulated system⁽⁴⁾. Last but not least, *P. pastoris* is not pathogenous, and can be used as part of this project.

The aim of this project is to produce, purify and study the PEP2 protein in *P. pastoris*.

Material and methods

Media

Subculturing media composed with yeast nitrogen base without amino acids (YNB,291940 - BD Difco™). The production media contains 700ml of BMGY medium (containing 10g of yeast extract (Sigma-Aldrich, cat.Y1625-250G) and 20g of peptone (Grosseron, cat. 211677)), 100mL of potassium phosphate buffer 1M, pH 6.0 (Sigma-Aldrich, cat. P5504-100G), 100mL of yeast nitrogen base without amino acids (YNB,291940 - BD Difco™), 100ml of glycerol 10% (Sigma-Aldrich, cat. G5516) and 2mL of biotin

(2mg in 1ml absolute ethanol with 9ml of H₂O, ref: Sigma-Aldrich, cat.B4501). Inducing media BMMY has the same composition of the BMGY except that methanol 5% (Sigma-Aldrich, 322415) is added instead of the glycerol.

Subculture of *P. pastoris*

P. pastoris strains were taken from -80°C stock and subcultured on YPD (yeast extract, peptone, dextrose) plates. Inoculated YPD plates were grown at 30°C for 48 hours. After incubation, colonies grew on YPD plates and subcultures are ready to be cultured (not shown).

Culture of *P. pastoris*

After subculturing the strains, colonies were picked up and precultured in 10 ml of BMGY medium for 24 hours at 30°C 200 RPM. Then precultures were cultured in 250 ml of BMGY for 48 hours at 30°C 200 RPM.

Induction of *P. pastoris*

Grown cultures of *P. pastoris* were harvested (5 000 RPM for 5 minutes) and the supernatant thrown away. Pellet was resuspended in 50 ml of BMMY and incubated at 30°C 200 RPM for 72 hours. 1 ml of pure methanol is added every 12 hours. A non-induced control is cultured by incubating it in 50 ml of BMGY instead.

SDS-page analysis:

Stacking gel is composed of Acrylamide/Bisacrylamide 30% (Sigma-Aldrich, cat. A3574), TrisHCL 1.5M, pH 8.8 (Sigma-Aldrich, cat. T1503), SDS 10% (Sigma-Aldrich, cat. 1614363), H₂O, TEMED (Sigma-Aldrich, cat. T9281) and Ammonium Persulfate (Sigma-Aldrich, cat. A3678). Separation gel is composed of Acrylamide/Bisacrylamide 30%, TrisHCL 0.5M, pH 6.8 (Sigma-Aldrich, cat. T5941), SDS 10%, H₂O, TEMED and Ammonium Persulfate. Samples were denatured by heating them for 5 minutes at 95°C. Gel was run at 100 V for an hour with a BioRad PowerPac™ HC High-Current Power Supply in a Mini-PROTEAN® Tetra Vertical Electrophoresis Cell. Coloration solution is composed of Isopropanol (Sigma-Aldrich, cat. I9516), Acetic Acid (Sigma-Aldrich, cat. 71251), Coomassie Blue (Labomoderne, cat. CL801450) and water. Decoloration solution is composed of acetic acid 10%.

Protein purification:

Protein solution was purified using ProBond™ Nickel-Chelating Resin (cat. R80115). The marbles were washed three times with Buffer B 1X (buffer B 10X contains Sodium Phosphate Dibasic 200mM

(Sigma-Aldrich, cat. 1551128-5G) and NaCL 5M (Sigma-Aldrich, cat.S7653-250G) and centrifuged without the brake at 900 RPM after each wash. *Pichia* supernatant were mixed with the marbles and incubated on slow gear at room temperature (RT) for 3h. After centrifuging at 900 RPM, the beads were washed three times with Wash Buffer pH 6 (75% wash buffer A 1X (10X contains sodium phosphate monobasic 200mM and NaCL 5M) a 25% Wash buffer B 1X). In order to pick up proteins, the pellet was mixed with Histidine 50mM (Sigma Aldrich, cat. H0750000) twice and put on slow gear for respectively 45 minutes and 20 min at RT.

Protein dosage with Bradford assay:

Bio-rad Protein Assay Dye Reagent concentrate (cat. 5000002) was used. Five dilutions were prepared of BSA for the standard range. 200 µl of each standard solution was put into separate microtiter plate wells and 50 µl of 5X dye reagent was added to each well. After incubation, the absorbance was measured at 595nm.

Genomic DNA extraction:

One yeast culture was suspended in a solution of Lithium acetate 200mM (Sigma-Aldrich, cat. 517992) and 1% SDS. After 5 minutes incubation at 70°C, ethanol 100% were added before centrifuging at 15 000g for 3 minutes. The supernatant was then removed and the pellet was washed with ethanol 70%. After removing as much ethanol as possible. The pellet was dissolved in water and centrifuged. The supernatant was then stored at 4°C for the PCR. Protocol from Løoke *et al.*, 2011⁽⁵⁾.

PCR primers:

PCR primers (Table 1) were designed on an online tool called Primer Blast.

PCR:

Presence of DNA in the pellet was checked by electrophoresis and quantified with the Thermo Scientific Micro-drop Plate. To calculate the concentration of DNA : $A_{260} / A_{320} \times \text{dilution factor} \times 50\mu\text{g/mL}$. Each tube was filled with 5µl OneTaq standard reaction buffer (NEB, cat.B9022S), 0.5µl of dNTPs (NEB, cat. N0447S), 0.5µl of each primers, 0.25 µl of OneTaq DNA polymerase (NEB,M0480L), less than 1µg of DNA and QSF to 25µl of water. We launched the PCR machine with a specific program (Figure 1).

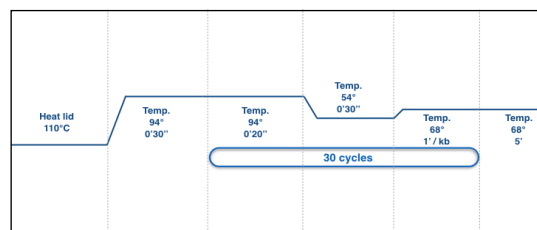


Figure 1: PCR program.

Table 1: PCR primers characteristics.

Name	Sequence	% GC	TM (°C)
Forward Primer 1 (oAP1)	ACATTGACGCTCATGTTTCGC	50	59,56
Reverse Primer 1 (oAP2)	CTTTCGCAGGAAAGCGTCTC	55	59,56
Forward Primer 2 (oAP3)	CTTTCTTCCTCTAGGGTGTCGT	50	59,17
Reverse Primer 2 (oAP4)	CTGGGCTGGTTTTTCGATGG	55	59,47

Results

Growth of *P. pastoris*

After the incubation of *P. pastoris*, the solution was murky with a very dark yellow-brownish color. It also had a harsh smell of yeast, all of this is due to the saturation of *Pichia* inside the solution (not shown).

Induction of *P. pastoris*

Induction of the cultures was carried out and verified with a SDS-PAGE. The gel clearly shows a single stripe standing for a protein around between 20 and 15 kDa for each sample of induced culture (Figure 2). Non-induced culture showed no band (not shown).

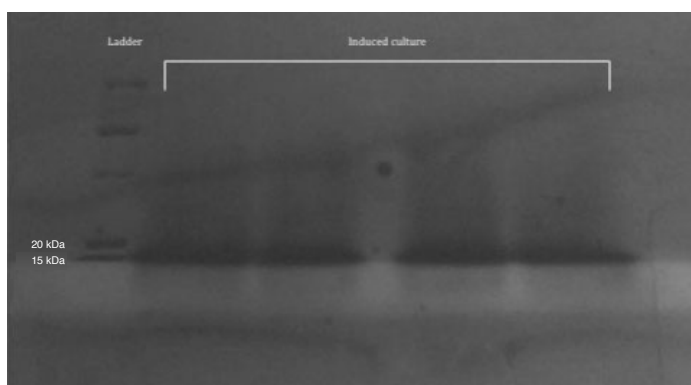


Figure 2: SDS-PAGE carried out with samples of four induced cultures. A stripe is visible for each sample between 20 and 15 kDa.

Purification of recombinant proteins

Purification with nickel bonds was performed three times and same results were obtained. No protein stayed in the purified elution but washing bonds supernatants were full of them, as shown on the SDS-PAGE verification gel (Figure 3). Furthermore, a Bradford assay confirmed those results (not shown).

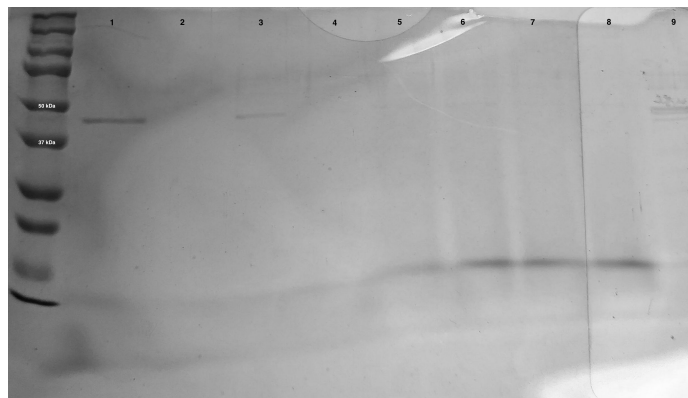


Figure 3: SDS-PAGE carried out after purification. 1, 2, 3, 4: purified protein samples. 4, 5, 6, 7: supernatants kept after bead washing. 8: purified sample containing the recombinant PEP2.

DNA

First, plasmid DNA was extracted and purified, and an electrophoresis on agarose gel was carried out. No DNA was obtained at all (Figure 4A), even when digested (Figure 4B). Afterwards, genomic DNA extraction was carried out and an electrophoresis on agarose gel confirmed that it had been correctly extracted (Figure 4C).

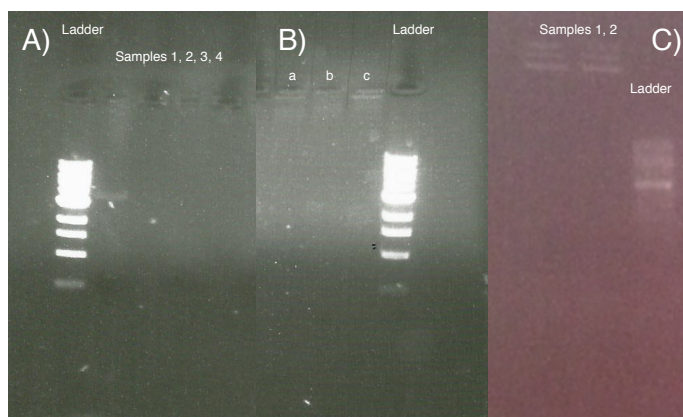


Figure 4: Electrophoresis on agarose gels. A) plasmic DNA extraction from the four induced liquid cultures B) digestion of the extracted plasmic DNA, a: NotI-HF b: EcoRV-HF c: both C) genomic DNA extraction.

PCR

PCR was performed to amplify the region of interest to verify the genetic construction. An electrophoresis on agarose gel was performed with PCR products, and nothing appeared on the gel (not shown).

Discussion

Culture and induction of *P. pastoris* gave satisfying results. Production of recombinant PEP2 has been verified by different gels. The stripes at 15-20 kDa, whereas the protein is 39 kDa, could not be explained. Indeed, those stripes appeared every time the production was performed and verified.

But as it is known that no other protein is secreted by *P. pastoris*, this stripe stands for the PEP2 protein. However, the purification process did not succeed. First, despite several attempts, no protein was present after the purification protocol. It had been supposed that the beads may not be efficient anymore, they could have been stored in a wrong way, or that a manipulation mistake was made. The whole process of production, induction and purification was made another time, and exact same results were obtained. By comparing those results to the ones of the previous group, which purification step succeeded, those results seem to be very surprising. All those arguments led us conclude that the beads, that arrived in the laboratory two years before, were defective and do not allow anymore a correct purification of His-tagged recombinant proteins.

Perspectives

Optimization of the genetic verification is a major research axis. PCR has to be improved, and annealing temperatures have to be enhanced to better correspond to the different couples of primers. The purification step, with new beads, has to be carried out again from the beginning in order to get a sample of purified proteins. Once those recombinant PEP2 proteins will be purified, coupling this protease with antibiotics could lead to a better efficiency of those substances. For instance, *Staphylococcus aureus* has many proteins on its surface⁽⁶⁾, and a synergistic effect of the PEP2 and the antibiotic could be easily thought of. That would lead to limit the use of antibiotics, or improve their efficiency.

Conflict of interest

The authors do not have any actual or potential conflicts of interest that would inappropriately influence, or be perceived to influence, the work presented here.

Acknowledgment

We would like to sincerely thank Jacqueline Bert for her availability, her patience and her kindness. It has been a real pleasure working with her during this project.

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Development of a thermo-responsive plasmid for a cold and hot temperature protein expression.

G. CHRISTIEN, M. FOLLENFANT, M. LHUISSIER, P. LUBRANO, A. MAGNANON

In this article we propose a plasmid construction harboring two gene-encoded proteins. Each protein is regulated by a thermo-inducible promoting system: the promoter of Cold-Shock protein A (CspA) and the p(L) promoter of the heat protein associated with its thermo-labile repressor cl857. Colorimetric and kinetic tests allowed us to characterize the construction that responds to low and high temperatures. The cold protein expression was successfully repressed at 37°C, and gradually increased from 20°C to 12°C. Despite the satisfactory results obtained for this part, several steps remains in order to characterize the construction that respond to high temperatures (above 40°C) and finally assemble the two constructions on the same plasmid in order to get a “2 in 1” system expressing different proteins depending on the temperature. The “2 in 1” system will allow to switch the production of one protein to another according to the temperature.

INTRODUCTION

Development of thermo-induced biological systems have been a subject of interest since many years, yielding various results for different applications.

Temperature stress is notably responsible for many losses of crops yield around the world (10), hence designing a system that adapts to temperature variation for plant protection offers interesting perspectives.

Here will be described the development of a thermo-responsive plasmid (TRP) carrying a “2 in 1” response interesting for plant protection against temperature stresses. The TRP will carry two genes: one expressing a heat protector only at high temperatures (above 30°C), and one expressing an anti-freeze compound at only low temperature (below 15-20°C).

Cold-response. The cold-induced expression is inspired by a family of proteins called Cold-Shock Proteins (CSPs). This wide (8) family of proteins is composed of nucleic acid chaperones that are mostly expressed in response to cold temperature stress.

During cold stress, survival and growth of organisms such as *E.coli* can be significantly impaired. Cell membranes lose fluidity, and mRNAs tend to form secondary structures which impair ribosomal translation, and global enzyme activity decreases (1). To counter the effect of cold temperature on mRNA, the cell expresses CSPs.

CSPs help translation to proceed at low temperatures (11). Despite all the biological processes being slowed or stopped at low temperatures, CSPs are still being expressed by cells. This has been explained by different features, at both transcriptional and translational levels, as studied in *E.coli* Cold Shock Protein A (CspA) (9). This protein is referred to as being the major Csp of *E.coli* and has been chosen as a model for the cold shock response of our TRP.

At the transcriptional level the CspA promoter (CspAp) responds to temperature downshift and has been shown to be stimulated by low temperatures up to three-fold as compared to its activity at 37°C (7). The promoter is supplemented by an upstream element called UP element whose implication in cold induction is still being researched (15, 17).

More importantly, the regulation of CspA expression occurs at a translational level due to its mRNA secondary structure. Indeed, at temperatures close to 37°C, the CspA mRNA is stable only for a few seconds, therefore limiting considerably its translation by the ribosomal machinery (7).

However, at 15°C, the CspA mRNA is much more stable and as a result CspA is expressed at a higher level (5, 6). Indeed, analysis of the *E.coli* protein pool has shown that CspA can represent around 10% of all the protein expressed by the species at 15°C, whereas being very poorly expressed at 37°C (7).

The master regulator of this translational regulation is the CspA mRNA's long 5' Untranslated Transcribed Region (5'UTR). It is an

unusually long sequence that is common to all the cold-shock protein genes (with slight variations). The 5' UTR contains a regulatory sequence called "Cold Box" that plays an important role in the stabilization of the mRNA at low temperature (ATTAAA) (15).

The structure of this 5'UTR is extremely unstable at high temperature, as it is degraded by an RNase, leading to a 12 sec mRNA stability at 37°C against 20 minutes at 15°C (7), temperature at which the RNase is no longer active.

Additionally, there is a regulatory sequence motif that appears within the CspA coding region, after the start codon, called the DownStream Box (DS box) which also contributes to the increased mRNA translation at low temperatures (15). The DS box is a cis-acting mRNA element enabling assembly of the translation pre-initiation complex, likely through interaction with ribosomal protein S1 (19).

All these elements; the CspAp, the CpsA 5'UTR and the CspA DSbox act synergistically to promote cold-temperature specific expression (Figure 1). Our strategy to express any compound of interest at low temperature only will be to use the three regulatory elements of the CspA gene and replace the CspA coding sequence by the coding sequence of the gene of interest. We will also insert the DSbox in the coding sequence of the gene of the compound intended to be expressed at low temperatures (Figure 2).

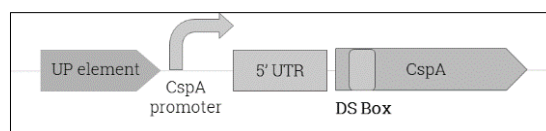


FIG 1: WT CspA genetic structure

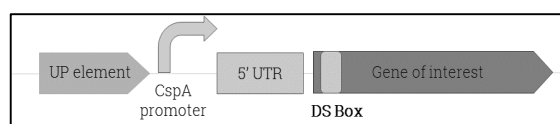


FIG 2: iGEM CspA genetic structure

Heat-response. To achieve protein expression only at high temperature we used the pL promoter and the λ cl repressor. Originally, those components serve as a genetic switch between the lysogenic and lytic phases of the bacteriophage λ life cycle. The creation of the cl857 mutant conferred its thermolabile property, and is widely used for industrial production of recombinant proteins in *E. coli* under up-shift in temperature (13).

Normally, cl repressor activity determines the phage life cycle pathway by selectively inhibiting pL promoter activity. Lysogenic growth is maintained through the binding of the cl dimer to the promoter operator oL/oR regions, preventing

the expression of lytic genes and thus the host lysis. The temperature sensitive cl857 mutant retains those inhibitory properties at low temperatures, but its dimeric conformation becomes highly unstable when temperature is elevated above 37°C (12) (Figure 3).

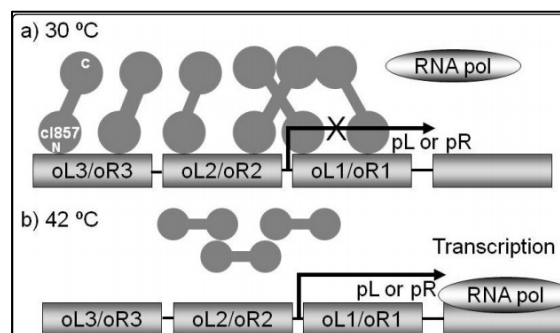


Fig. 3: Representation of the pL promoters controlled by the cl857 repressor (Norma A Valdez-Cruz et al, 2010)

This thermo-induced expression system is of great biotechnological interest as it allows a better culture handling and optimized production yields. Its use is based on the creation of recombinant plasmid vectors in which the gene of interest is inserted downstream of the pL promoter in presence of the thermolabile cl857 repressor (16).

Our strategy to induce the expression of any gene of interest at high temperature only will therefore be to create a recombinant plasmid containing the sequence coding for our reporter gene inserted downstream the pL promoter in presence of the cl857 repressor (Figure 4).

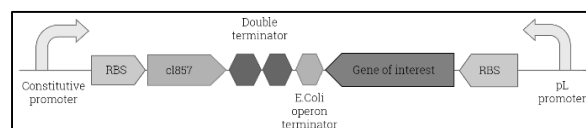


Fig. 4: iGEM pL/cl857 genetic structure

The combination of both cl857/pL system along with the Csp mimicry on a single plasmid with one compound expressed will constitute our TRP, and will allow us to have a polyvalent temperature specific gene expression applicable for various cases.

MATERIALS AND METHODS

Chassis. *E.coli* was used as a chassis for the plasmid construction because its genome is well studied, and has several strains useful for molecular and cellular biology studies. The DH5 α and BL21 strains were supplied by NEB (New England Biolabs) from the partnership with the iGEM Foundation of Boston. The DH5 α strain was used for plasmid construction and cloning procedures because of its high transformation efficiency. The BL21 strain was used for expression assays because of its high levels of protein expression. All our designed sequences have been optimized and sequenced for *E.coli* strain DH5 α .

Plasmid backbones. Our designed sequences were produced on the pSB1C3 and pSB1A3 plasmid backbones. Our final constructions were delivered on the pSB1C3 per iGEM standards. Plasmid pSB1A3 was used for the propagation of BBa_K2282011 sequence. This BioBrick corresponds to the genetic construct permitting cold-expression of AmilCP (blue chromoprotein) based on the use of the Cold-shock protein genetic pattern. The pSB1A3 backbone includes the ampicillin resistance gene and pSB1C3 includes chloramphenicol resistance and a high copy replication origin, which allows a high copy number per cell, facilitating DNA purification.

Primer Design. Primers were designed using the BioBricks suffix and prefix of the backbone pSB1C3. Sequences containing the EcoRI and PstI restriction sites in a reverse form from the BioBricks were extracted using Geneious. The sequences had a size between 20 and 30bp with a T_m around 50°C and 60°C and a GC ratio of 50%. The sequences were then converted into primers. A "TATATA" sequence was added at the beginning of the forward primer and at the end of the reversed primer for the DNA polymerase adhesion. All our sequences were design using Geneious by adding single bricks of DNA to form the whole DNA construct. Each sequence is formed by a BioBrick prefix and suffix at the beginning and the end of the sequence to which the primer binds. Each element was then introduced (Promoter - RBS- Gene of interest- chromo protein- terminator). Geneious was used to detect the restriction sites, design our primers, choose the best T_m for PCR, and visualize our system, while adhering to the pSB1C3 backbone assembly standard.

Design and construction of sequences. Sequences were ordered and synthesised by Integrated DNA Technologies (IDT) from the partnership with the iGEM Foundation of Boston. We imported the sequences from Geneious to IDT. IDT must test the complexity of the sequences before ordering it. The sequence BBa_K2282013 had to be modified from its first

draft due to its repetitive sequence content (arising from two double terminators). We integrated the last part of the sequence in a reverse form and we add a different terminator from the first part of the sequence to avoid concatemers.

Part	Type	Status	Name
BBa_K2282001	Basic	Submitted	AmilCP (from BBa_K592009) containing the DS box sequence at the N terminus
BBa_K2282002	Basic	Submitted	5' Untranslated Transcribed Region of the CspA gene
BBa_K2282003	Basic	Submitted	The UP element followed by the CspA promoter
BBa_K2282004	Basic	Submitted	Phage lambda promoter regulated by the cI857 repressor
BBa_K2282005	Composite	Submitted	Constitutive expression of amilCP
BBa_K2282006	Composite	Submitted	AmilCP with DSbox
BBa_K2282007	Composite	Submitted	5'UTR + AmilCP
BBa_K2282008	Composite	Submitted	5'UTR + AmilCP with DSbox
BBa_K2282009	Composite	Submitted	AmilCP under Upelmt/CspA promoter
BBa_K2282010	Composite	Submitted	AmilCP with DSbox under Upelmt/CspA promoter
BBa_K2282011	Composite	Submitted	AmilCP with DSbox under Upelmt/CspA promoter + 5'UTR
BBa_K2282012	Composite	Submitted	mRFP under pL promoter
BBa_K2282013	Composite	Submitted	mRFP under pL/cI857 system, heat inducible
BBa_K2282014	Basic	Submitted	<i>E.coli</i> K12 MG1655 his operon terminator
BBa_K2282015	Basic	Deleted	<i>E.coli</i> K12 MG1655 his operon terminator (reverse)
BBa_K2282016	Basic	Deleted	mRFP reverse (BBa_E1010 reverse)

Table 1: ID and description of the designed parts.

Cloning strategy. All used sequences were digested with EcoRI-HF and PstI from NEB with a final concentration of 40U/ μ L and the digestion mix was incubated for two hours at 37°C. We used the same protocol for the pSB1C3 backbone. Each digested sequence was ligated with digested pSB1C3 using T4 DNA ligase from NEB with a final concentration of 40U/ μ L. Competent *E.coli* DH5 α were transformed with the ligation product into SOC medium, using the classic heat-shock technique, and then spread on solid LB medium with chloramphenicol and incubated at 37°C for 24 hours. Constructs were checked by colony PCR.

Constructs testing

Colorimetry: Cold and heat response were characterize through colorimetry to study the color intensity emitted by bacteria due to the chromo proteins responsive to the responses. The test was run for three days.

The first day, cold and heat system bacteria were incubated at 37 °C overnight and next day the OD600 was taken every hour until reaching an OD600 = 0.5. After reaching an OD of 0.5, cultures were split in half and held at temperatures accordingly to their genetic construction to be tested:

-For the Heat-response system, one half at 37°C and the other at 27°C

-For the Cold-Response system, one half at 12/15/20/27°C and the other at 37°C

Incubation at 37°C was necessary to obtain a control.

Other controls were used as well for the heat and cold-response system based on the constitutive

expression of AmilCP and mRFP (see BBa_K2282005 and BBa_K2282012). Incubation temperatures were chosen based on bibliographic studies. The assay protocol was inspired by the Takara pCOLD cold-expression vector method (21).

Kinetics

This test was carried out to study the light intensity emitted by bacteria for each response at different times.

Tests were performed on 96 well plates. We prepared different media to test our sequences. Colonies from plates were scraped and inoculated in an Erlenmeyer of 250ml containing 50ml of the LB medium with antibiotics for each DNA sequence. BBa_K2282011 was prepared with ampicillin, BBa_K2282012 and BBa_K2282013 with chloramphenicol and pCOLD (iGEM Tokyo 2016 and 2017) with kanamycin. 1 000x and 10 000x dilutions were carried out in each well to reach OD_{0,01} at time zero. The plates were then incubated at 37°C for 2-3 hours. They were later incubated at 37°C and 15°C, then measurements were carried out every hour over 24 hours for mRFP fluorescence (BBa_K2282013, BBa_K2282012 OD607 and BBa_K1949000) at OD₅₁₁ and the blue light emitted by BBa_K2282011 (amilCP) at OD₅₈₉. Measurements were carried out with the TECAN Infinite 200pro.

RESULTS

Cold response

A. Influence of the DS box on amilCP expression

Impact of the DS on amilCP visual aspect.

The first test was performed by incubating BBa_K2282006 and BBa_K2282005 at 37°C (Figure 5) during 3 days at 37°C. The color was visually observed and compared to the native amilCP reference. The blue coloration appears similar in both conditions. This first result suggests that the presence of a DS box inside the amilCP sequence does not have any impact on the protein color.

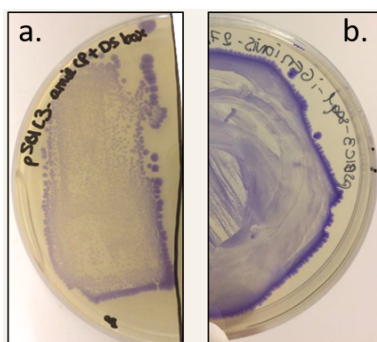


Fig. 5: DH5- α *E.coli* cultivated on LB agar plates supplemented with chloramphenicol at 37°C for 3 days

with and transformed with (a). pSB1C3-BBa_K2282006 expressing amilCP+DSbox, (b). pSB1C3-BBa_K2282005 expressing WT amilCP,

Incubated into liquid cultures at either 15°C or 37°C, two bacteria samples were collected and pelleted by a 1 min centrifugation at 13000 rpm. The resulting pellets were observed after 18h and 42h and the blue coloration visually compared (Figure 6).

The results tend to show a lighter blue color of the amilCP with the DS box compared to the native protein, irrespective of the incubation temperature or duration. This color difference was not observable on Petri dishes and bacterial suspensions. We therefore had to further explore the amilCP properties in both cases to verify that the same absorbance properties were conserved with the BBa_K2282006 part. Also, the lighter coloration at 15°C compared to 37°C in both cases suggests that the DS box alone does not have any enhancing activity on the protein expression at cold temperatures.

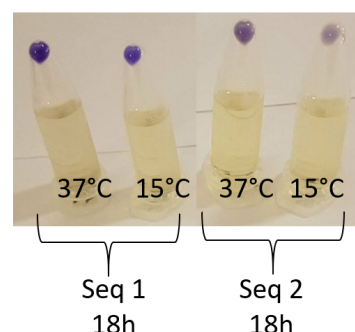


Fig. 6: Liquid culture (LB + chloramphenicol) pellet of DH5- α *E.coli* transformed with pSB1C3-BBa_K2282006 (Seq 2) expressing amilCP+DSbox and pSB1C3-BBa_K2282005 (Seq 1) expressing WT amilCP at 15°C and 37°C

Impact of the DSbox on the amilCP absorbance spectrum. After an overnight bacterial culture at 37°C, the protein content was extracted, and the absorbance spectrum of the obtained solution was measured. We corrected the values obtained with a blank consisting of wild-type culture extracted proteins (Figure 7A).

The native amilCP as well as the amilCP including the DS box both give the same absorbance peak at 588nm. This is consistent with the existing data on native amilCP maximal absorbance (BBa_K592009), and shows that the DS box does not alter the chromoprotein absorption profile, and therefore its color properties (Figure 7B).

This is in accordance with the DS box location outside the chromophore domain. We therefore made the hypothesis that the slight color

difference observed between pellets might be due to a lower protein expression rather than a color variation. This was verified through a kinetics measure of OD588, representative of the amilCP expression, at different temperatures.

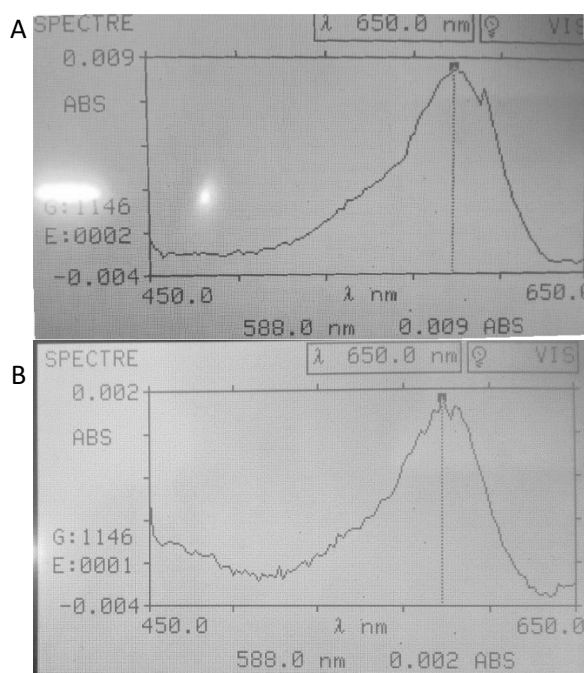


Figure 7: A. Absorption spectrum of the DS box-added AmilCP from part BBa_K2282006. B Absorption spectrum of the WT AmilCP from part BBa_K2282005.

Impact of the DS on amilCP expression rate.

E. coli BL21 were transformed with either BBa_K2282005 coding for native amilCP expression, or BBa_K2282006 coding for the amilCP+DSbox. They were cultivated at 37°C or 18°C for 60 hours and OD588 was measured approximately every 10 min. We took a wild-type, non-transformed bacteria batch as a negative control. From raw OD588 values were withdrawn the background noise corresponding to photon deviation instead of photon absorption. This was determined by calculating the linear function linking OD588 as a function of OD800. This latter was taken as a bacterial growth indicator, as the commonly used OD600nm was too close to our wavelength of interest and could therefore interact in the results (Figure 8).

The increasing of the signal at OD588nm over time for both transformed bacterial batches shows the expression of functional amilCP compared to the non-transformed ones. This confirms that the presence of the DS box does not impact the protein absorbance properties. The same protein expression pattern is observable at both temperatures, with an overall lower expression at 18°C. This predictable result suggests the absence of enhancing properties on protein expression at 18°C from the DS box alone.

Despite the presence of the protein in both cases, these curves also show a slightly better expression without the DS box, the difference being more pronounced at 37°C. This result is in accordance with our hypothesis that the presence of the DS box influences the amilCP expression rate, and may be explained by a perturbation of amilCP mRNA folding, but this statement still requires a mathematical analysis. Originally, the DS box is suggested to enhance mRNA translation during cold shocks (4). However, those results tend to show that without the whole cold-inducing machinery, the DS box does not seem to operate its enhancing action. Further characterization of our complete cold-shock plasmid showed its efficiency through a good protein expression at 15°C and not 37°C. This proves that a lower expression rate possibly induced by the presence of the DS box is not problematic when combined with the CspA promoter and 5'UTR sequence, allowing its activation. Finally, we attempted to sequence this part but did not obtain any satisfying result as the sequencing failed.

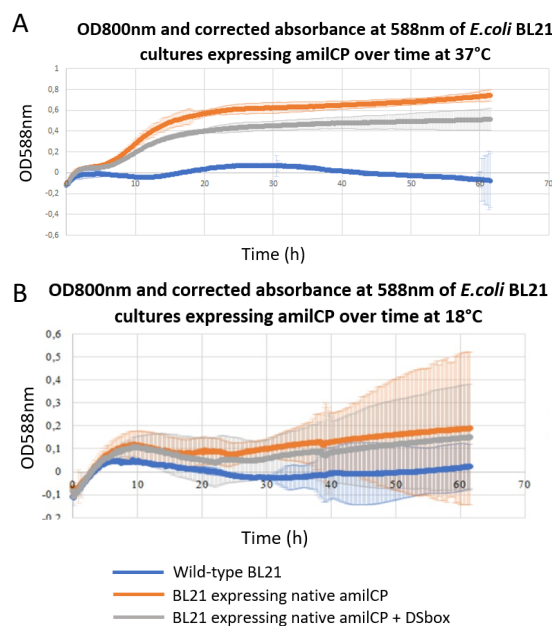


Fig. 8: Expression levels of AmilCP in *E. coli* BL21 over time at **A.** 37°C. **B.** 18°C. Difference of expression between amilCP and amilCP with DSbox is stronger at 37°C, displaying the influence of the DSbox at this temperature.

B. Efficiency of the complete cold-responsive system

Absence of amilCP expression at high temperatures. Transformed bacteria were pre-incubated at 37°C until the OD600nm reached 0.5. Then they were cultured at different temperatures (12°C, 15°C, 20°C, 27°C) with a control at 37°C for different incubation times

(18h, 20h, 42h). The results show a blue color at 15°C but not at 37°C, which mean that the system worked (Figure 9A). The minimal temperature for which a blue color has been obtained is 12°C (Figure 9.B).

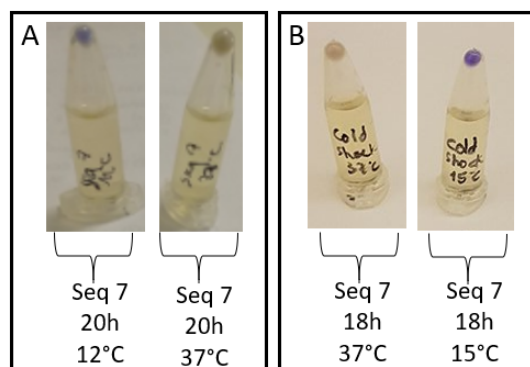


Fig. 9: Pellets of bacteria transformed with BBa_K2282011, after: **A:** incubation 18h at 37°C (on the left) and 18h at 15°C (on the right). **B:** incubation 20h at 12°C (on the left) and 20h at 37°C (on the right)

As most of the experiments on Cold-shock proteins= were carried out only at 15°C and 37°C, we wanted to get a better insight into the cold-shock expression pattern by using other temperatures such as 20°C and 27°C. At 20°C incubation time bacteria transformed with BBa_K2282011 were also blue compared to the wild type bacteria. This result shows the cold response system doesn't react at a precise temperature. Its efficiency likely increases as much as temperature decreases (Figure 10). At 27°C incubation time bacteria were slightly blue and almost uncolored at 37°C. Despite the presence of a light coloration, this results shows the significant expression reduction at 27°C compared to 20°C (Figure 11).

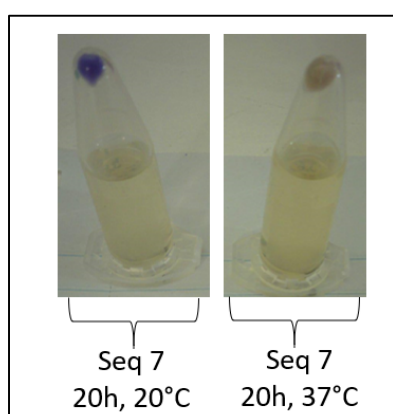


Fig. 10: Pellets of bacteria transformed with BBa_K2282011, after incubation 20h at 20°C (on the left) and 20h at 37°C (on the right).

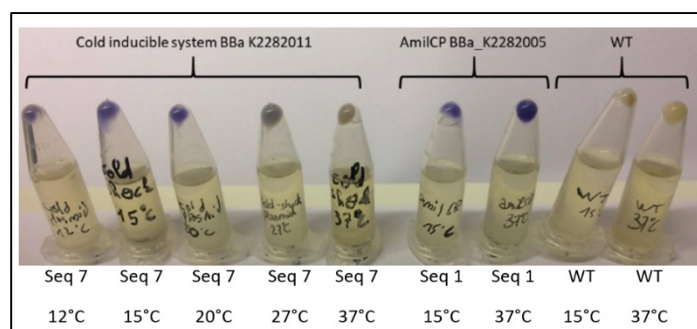


Fig. 11: Pellets of DH5α transformed with BBa_K2282011, after incubation 20h at respectively 12°C, 15°C, 20°C, 27°C and 37°C (on the left), pellets of DH5α transformed with BBa_K2282005 after incubation 20h at respectively 15°C and 37°C (middle), and pellets of wild type DH5α (on the right).

Visual observations confirmed that our part BBa_K2282011 works as expected and allows efficient proteins production only at temperature starting from about 20°C, until low temperature as 12°C. Moreover, the cold response system blocks greatly but not completely the protein expression at higher temperature above 27°C.

According to our characterization results, the mRNA stability seems to increase progressively as the temperature decreases. This observation is in line with a degressive pattern more than an absolute expression switch-off system under a certain threshold. This is in accordance with the existing data on the CspA cold-shock system, which suggests a very short mRNA stability at high temperatures. It is then possible that a low translation still occurs during this time-lapse (1). Additionally, those multiple results were interesting in the sense that we did not find any previous data on amilCP expression between 15°C and 37°C.

As we didn't have any fluorimeter in our laboratory, amilCP was a useful reporter gene for observing the results with the naked eye. But it was difficult to perform OD measurements without extracting proteins because amilCP has its maximum of absorbance at 588nm near to 600nm, the wavelength used to measure the concentration of bacteria. This proximity is prone to alter the actual signal corresponding to the amilCP protein.

Kinetics of amilCP expression with BBa_K2282011. *E.coli* BL21 were transformed with either BBa_K2282011 coding for the cold-dependent amilCP expression, or BBa_K2282005

coding for constitutive amilCP expression. They were cultivated at 37°C for 60 hours and OD588 was measured every 10 min approximately with a Spark 10M Tecan. The improvement of our characterizations would require another kinetics experiment with an incubation at 18°C. From raw OD588 values were withdrawn the background noise corresponding to photon deviation instead of photon absorption. This was determined by calculating the linear function linking OD588 as a function of OD800. This latter was taken as a bacterial growth indicator, as the commonly used OD600nm was too close to our wavelength of interest and could therefore interact in the results.

The results show that the cold shock plasmid (BL21 seq7 abs588/OD800) induces lower expression at 37°C compared to the constitutive one (BL21 seq1 abs588/OD800) (Figure 12). It is in accordance with the CspA construction and the previous picture.

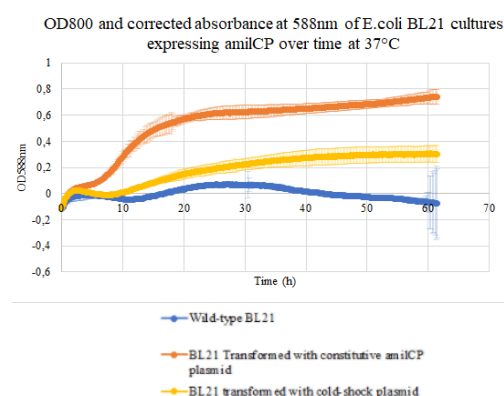


Fig. 12: OD800 and corrected absorbance at 588nm of *E. coli* BL21 cultures expressing amilCP over time at 37°C.

Heat response

After bacterial transformation of DH5- α *E. coli* cells with the heat-shock and the positive control plasmids, we incubated them into Petri dishes added with solid medium at 30°C and 37°C. At 37°C, both bacterial populations appeared strongly red. Surprisingly, at 30°C, the populations also displayed a slight red color. Because of this non-optimal growth temperature, the results were not conclusive. To get a better insight into the mRFP expression level at low and high temperature, we incubated our transformed bacteria in liquid culture and did a kinetics study to monitor the mRFP expression over time at 37°C (Figure 13) and 18°C (Figure 14).

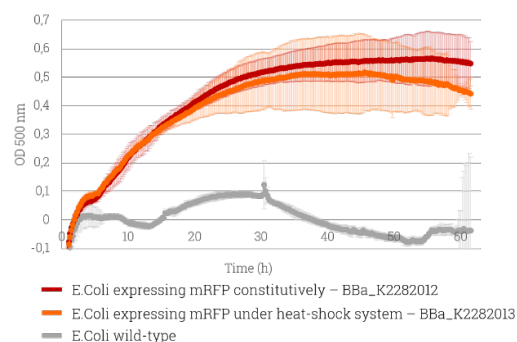


Fig. 13: Expression of mRFP induced by the heat-responsive plasmid compared to the positive control over time at 37°C.

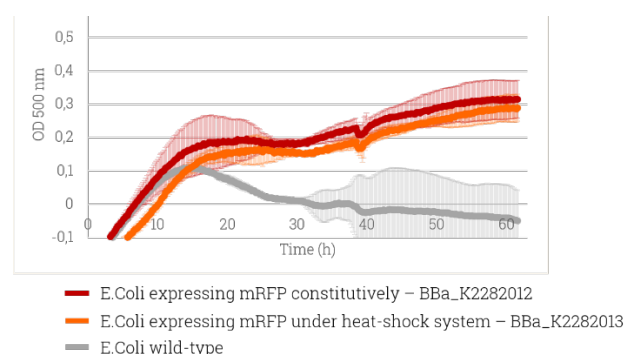


Fig. 14: Expression of mRFP induced by the heat-responsive plasmid compared to the positive control over time at 18°C.

As expected, the graph obtained at 37°C shows that mRFP was expressed at a similar level under the expression of both genetic constructions. The results at 18°C globally show a lower protein expression, explained by a non-optimal growth temperature. Surprisingly, both curves do not differ and suggest a same expression level in presence and absence of the cl857 repressor. The considerable standard deviations and the apparent lowering of the wild-type DH5- α curve was probably due to little evaporation of the culture medium.

DISCUSSION

Cold-response results. In this article we demonstrated a proof of concept using the color reporter AmilCP that our cold-inducible system works as expected. Even though the results we got are limited because we relied on a colorimetric assay, this shows the potential of the mechanism for many applications. The temperature starting point of the high expression activation seems to be around 20°C. This is interesting regarding scientific literature since most of the articles we reviewed about CspA relies on the 15°C threshold rather than the 20°C one (1). The appearance of the gradient indicates clearly that the mechanism is not a strict switch mechanism but a progressive one. The modified

mRNA of AmilCP shows high stability at low temperature, low stability at high temperature, and an intermediate stability between both. What we wanted to achieve with the CspA mechanism is the deletion of the protein expression under its regulation at a temperature close to the activating temperature of our heat response (30°C). At 27°C, the grey color shows low expression of AmilCP and gives us promises about the validation of our strategy.

The promises given by our results to validate the CSPs mimicry as being a reliable mechanism for cold-inducible protein expression are interesting. Perspectives of characterization for the future are numerous. We designed several parts for improving the characterization of our cold-response system. Although we did not have the time to use them in the iGEM competition, they can potentially be used by other teams:

BBa_K2282007: to check the influence of 5'-UTR (5' UnTranslated Region) on the amilCP expression.



Fig. 15: Anderson Constitutive promoter (BBa_J23100) - 5'UTR (BBa_K2282002) - AmilCP (BBa_K592009) - Double terminator (BBa_B0015)

BBa_K2282008: to check if both DSBox and 5'-UTR have an influence on the amilCP expression at different temperatures.



Fig. 16: Anderson constitutive promoter (BBa_J23100) - 5'UTR (BBa_K2282002) - AmilCP & DSBox (BBa_K2282001) - Double Terminator (BBa_B0015)

BBa_K2282009: to check the influence of the UP element and CspA promoter on the amilCP expression (at a transcriptional level)

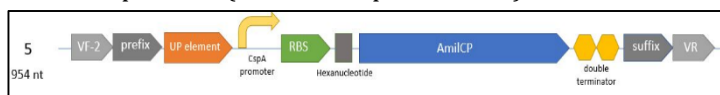
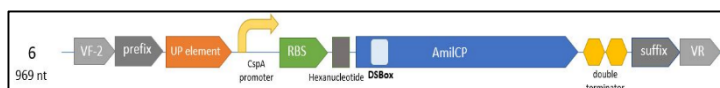


Fig. 17: UPelmt + CspA promoter (BBa_K2282003) - RBS (BBa_B0034) - AmilCP (BBa_K592009) - Double terminator (BBa_B0015)

BBa_K2282010: Checking the influence of the



UP element, the CspA promoter and the DSBox on the amilCP expression

Fig. 18: UPelmt + CspA promoter (BBa_K2282003) - RBS (BBa_B0034) - AmilCP & DSBox (BBa_K2282001) - Double terminator (BBa_B0015)

Heat-response results. The results obtained at 37°C shows that mRFP under the expression of the pL promoter was expressed at a similar level with or without its thermo-labile cI857 repressor. This is consistent with what we expected, as the pL promoter is constitutively activated in both cases at this temperature. Indeed, at this temperature, cI857 is denatured and cannot dimerize. It can no longer bind the pL promoter, allowing a regular mRFP expression (12).

The results at 18°C are not consistent with our theoretical expectations, and confirm the visual results previously obtained. Indeed, the heat-shock plasmid induced a similar mRFP expression compared to the positive control, whereas this genetic construction was supposed to trigger gene expression above 37°C only. This lack of cI857 inhibition could have several explanations. First, the BioBrick construction itself could be questioned. The transcription of mRFP may have induced a conformational change in the double terminator we used, preventing the transcription of cI857 or considerably lowering it to a point where the repressor cannot perform its inhibitory action anymore.

As the two RBS used are the same, we could also question the relative strength of both promoters used. The promoter used for cI857 expression comes from a small combinatorial library of constitutive promoters derived from a consensus sequence by Chris Anderson. Among the different promoters existing in the collection, displaying variable strengths, we chose J23100 which induces the highest relative expression after the consensus promoter (iGEM Registry of Standard Biological Parts). The challenge is then to compare its relative strength with the pL promoter, derived from a phage, in standardized units. To test this hypothesis, we could design a new sequence coding for the mRFP expression under the Anderson promoter used. Measuring the relative expression of the same reporter gene under both promoters will allow us to assess if a significant difference can be observed.

Finally, this result could come from a lower affinity of the Anderson promoter for the RNA polymerase. Its systematic binding to the pL promoter would explain the absence of difference between our constitutive control and the sequence in which the repressor has been added. Several axes of research are then opened to test our hypothesis: we could first use another

promoter for the expression of cl857, known to have a similar affinity for the RNA polymerase as the pL promoter. To assess if the promoter affinity is involved, another solution would be to prevent the “competition” for the RNA polymerase binding between both promoters by designing two separate plasmids instead of one, respectively coding for the cl857 constitutive expression and for the mRFP expression under the pL promoter. Finally, an alternative would consist in constructing another BioBrick by using standard iGEM parts and test the same construction with the pR promoter. Indeed, the system based on pR promoter regulated by cl repressor under the Anderson J23114 promoter has already been successfully tested by previous teams (Part BBa_K608351).

PERSPECTIVES

These parts can provide a solid background concerning the behavior of each elements of the CSP system when characterized separately and together with the DS box. What could also be interesting is to use a GFP rather than AmilCP for protein quantification. We chose AmilCP in our lab due to the lack of a fluorimeter.

Once the system is completely characterized, the next step would be to replace the reporter gene in the construction by a protein of interest to be expressed at low temperature only. This will require further modeling as for the effect of the DS box insertion in the new protein and its expression at low temperature as well. Based on the protein of interest, a Type III expression system might also be needed to enable secretion of the protein. This will also need characterization because secretion system at low temperature might be difficult to implement for a given protein.

The negative results obtained from our heat-inducible system are not a significant drawback for TRP development. The system using the pL/pR promoters with the cl857 repressor has already been widely characterized and there are several possible reasons for its failure in our experiments: Our genetic construction might have simply been wrong: the reverse orientations of both cl857 and mRFP genes might have accounted for the negative results. What is possible is to test the system with a clean BioBrick standard assembly R10, using cl857 and an mRFP gene regulated by either pL or pR. If the mechanism persists in being non-functional, we could test on other high temperature induced mechanisms. For example, the 5'UTR of *P. aeruginosa* virulence factors has been shown to regulate its mRNA stability, much like the 5'UTR of CspA but at high temperature.

In summary, one of the two mechanisms tested worked as expected and we provided a great basis for other projects to rely on for the future. Many things need to be done but we are confident that the TRP development can be achieved and used for many different applications.

Each year plants are subjected to temperature stresses such as ice crystal formation at low temperatures and desiccation at high temperature, leading to yield and economic loss. Transforming a microorganism with the TRP to make it express a specific protectant at the plant surface depending on the outside temperatures could be of great value.

Below 15°C, ice-binding proteins will interact with ice crystals to either inhibit their growth (antifreeze proteins) or favoring the nucleation process (ice-nucleation proteins). Despite their opposite functions, both strategies could be of great help to prevent frost damage in their own way and small-scale tests are required to make a final choice. Above 37°C, light-reflecting compounds will limit evapotranspiration by creating a reflective layer. Once applied on crops, the solution will possess a double protection: anti-drought and anti-frost (Figure 19).

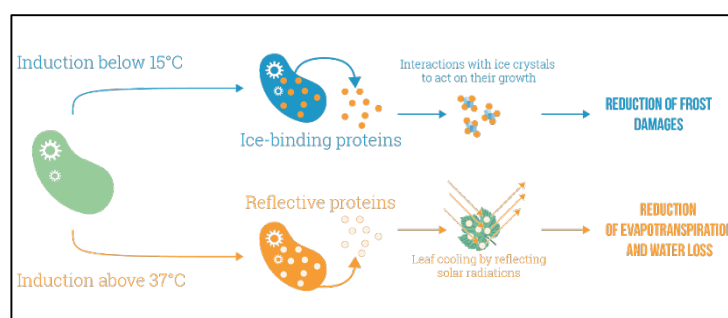


Fig. 19: TRP plant application graphic

Plants “record” the daily average temperature and store it in a total accumulated sum called “thermal time”. As soon as the needed thermal time is reached, the plant can go to the next stage of development. An organism could be transformed with TRP to accumulate a red pigment above a certain temperature. A red progression would then be visible as a function of the heat recorded over time. Alternatively, we could use a TRP-transformed organism to record the average temperature every day, by comparing the color obtained to a reference scale of colors (like pH paper). Being able to visualize thermal time could be useful to predict crop development in a particular location or retro-calculate the best sowing date.

Thermoregulation solutions can be costly to implement for industrial or research

applications. What could be interesting is to make a thermo-responsive organism induce an exothermic reaction as the temperatures decrease, to reheat the medium, and an endothermic reaction as the temperatures increase in order to cool it down.

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Author affiliation : Sup'Biotech

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Prevention of biofilm formation using structural improvement of surfaces in industrial conditions

Quentin BERNARD, Pierre TOURAME, Paul CAROEN

Nowadays, many industrial processes are the object of potential contaminations by natural biological agents (yeasts, bacteria, fungi). Upon stressful environments, these biological agents produce biofilms to be able to survive. These biofilms induce modifications of the biological and biophysical properties of the contaminated surfaces, therefore disturbing the efficiency and/or the yield of industrial processes. Because most bio-chemical solutions are not able to completely destroy biofilms and lots of these chemical agents are now forbidden by the “REACH” European regulation, industrials need new solutions to prevent biofilm formation. Directly inspired from animals and plants, some geometrical structures were found to prevent biofilm formation. By doing a technology transfer from nature to industrial surfaces, we could inhibit biofilm formation by preventing bacteria’s adhesion. This solution would be mechanical, therefore in total respect of European regulation. Improvements on the anti-biofilm structuration may induce interesting properties such as self-cleaning or hyper-hydrophobia. The two tested surfaces showed significant anti-biofilm properties for both motile and non-motile organisms. These results were the first step to build a full attractive proof of concept for industrials.

Microorganisms have the capacity to adhere on biotic and abiotic surfaces. If they are not eliminated and if the conditions are optimal, these cells will form a biofilm and multiply. Cells that constitute a biofilm are known to have a better stress resistance. This biofilm contamination and resistance to treatment has been a major issue in industrial processes since many years [1, 2, 3]. For instance, persistent bacteria on industrial surfaces can lead to a cross contamination of products in food-processing industry [4]. Industrial use chemical products to prevent contamination but are limited into action by the Registration, Evaluation, Authorization and Restriction of Chemicals (REACH). REACH is a European regulation created in 2007 to protect human health and environment, but also to encourage alternatives to chemicals [5]. This regulation laws forbid the use of chemical agents for both industries and private individuals, in which biocides massively used by industrial are included. Therefore, industrials need to find other ways to deal with biofilms. To find an alternative way for fighting biofilm

formation, it is important to understand how a biofilm develops. Biofilm formation is a natural process following multiple different steps [6]. The first step is the surface conditioning made by absorption of ions and macromolecules. Then microorganisms need to enter in contact with the surface, where they will stick. If microorganisms cannot stick to the surface, they will not create a biofilm. Then when a large enough number of microorganisms is stuck to the surface, the biofilm will enter in maturation and leads to complex cell matrix (figure 1). Microorganisms adhesion to the surface is a key step because biofilm process is still reversible at this stage, which is not necessarily the case later. It depends on Van der Waals interactions, electrostatic interactions, acido-basic interactions but also specific interactions between specific surface properties and cells (by cells *pili*, cells adhesines or exopolysaccharides in medium) [7]. So, these interactions are highly dependents on the surface type and the nature of the microorganism [8]. Our project is to make a structural modification of the industrial surfaces that prevent any biofilm

formation by making impossible the cells adhesion. Multiple structural modifications of

these surfaces could be tested, each one with their own properties and methodologies.

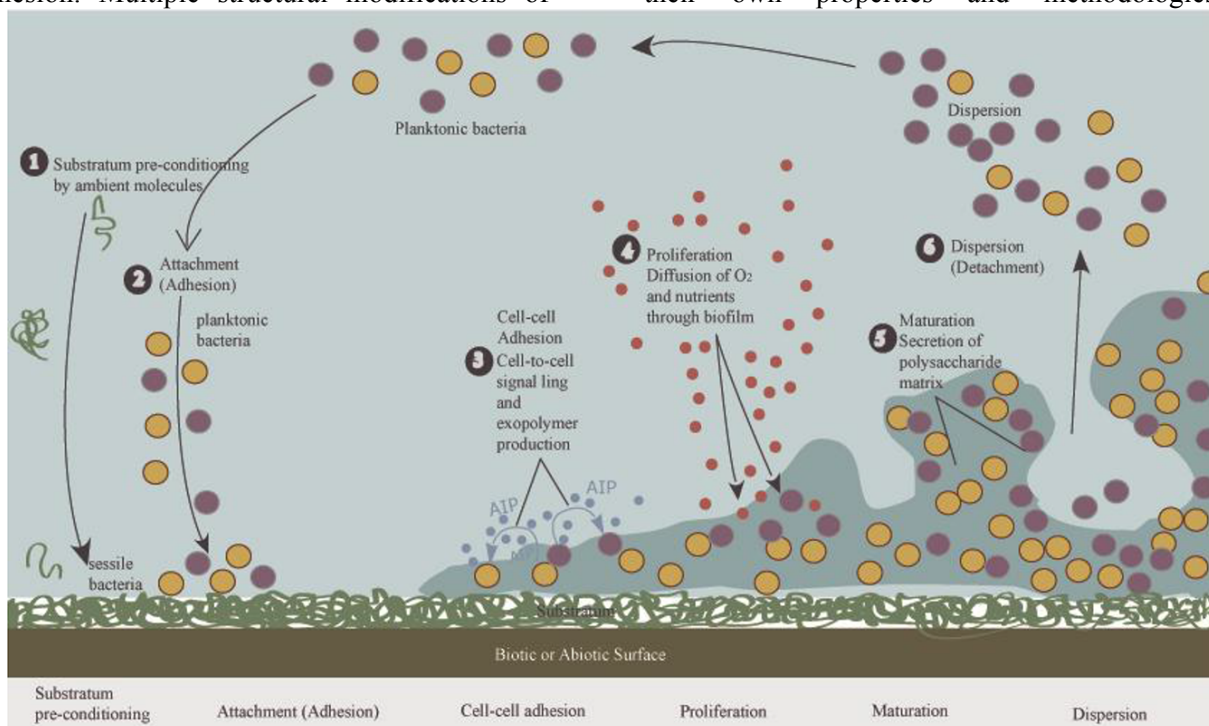


Figure 1: Schematization of biofilm formation [9].

The choice of the structure is an important step and depends on the type of microorganisms responsible for the contaminations and on industrial requirements for other material properties such as anti-oxidation or self-cleaning characteristics. Finally, resistance to specific micro-environmental conditions (temperature, PH, mechanical constrain, nutriments, oxygenation) is sometimes required by industrials and needs to be tested. In this review, our goal was to study two different structuration effects on biofilm formation, with one material and two different microorganisms.

MATERIALS AND METHODS

Surface modification. The surface was modified using plasma etching by oxygen and argon gases. Clean surfaces were loaded inside the vacuum chamber. Then the chamber was evacuated up or down to a pressure of 1.0×10^{-1} Torr. Surface modification was made using radiofrequency at 13.56 MHz to excite the plasma. The flow rates of both gases and the temperature were controlled in the

chamber. Plasma treatment was employed to fabricate three different surfaces with different parameters. Surfaces were incubated with high humidity for 10 days, and pressured by different weights (5g, 50g, 500g, 2kg) to test mechanical resistance. Scanning electron microscopy (SEM) was used to picture the results.

Biofilm formation. 12-well plates (SIAL0513 SIGMA) were used to culture both microorganisms to verify their ability to form biofilm. Each microorganisms (4 ml per well) was cultivated in their respective culture conditions for 90 minutes with no shaking. Then the wells were emptied and washed 3 times using distilled water. After the last wash, 4 ml of medium were added and incubated for 6, 24 and 48h without stirring.

Strains and media. Two strains obtained from the Deutsche Sammlung von Mikroorganismen und Zellkulturen (DSMZ) in freeze dried conditions were used for this study, Bacteria 1 and Yeast 1. Bacteria 1 was cultured at 37°C in

Luria Bertani broth containing Yeast extract (2 g) (Y1625 SIGMA), Tryptone (4 g) (T7293 SIGMA-ALDRICH) and NaCl (4 g) (S7653 SIGMA-ALDRICH) per liter. Yeast 1 was cultured in a special broth. After reception, strains were grown in 5 ml of their respective medium for 12-18h in shaking conditions. Then 1 ml of each sample was transferred in 500 ml their respective culture condition prior to use.

Scaffold fabrication. Pools were made by taping 5 squares of Material 1 together to form a cube with the structures oriented towards the inside of the cube. Mastic sealant was then used to seal, from the inside, each edge. Non-structured samples were used as controls. Material 1 (structured and not-structured) was obtained from a partner laboratory.

Crystal violet analysis. A solution of 0.5% crystal violet (1 ml) was added to each well which were then incubated at RT for 20 minutes. Four washes were performed using tap water to remove excess dye, after each wash the plate was emptied and inverted on a sheet of filter paper and tapped gently to remove any remaining liquid and allowed to dry at RT for 2h. After cell coloration with crystal violet, methanol (200 μ l) was then added to each well which were incubated at RT for 20 minutes in shaking conditions at 20 rpm. The purpose of methanol is to rinse cells stained with crystal violet so that the dye initially staining the cell can be retrieved and solubilized in methanol. The amount of crystal violet is therefore representative of the number of cells. Crystal violet concentration was assessed at OD570nm for each well and sterile well as control. The average OD570nm of non-stimulated cells was set to 100%, and the number of viable cells was calculated as the difference between stimulated and non-stimulated cells. The mean and standard error of the mean was obtained from at least 3 independent experiments. [10]

XTT solution. A Stock solution (5 mg ml⁻¹) was made by dissolving 5 mg of XTT in 1 ml of pre-warmed PBS. XTT was diluted at 1:5 (vol/vol) in pre-warmed PBS before use at 37°C in a water

bath. Stock solution of XTT was stored for a week at -20°C. Because XTT is toxic, we referred to MSDS and took proper care to handle it.

XTT assay for biofilm quantification. The substrate containing biofilms was first transferred to individual wells of a clean 12-well plate containing of PBS (4 ml) in each well. Special care is to be taken during transfer of biofilm. A 1mg/ml solution of XTT (50 ml) was added to each well containing biofilms formed on the substrate. XTT assay is light sensitive thus the test was carried out in the dark (in a biological safety cabinet). Then 1 mM menadione (4 ml) was added to each well. The plate was gently mixed (to avoid disturbing the biofilm). Next the plate containing biofilms was covered with aluminum foil and incubated at Y°C for X h in an incubator (with rotor at low speed). X and Y varied for each microorganism between 36 and 40 °C for temperatures and between 12h and 72h for incubation time. Then blank plates containing the same constituents except biofilms and substrate were prepared, covered with aluminum foil, and incubated along the plates containing biofilms. After incubation biofilms were scraped into the wells using sterile cell scalpels. Using a 10-ml sterile pipette, the mixture containing biofilm was transferred to a sterile 15 ml polypropylene tube. Then centrifuged at 3,500g for 5min at 4°C. Both supernatant and pellet were retained (this cell pellet can be used for dry weight (biomass) determination) and placed of supernatant solution (1 ml) into a 1 ml-cuvette to finally measure absorbance at a wavelength of 492 nm using a spectrophotometer.

RESULTS

Our surfaces were successfully obtained and characterized. Figure 2 A-C shows the schematic, photos, and SEB images illustrating the fabrication result of our nanostructures. Two different structures were obtained using two different lithography techniques. Having different structures type allow to explore their specific effects on biofilm formation on surfaces and may help to comprehend what structuration type should be used to prevent biofilm formation in

specific conditions. The first lithography result shown non-patterned spherical structures, and the second one patterned square structures.

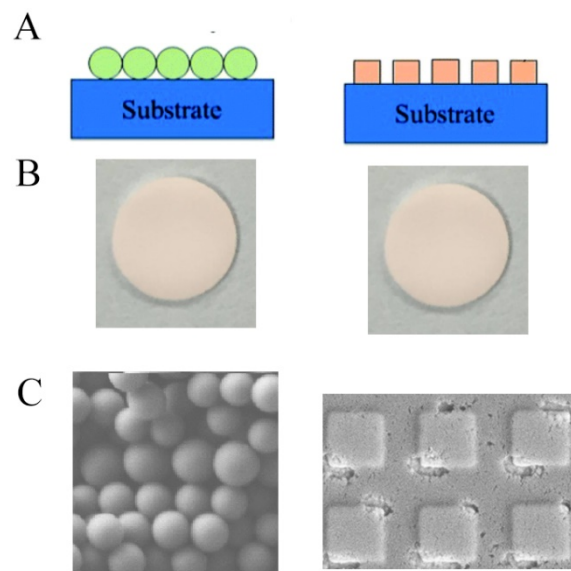


Figure 2: Schematic (A), photographs (B), and SEB images (C) of the nan-structured samples.

The firsts biofilms obtained on 12 wells plates for our both microorganisms are represented in figure 3. Motile and non-motile microorganisms present a clear difference for the location of biofilm formation. Motile microorganisms seem to be able to move where the environment is more suitable to develop a biofilm. In opposition, non-motile microorganism biofilm just develops in the bottom of the well.

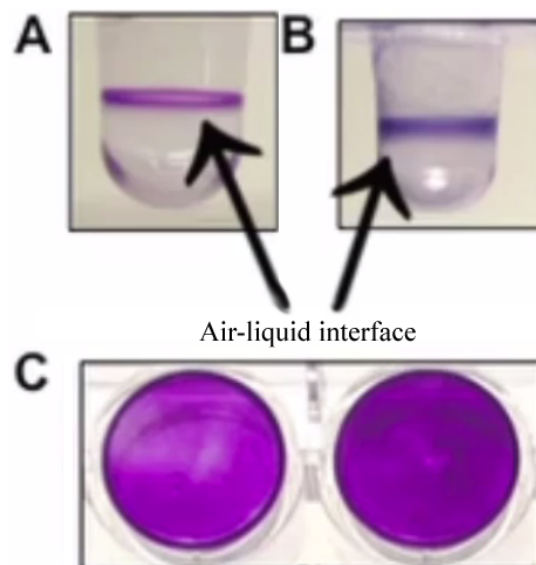


Figure 3: Biofilm after washing with crystal violet, A for motile microorganism with agitated medium, B for motile microorganism with static medium and C for non-motile microorganism cultured in agitated (left) and static (right) media.

The results of biofilm viability studies in different conditions are shown in the figure 4. Both surfaces were able to significantly reduce biofilm formation in agitated and static medium and for both microorganisms. The first surface show better results for preventing biofilm growth in agitated conditions while the second one show better results for static medium. No significant differences were observed between motile and non-motile organisms. No significant differences were observed between crystal violet and XTT assay.

	Crystal Violet			
	Bacteria		Yeast	
	Agitated medium	Static medium	Agitated medium	Static medium
Structured surfaces 1	< 5 %	< 10 %	< 5 %	< 10 %
Structured surfaces 2	< 10 %	< 5 %	< 10 %	< 5 %
Wild Type surfaces	> 90 %	> 90 %	> 90 %	> 90 %
	XTT			
	Bacteria		Yeast	
	Agitated medium	Static medium	Agitated medium	Static medium
Structured surfaces	< 5 %	< 10 %	< 5 %	< 10 %
Structured surfaces 2	< 10 %	< 5 %	< 10 %	< 5 %
Wild Type surfaces	> 90 %	> 90 %	> 90 %	> 90 %

Figure 4: table containing % of biofilm development after 48h in the different conditions tested with XTT and Crystal Violet.

DISCUSSION

This study began with the selection of micro-organisms to be tested. Criteria of selection must match expectancies from potential future users of the surfaces. Problem is that all concerned industrials are not willing to reveal what micro-organisms infect their production sites. Based on the advice we were given from former employees, we oriented our first choices to one bacteria and one yeast that are often found in biofilms. With these two candidates, we could investigate if our surfaces have a different effect on two different kind of living organisms. Further investigation on the type of bacteria, according to its biological characteristics (pilis, positive gram, negative gram, etc.) should be done.

Results confirmed that structured surface has an impact on biofilms development. Obviously, other structures must be tested. As shown, biofilms formation is based on cells adherence thus different patterns influence the micro-organisms capacities to attach to.

Micro-organisms, as we all know, are not fixed but highly motile. Lack of knowledge on this

aspect cannot inform us how in real conditions, the yeasts and bacteria will move around the surfaces. In some specific conditions, when a liquid is present for example, motile bacteria can freely develop into the liquid medium. Further studies should be performed to see if structured surfaces have an impact on this development.

Even though making biofilms grow was not hard to achieve, constructing our pools, on the opposite, was much more complicated. Raw materials selection (having the same composition than our surfaces) which could be transformed in pool forced us to redesign the form of those pools. We opted for a cubic shape rather than circular as expected in regular and well-known well. In addition, tape was used to maintain the structured pool and to avoid leakage. Small residues of biofilms were observed in the corners where the tape was in contact with the liquid. It highlights the need of construction of a simpler system avoiding this process to be free from any impurities. Initial investigation on the effect of the surface was structuration-based. However, selection of other materials to structure could be further investigated. Composite materials, presenting biocide properties, combined to the structuration could present much more effective results compared to ours. Nevertheless, selection of such materials must be compatible with the criteria of the industry and their related laws.

Scalability is a limit to this study. In low quantity, micro-organisms were not able to develop. But it could be due to the quantity of available micro-organisms or to the growth medium. Further investigation should be done with much materials and more growth medium. There are chances for the selected micro-organisms to ignore the structures if they are in high quantities. Adherence would be done on residual wastes which would not be destructed by the structures and would accumulate and eventually fully cover and mask the structures. This is why it is highly probable that users of those surfaces would be recommended to sprinkle their surfaces to avoid previous exposed issue. Scalability is directly linked to mechanic properties of the surfaces which are to be improved and characterized. Further investigations should be completed on mechanic

and thermal resistance of the surface. Our selected material is inert chemically and resistant in usual temperature, but in specific ones (ex: gas tank of an airplane) where temperatures can rise, studies should be performed to see if structures remain intact after treatment and repeated in time. The potential use of the surfaces is for years or even decades but should be confirmed with specific tests.

PERSPECTIVES

This first proof of concept confirmed the potential application of such surface structuration in industrial processes, however more materials need to be tested, especially the ones that are used in the targeted market. In fact, any industrial order for a specific structure will need to set up a specific mission statement where organisms, materials, and conditions are described in order to develop a more specific proof of concept. Another way to explore this technology could be to combine this anti-biofilm property to self-cleaning and hyper-hydrophobic effects. Furthermore, specific mechanical tests for the surfaces need to be conducted to be more appealing for industrials.

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